
Phase 1 PDAC IND: AI Generation Draft 1.0

10.5281/zenodo.xxxxxxxx | ORCID 0009-0007-5457-8667
CEO Kevin Kawchak, ChemicalQDevice | kevink@chemicalqdevice.com

IND version: v1.0 **Repository version:** v4.3.0 **Date:** July 1, 2026 **San Diego**
Repository: kevinkawchak/physical-ai-oncology-trials/tree/main/trial-ind

Independent research paper and practical adoption guide. It is not medical or regulatory advice and is not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor. All figures derive from the author's repository sources and are illustrative unless tied to a cited reference. Disclaimer: this work is independent and is not endorsed or sponsored by any trial sponsor, CRO, site, IRB, regulator, or medical society; and was adapted using Claude Code Opus 4.8. The DOI placeholder 10.5281/zenodo.xxxxxxxx is filled at deposit.

Cover Letter

ChemicalQDevice
San Diego, California
July 1, 2026

Food and Drug Administration
Center for Drug Evaluation and Research
Division of Oncology 1
5901 to 5913 Ammendale Road
Beltsville, Maryland 20705 to 1266

Re: Initial Investigational New Drug Application, Serial 0000 - Daraxonrasib (RMC-6236), Phase 1, LLM-Directed Robotic Whipple in KRAS-Mutated PDAC

Dear Division of Oncology 1 Reviewer:

On behalf of ChemicalQDevice, and in my capacity as sponsor-investigator, I am submitting this initial Investigational New Drug Application (IND v1.0) under 21 CFR §312.23 for a Phase 1, first-in-human, prospective open-label single-arm study of daraxonrasib (RMC-6236), an oral once-daily RAS(ON) multi-selective (pan-RAS) small-molecule inhibitor, administered in the perioperative setting of a large language model (LLM) directed eight-arm robotic pancreaticoduodenectomy (Whipple) for patients with resectable or borderline-resectable, KRAS G12 mutated pancreatic ductal adenocarcinoma (PDAC) [1, 2]. This application is assigned serial number 0000 as the initial submission and is transmitted with FDA Form 1571 as the cover sheet for the submission and the sponsor agreement, together with FDA Form 3674 (clinicaltrials.gov certification) and FDA Form 1572 [3].

This submission is combined in nature. It is concurrently an IND under 21 CFR Part 312 for the investigational drug daraxonrasib and an Investigational Device Exemption (IDE) under 21 CFR Part 812 for the significant-risk, Class II collaborative on-premises robotic surgical device that directs the operative procedure as an early-feasibility study [2, 4]. The LLM functions strictly as a hash-pinned, second-opinion advisory oracle isolated outside the vendor kinematic stack; full device autonomy is prohibited

in accordance with 21 CFR §312.21(e), and a verification-before-generation ten-gate suite governs every advisory output consistent with H.R. 9510, the Verification Before Generation in Physical AI Oncology Trials Act of 2026 [5, 6]. The dual regulatory pathways are presented in an integrated dossier so that the drug and device safety considerations are reviewed together.

We respectfully request the standard 30-day safety review period under 21 CFR §312.40, after which, absent a clinical hold, the sponsor-investigator intends to initiate the study at a single academic site [7]. The dose-escalation design follows a 3+3 schema across three oral once-daily dose levels (DL1 160 mg, DL2 220 mg, DL3 300 mg, the RASolute 302 recommended Phase 2 dose) with a 28-day dose-limiting toxicity window, enrolling up to 18 treated subjects [1].

Contents enclosed. The following components are submitted in the ReGARDD IND order, with the Cover Letter and FDA Forms 1571 and 3674 preceding the generated Table of Contents:

Item	Component
1	FDA Forms 1571 and 3674 (3674 Box B selected; PHS Act 402(j) registration requirements addressed)
2	Table of Contents (generated)
3	Introduction, including drug name, pharmacological class, structure, formulation, route, and duration
4	General Investigational Plan, including the 3+3 escalation rationale and first-year plan
5	Investigator Brochure
6	Proposed Clinical Research (study protocol v1.0.0, informed consent, Form 1572, financial disclosures)
7	Chemistry, Manufacturing, and Control information (drug substance, drug product, placebo, labeling, environmental assessment)
8	Pharmacology and Toxicology information, including the integrated computational evidence base
9	Previous Human Experience (RASolute 302, RASolve 301, Breakthrough Therapy designation)
10	Additional Information (drug dependence, radioactive drugs, pediatric studies)
11	Relevant Information and References with Back Matter

Table 1. Enclosed components of the initial IND, in ReGARDD order, with the Cover Letter and FDA Forms preceding the generated Table of Contents.

Sponsor-investigator and contact. The sponsor-investigator is Kevin Kawchak, Chief Executive Officer, ChemicalQDevice, San Diego, California, who may be reached at kevink@chemicalqdevice.com. The sponsor-investigator's financial interest is disclosed under 21 CFR Part 54 and mitigated by a three-tier independent oversight structure comprising a Data Safety Monitoring Board with halt authority, an Independent Safety Monitor with real-time per-procedure stop authority, and a Physical AI Safety Review Committee operating on a cadence of no more than 90 days [8].

AI-assisted assembly and real-time monitorability. This IND was assembled under a single master prompt that executed a four-stage mermaid to draft to full to final build, with each generated file committed and pushed to one continuously updated GitHub pull request in real time, so that the complete authoring history of the submission is human-monitorable as it is produced and is reproducible

from a hash-pinned deterministic commit (repository v4.3.0) [9, 10]. Figure 1 shows the single-prompt build pipeline and Figure 2 shows the ReGARDD Table-of-Contents architecture that maps each numbered IND section to exactly one source file. The repository is available at <https://github.com/kevinkawchak/physical-ai-oncology-trials/tree/main/trial-ind>.

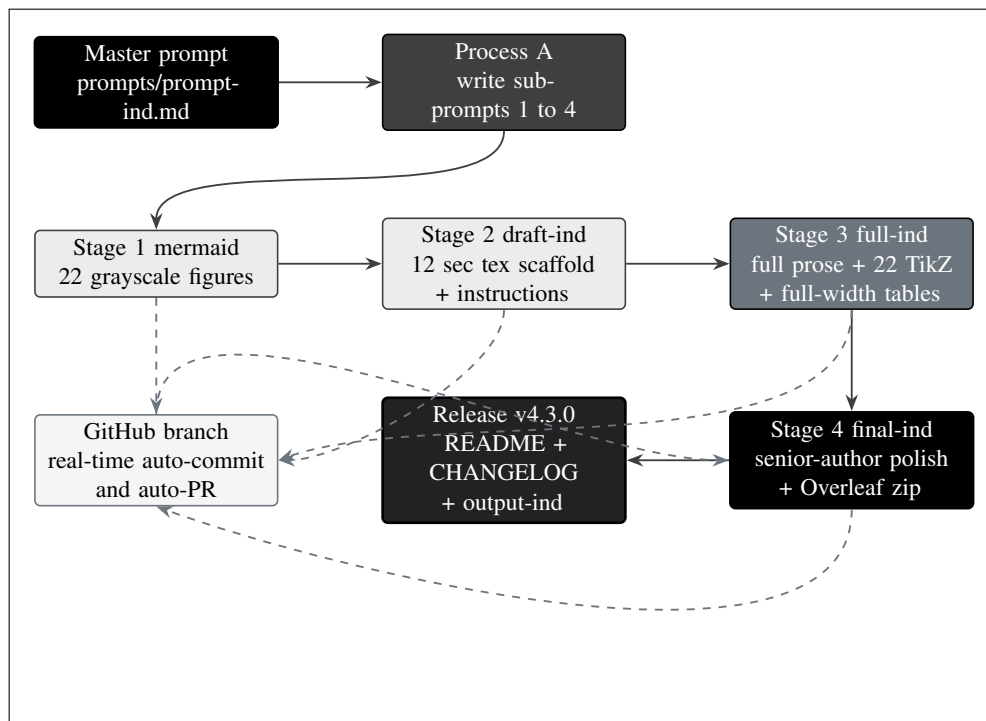


Figure 1. The single-prompt mermaid to draft to full to final build pipeline for the Phase 1 PDAC IND. One master prompt drives Process A (write the four sub-prompts) and then Process B (execute them as Stages 1 to 4). Dashed edges show that every generated file is committed and pushed to one continuously updated pull request in real time, and the feedback edge shows author review returning into the final stage, all without manual intervention.

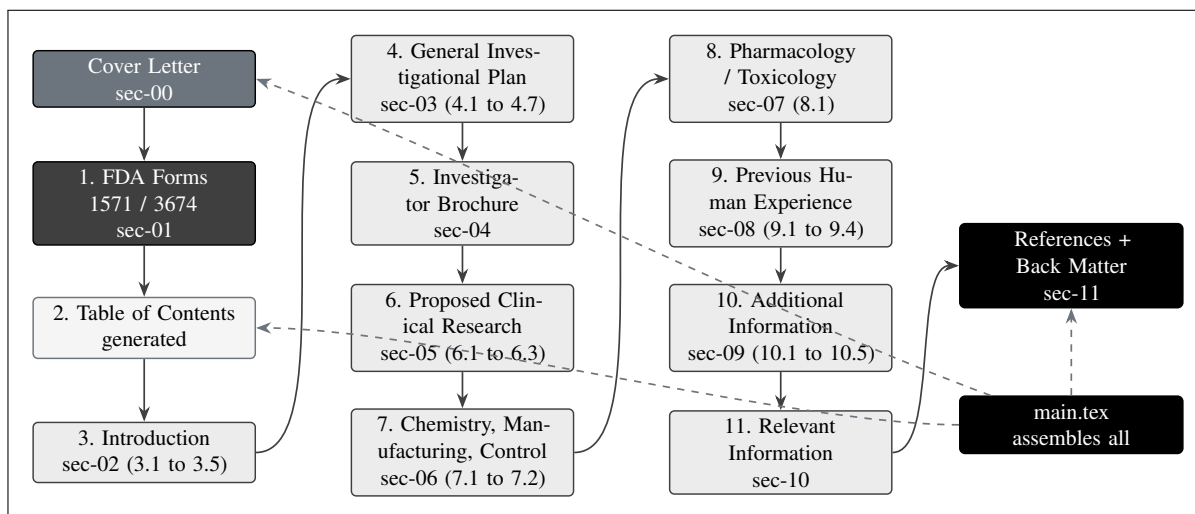


Figure 2. The ReGARDd IND Table of Contents reproduced as the file architecture of the build. The Cover Letter and the FDA Forms 1571 and 3674 precede the generated Table of Contents; the eleven numbered IND sections and the References and Back Matter each become exactly one sections/sec tex file assembled by main.tex, so the document structure and the repository structure match one-to-one.

We appreciate the Division's review and welcome any questions during the 30-day period. The sponsor-investigator is available for a teleconference at the Division's convenience and will respond promptly to any request for additional information.

Respectfully submitted,

Kevin Kawchak
Chief Executive Officer and Sponsor-Investigator
ChemicalQDevice, San Diego, California
kevink@chemicalqdevice.com
IND v1.0; repository v4.3.0

1 FDA Forms 1571 and 3674

The administrative gateway to this initial Investigational New Drug (IND) application is the pair of executed Food and Drug Administration (FDA) forms that open every Center for Drug Evaluation and Research (CDER) submission: FDA Form 1571 (Investigational New Drug Application) and FDA Form 3674 (Certification of Compliance, under 42 U.S.C. §282(j)). These two forms are positioned, together with the Cover Letter, ahead of the Table of Contents so that a reviewer encounters the sponsor agreement, the serial number, the contents map, and the clinicaltrials.gov certification before any scientific module. They are the regulatory envelope around the LLM-Directed PDAC Robotic Darax-onrasib Phase 1 protocol (protocol v1.0.0), a first-in-human, prospective, open-label, single-arm, early-feasibility study that combines an IND under 21 CFR Part 312 with an Investigational Device Exemption (IDE) framework under 21 CFR Part 812 for an on-premises, LLM-directed, eight-arm robotic pancreaticoduodenectomy (Whipple) device used together with oral daraxonrasib (RMC-6236), an oral RAS(ON) multi-selective (pan-RAS) inhibitor [1, 2]. This section summarizes the field-by-field content and the governing logic of each form; the executed, signed PDF versions of FDA Form 1571 and FDA Form 3674 are assembled with the physical submission and are not reproduced inside this LaTeX source, in keeping with FDA's expectation that the original signed forms accompany the package [7, 3].

Figure 1 traces the cover-sheet flow from the Cover Letter through Form 1571 (its serial logic and dual purpose) into Form 3674 (its three mutually exclusive certification boxes) and onward to the assembled IND that starts the 30-day FDA review clock under 21 CFR §312.40.

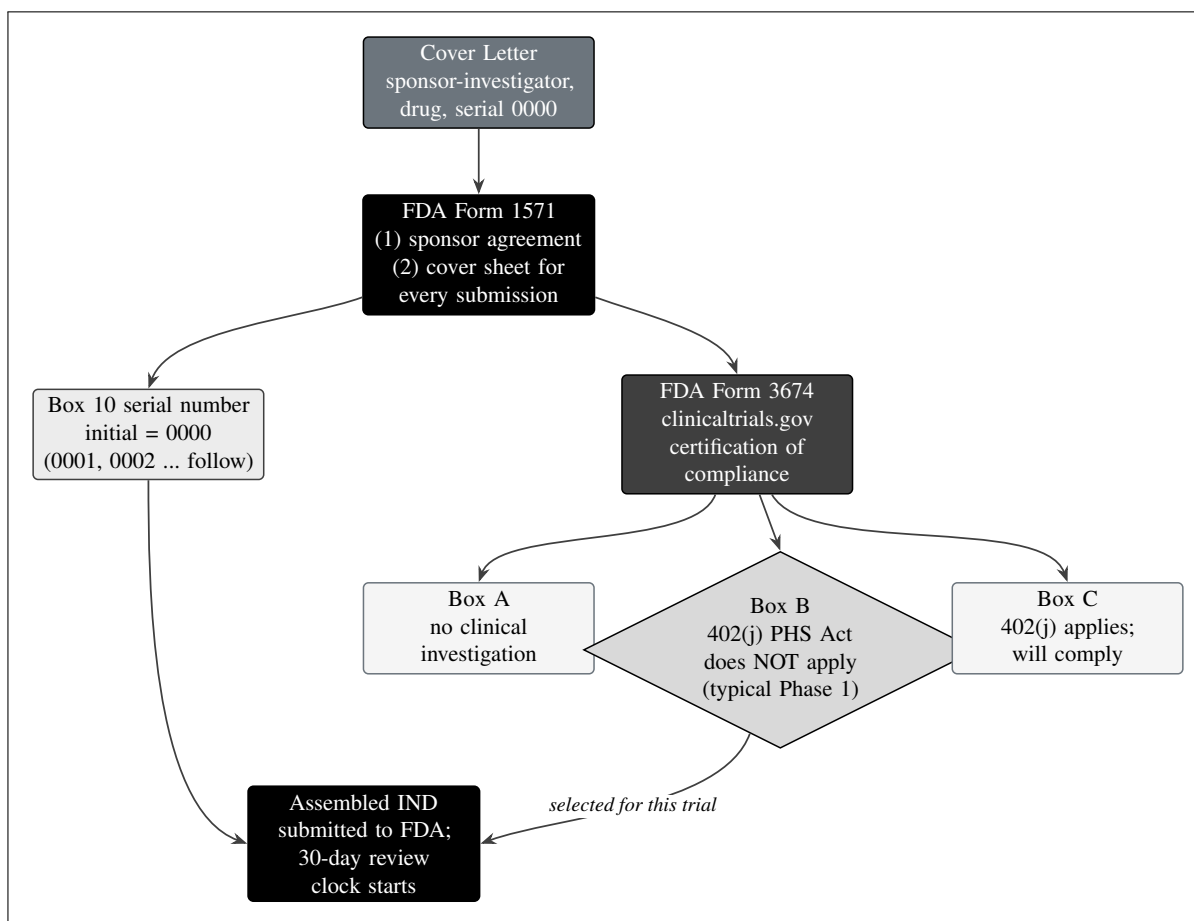


Figure 1. The Cover Letter, FDA Form 1571, and FDA Form 3674 cover-sheet flow. Form 1571 serves the dual purpose of a sponsor-investigator agreement and a cover sheet for every submission; the initial serial number in box 10 is 0000. Form 3674 certifies clinicaltrials.gov compliance through one of Box A (no clinical investigation), Box B (an investigation to which 42 U.S.C. §282(j) / section 402(j) of the PHS Act does not apply, the typical Phase 1 selection), or Box C (the requirements apply). The Cover Letter and the FDA forms precede the Table of Contents.

1.1 FDA Form 1571 - Investigational New Drug Application

FDA Form 1571 serves a dual purpose that defines its central place in the submission. First, it is the binding sponsor-investigator agreement: by signing box 13, the sponsor-investigator, ChemicalQDevice (Kevin Kawchak, Chief Executive Officer, as the named sponsor-investigator), commits that the investigation will not begin until 30 days after FDA receives the IND (unless notified sooner that the studies may proceed), that a reviewing Institutional Review Board (IRB) compliant with 21 CFR Part 56 will be responsible for the initial and continuing review of each study, that the investigation will be conducted in accordance with all applicable regulatory requirements, and that an investigator will obtain informed consent under 21 CFR Part 50 [11, 12]. Second, and on a recurring basis, Form 1571 is the cover sheet that accompanies every subsequent submission to the IND file, including protocol amendments, information amendments, IND safety reports, annual reports, and general correspondence; each such submission carries its own Form 1571 bearing the next sequential serial number so that the agency can reconstruct the complete history of the file [7].

The form opens with sponsor identification and contact fields. Box 1 (Name of Sponsor) and boxes 2 through 5 (Address, including street, city, state, and ZIP) carry the sponsor-investigator entity ChemicalQDevice and its San Diego, California place of business; box 6 (Telephone Number) and the associated electronic-contact information complete the routing so that the assigned project manager and the safety contact can be reached. Box 7 captures the name(s) of any contract research organization, which for this single-academic-site study is limited to the named clinical site and the entities providing investigational product and device support. The IND Number field is left blank on the initial submission because the number is assigned by FDA upon receipt; once assigned, that pre-assigned or receipt-assigned IND number is carried on the Form 1571 of every later serial submission. Box 9 (Name of Drug) records daraxonrasib (RMC-6236), the oral RAS(ON) multi-selective small-molecule inhibitor that binds the GTP-bound active state of mutant RAS in complex with cyclophilin A and suppresses downstream effector signaling, administered once daily [1, 13].

The general-information block describes the proposed research and the phase of investigation. The form indicates that this is a Phase 1 investigation (the first-in-human checkbox), names the indication as histologically or cytologically confirmed resectable or borderline-resectable pancreatic ductal adenocarcinoma (PDAC) with a documented KRAS G12 mutation (G12D, G12V, or G12R) in patients who are candidates for pancreaticoduodenectomy, and lists the protocol title and the combined IND and IDE nature of the application. The contents-of-the- application checkboxes (box 12) are marked to indicate which items of 21 CFR §312.23 are included in this serial submission: the Table of Contents, the Introductory Statement and General Investigational Plan, the Investigator's Brochure, the study protocol(s), Chemistry, Manufacturing, and Controls (CMC) information, pharmacology and toxicology information, previous human experience, and additional information. Box 11 designates the individuals responsible for monitoring the conduct and progress of the clinical investigation and the person(s) responsible for review and evaluation of information relevant to the safety of the drug; for this study those responsibilities map onto the three-tier oversight structure (the Data and Safety Monitoring Board, the Independent Safety Monitor, and the Physical AI Safety Review Committee) described in the protocol [4, 6].

Box 10 is the serial number. FDA requires every submission to an IND to be numbered consecutively, and the initial submission is serial number 0000. Each subsequent submission to this IND file is numbered sequentially as 0001, 0002, 0003, and so forth, in the chronological order in which the agency

receives it, regardless of the submission type. For this initial IND the serial number is therefore 0000, and that value appears both in box 10 of Form 1571 and in the Cover Letter so that a reviewer can immediately confirm that this is the originating submission that starts the 30-day review period under 21 CFR §312.40 [7, 3]. Table 1 summarizes the principal Form 1571 fields and the value or status of each for this initial submission.

Form 1571 field	Description and content	Value / status (serial 0000)
Boxes 1 to 6: Sponsor and contact	Sponsor-investigator name, mailing address, and telephone for the responsible entity.	ChemicalQDevice; San Diego, California; sponsor-investigator Kevin Kawchak, CEO.
IND Number	Assigned by FDA on receipt; carried on every later serial submission.	Blank on initial submission (to be assigned).
Box 7: CRO / site	Contract research organizations and clinical site(s) supporting the study.	Single academic clinical site; no separate CRO.
Box 9: Name of Drug	Investigational drug and active ingredient.	Daraxonrasib (RMC-6236), oral RAS(ON) pan-RAS inhibitor.
Phase / indication	Phase of investigation and the disease studied.	Phase 1, first-in-human; resectable or borderline-resectable KRAS G12 PDAC.
Box 10: Serial Number	Consecutive numbering of every submission to the file.	0000 (initial); 0001, 0002 ... follow.
Box 11: Monitoring / safety	Persons responsible for monitoring conduct and for safety review.	DSMB, Independent Safety Monitor, Physical AI Safety Review Committee.
Box 12: Contents	Checkboxes for the 21 CFR §312.23 items included.	Introductory statement, general plan, IB, protocol, CMC, pharm/tox, prior human experience, additional information.

Form 1571 field	Description and content	Value / status (serial 0000)
Box 13: Sponsor agreement	Signed commitments (30-day wait, IRB review, GCP, informed consent).	Signed by sponsor-investigator; executed PDF assembled with submission.

Table 1. Field-by-field summary of FDA Form 1571 for the initial IND (serial number 0000). The executed and signed PDF of the form is assembled with the physical submission and is not reproduced in this source.

The dual purpose of Form 1571 has a practical consequence for the lifecycle of this Physical AI study. Because each protocol amendment, information amendment, and IND safety report under 21 CFR §312.32 will itself be filed under a new serial number on its own Form 1571, the form becomes the running index of the entire investigational record, including the Physical AI specific reporting events. The protocol defines six Physical AI reporting triggers under 21 CFR §312.32(g) (a serious Physical AI adverse event reported on the 7-day or 15-day clock; a system-drug interaction within 15 days; a cybersecurity incident within 15 days, or within 7 days where there is potential for patient harm; sustained AI model degradation over at least three consecutive procedures or 24 cumulative hours; sim-to-real divergence exceeding twice tolerance, that is a trajectory deviation greater than 4 mm or a force deviation greater than 1.0 N; and a digital-twin discrepancy), and each such report will enter the file as a serially numbered submission cover-sheeted by Form 1571 [4, 5]. In this way the form binds the administrative numbering scheme to the safety-surveillance scheme of the trial.

1.2 FDA Form 3674 - Certification of Compliance

FDA Form 3674 is the certification, required by section 402(j) of the Public Health Service (PHS) Act (codified at 42 U.S.C. §282(j)), that the sponsor has complied with the registration and results-reporting requirements of clinicaltrials.gov to the extent those requirements apply to the submission. The form presents three mutually exclusive certification boxes, exactly one of which must be selected, and the selection determines what the sponsor is affirming about the clinicaltrials.gov status of the studies referenced in the application [7]. The logic of the three boxes is summarized in Table 2.

Box	What the box certifies	Applicability to this Phase 1
Box A	The submission does not reference any clinical trial; that is, there is no clinical investigation at all.	Not selected. This submission references a Phase 1 clinical investigation.

Box	What the box certifies	Applicability to this Phase 1
Box B	The submission references a clinical trial, but the requirements of 42 U.S.C. §282(j) (section 402(j) of the PHS Act) do not apply to that trial.	Selected. This is a Phase 1 investigation; a Phase 1 trial of a drug is excluded from the "applicable clinical trial" definition that triggers mandatory 402(j) registration and results reporting.
Box C	The submission references a clinical trial that is an applicable clinical trial to which 42 U.S.C. §282(j) does apply, and the sponsor certifies it has complied (or will comply) with registration and results reporting.	Not selected for this initial Phase 1. Would apply to a later applicable (for example, Phase 2 or later controlled) trial.

*Table 2. The Box A / B / C certification logic of FDA Form 3674 and the selection for this initial Phase 1 IND.
Exactly one box is checked; Box B is selected with the rationale stated in the text.*

For this initial Phase 1 IND, Box B is selected. The rationale is that the study referenced in this application is a Phase 1, first-in-human investigation of daraxonrasib together with the LLM-directed robotic Whipple device, and a Phase 1 trial of a drug falls outside the statutory definition of an "applicable clinical trial" that triggers the mandatory clinicaltrials.gov registration and results-reporting obligations of 42 U.S.C. §282(j). Under that definition, the controlled clinical investigations of a drug or biologic that are subject to mandatory registration are generally those other than Phase 1 trials; because the present study is Phase 1 (up to n = 18 treated subjects, with approximately 36 screened, at a single academic site, using a 3+3 dose escalation across dose levels of 160 mg, 220 mg, and 300 mg once daily with a 28-day dose-limiting-toxicity window), the section 402(j) requirements do not apply to it, and Box B is therefore the correct and typical Phase 1 selection [7, 9].

Selecting Box B does not signify that the trial will go unregistered. The sponsor records that, notwithstanding the inapplicability of the mandatory 402(j) requirements, voluntary registration on clinicaltrials.gov is intended and is encouraged as a matter of transparency, and that any human clinical trial supported in whole or in part by National Institutes of Health (NIH) funding is subject to the NIH policy on clinical-trial registration and results information submission, which independently requires registration on clinicaltrials.gov regardless of the Form 3674 box selected. The first-in-human Physical AI nature of this study, and the public-trust considerations attached to an LLM-directed surgical device operating under verification-before-generation safeguards aligned with H.R. 9510 (the Verification Before Generation in Physical AI Oncology Trials Act of 2026), further support voluntary, prospective registration so that the design, the staged autonomy limits, and the safety endpoints are publicly visible before the first dose [5, 6, 14]. Should the program advance to a later applicable clinical trial (for example, a

Phase 2 study), Box C would then be selected on that submission's Form 3674 and the corresponding clinicaltrials.gov registration and results reporting would be certified.

As with Form 1571, the certification on Form 3674 is provided on the executed, signed PDF that is assembled with the physical submission; the signature attests that the information on the certification is true and complete and acknowledges that a knowing and willful false certification is a criminal offense under 18 U.S.C. §1001. Because Form 3674 must accompany an IND submission that references one or more clinical trials, a current Form 3674 is filed with the initial IND (serial 0000) and is re-filed, with the box selection re-evaluated, whenever a later submission adds or materially changes a referenced clinical investigation, keeping the clinicaltrials.gov certification synchronized with the evolving scope of the program [7, 8]. Together, the executed Form 1571 and Form 3674, placed ahead of the Table of Contents with the Cover Letter, complete the administrative envelope that initiates FDA's review of this LLM-directed PDAC robotic daraxonrasib Phase 1 application.

2 Table of Contents

Cover Letter	1
1 FDA Forms 1571 and 3674	5
1.1 FDA Form 1571 - Investigational New Drug Application	6
1.2 FDA Form 3674 - Certification of Compliance	8
2 Table of Contents	11
3 Introduction	13
3.1 Introductory Statement	13
3.1.1 Name of the Drug and All Active Ingredients	15
3.1.2 Pharmacological Class of the Drug	17
3.1.3 Structural Formula of the Drug	17
3.1.4 Formulation of the Dosage Form(s) to be Used	17
3.1.5 Route of Administration	18
3.1.6 Duration of the Proposed Clinical Investigation(s)	19
3.2 Summary of Previous Human Experience	20
3.3 Status of Drug in Other Countries	21
3.4 Overview of Preclinical Data	21
3.5 References	22
4 General Investigational Plan	24
4.1 Rationale	24
4.2 Indication to be Studied	27
4.3 General Approach for Evaluation of Treatment	28
4.3.1 Objectives and Endpoints	29
4.3.2 Dose-Escalation Approach	32
4.3.3 Device Readiness and Statistical Framework	34
4.4 Description of First Year Trial(s)	35
4.5 Number of Subjects to be Evaluated	36
4.6 Drug Related Risks	37
4.7 References	39
5 Investigator Brochure	40
6 Proposed clinical research	46
6.1 Study Protocol	46
6.2 Informed Consent	51
6.3 Investigator and Facilities Data	52
6.3.1 FDA Form 1572	53
6.3.2 Disclosure of Financial Interests	54
7 Chemistry, Manufacturing and Control Information	57
7.1 Chemistry, Manufacturing and Control	57
7.1.1 Drug Substance	58
7.1.2 Drug Product	59
7.1.3 Placebo Product	61
7.1.4 Labeling	61
7.2 Environmental Assessment	61
8 Pharmacology and Toxicology Information	67
8.1 Pharmacology and Drug Distribution	67
8.1.1 Pharmacology Summary and Conclusions	68

8.1.2	Toxicology: Integrated Summary	70
8.1.3	Toxicology: Full Data Tabulation	72
9	Previous Human Experience	74
9.1	Marketed experience	74
9.2	Prior Clinical Research Experience	76
9.3	Clinical Care Experience	77
9.4	References	80
10	Additional Information	81
10.1	Drug Dependence and Abuse Potential	81
10.2	Radioactive Drugs	81
10.3	Pediatric Studies	82
10.4	Other Information	82
10.4.1	Physical AI Governance and the Three-Tier Oversight Structure	83
10.4.2	Binding Halt Rules and Stopping Authority	84
10.4.3	Cybersecurity and System Integrity	85
10.4.4	The Seven Physical AI Record Types and 21 CFR Part 11	85
10.4.5	Quality Assurance, Re-Validation, and Decommissioning	85
10.5	Selected References	86
11	Relevant Information	88
11.1	Method-Level Grounding and Verification	88
11.1.1	The Five Verifications in Detail	89
11.2	Figure Grounding: Mermaid to TikZ	90
11.3	Real-Time Auto-Commit and Auto-Pull-Request Monitorability	91
11.4	Repository, Directory, and File Name Matching	92
11.5	Limitations of the AI-Generation Method	92
11.6	Compilation Statement	93
	References and Back Matter	95
	Acknowledgments	95
	Author and ORCID	95
	Ethical Disclosures	95
	Data Availability	97
	Rights and Permissions	97
	Cite This Article	97
	References	98

3 Introduction

3.1 Introductory Statement

This Investigational New Drug (IND) application (IND v1.0, repository v4.3.0) supports a Phase 1, first-in-human, combined IND and Investigational Device Exemption (IDE) clinical investigation of on-premises large language model (LLM) directed eight-arm robotic pancreaticoduodenectomy (Whipple) with perioperative daraxonrasib (RMC-6236) in patients with KRAS G12-mutated pancreatic ductal adenocarcinoma (PDAC). The investigation is filed under 21 CFR §312 for the investigational drug and is coupled to a significant-risk early-feasibility device study under 21 CFR §812 for the LLM-directed robotic platform; the two pathways are coordinated within a single integrated protocol because the device conducts the resection that generates the operative and pathology inputs to the daraxonrasib restart advisory [1, 2]. The sponsor-investigator is ChemicalQDevice (Kevin Kawchak, Chief Executive Officer), and the conduct, oversight, and reporting commitments described throughout this dossier are made in accordance with the applicable provisions of 21 CFR §312.23 and the FDA Phase 1 content and format guidance [7, 3].

The clinical problem this investigation addresses is severe and time-sensitive. PDAC is the third leading cause of cancer death in the United States and the fourth in the European Union, with total five-year overall survival below 13 percent [13]. The pancreaticoduodenectomy is the most technically demanding of all curative-intent abdominal oncology operations: it requires resection and reconstruction of four luminal structures (the pancreas, the duodenum, the common bile duct, and the stomach or pylorus) and dissection around five named vessels (the superior mesenteric vein, the portal vein, the hepatic artery, the celiac axis, and the superior mesenteric artery). The leading lethal postoperative sequence is the fistula-sepsis cascade: a clinically relevant pancreatic fistula that progresses to intra-abdominal infection, hemorrhage, sepsis, and failure to rescue. The 2025 Dutch nationwide cohort of 1000 robotic pancreaticoduodenectomies, which anchors the contemporary human baseline, reported a conversion rate of 10.1 percent, Clavien-Dindo grade III or higher complications of 41.3 percent, International Study Group on Pancreatic Surgery (ISGPS) grade B or C fistula of 24.4 percent, and 30-day mortality of 3.9 percent. Even expert, high-volume robotic surgery therefore leaves a substantial residual of conversions, high-grade complications, and clinically relevant fistulae. The intervention studied here pairs a precise, auditable, on-premises LLM-directed robotic Whipple with the RAS(ON) multi-selective inhibitor daraxonrasib, targeting an in-window R0 resection while layering hard cyber-physical safety limits and optimizing the perioperative drug-restart window.

The device that conducts the resection is an eight-arm parallel coelomic surgical platform carrying 56 mechanical degrees of freedom (eight arms with seven degrees of freedom each) and a 640-channel mixed-rate sensor stack (80 channels per arm), with force channels sampling at 100 kHz and all remaining channels at the 10 kHz command rate, positioning accuracy of 0.05 mm root mean square at a peak tip velocity of 1200 mm/s. An on-premises repository-pinned LLM acts as a second-opinion advisory oracle held strictly outside the robot vendor kinematic stack and hash-pinned to a deterministic seed and a specific commit, never as an autonomous controller; full autonomy is prohibited consistent with 21 CFR §312.21(e). The hard cyber-physical safety envelope comprises a per-arm tip-force cap of 3 N or less, a cumulative cross-arm tip-force cap of 18 N or less enforced at the heartbeat boundary, a 10 kHz heartbeat coordination bus broadcasting a 64-byte frame on a 100 microsecond per-arm watchdog deadline, an emergency arm park within 50 microseconds of a missed or out-of-parameter frame, a cross-arm emergency stop that halts the full platform in 3 ms or less, and a software-independent hardware emergency stop in 500 ms or less consistent with 21 CFR §312.404. A vascular no-fly gate samples at 10 kHz around five named vessels (the superior mesenteric vein, portal vein, hepatic artery, celiac axis, and superior mesenteric artery), with a soft-warning advisory zone at 3.0 mm, a no-fly

blocking zone at 1.0 mm for the superior mesenteric and portal veins and 1.5 mm for the hepatic artery, celiac axis, and superior mesenteric artery, and a hard-stop emergency-stop zone at 5.0 mm. In this Phase 1 the device operates only at the clinician-controlled and supervised-semiautonomous levels under continuous human oversight with immediate manual override, consistent with the staged-autonomy posture in which higher autonomy is reserved for future stages [1, 2, 6].

The eligible population is adults (18 years or older) with histologically or cytologically confirmed, resectable or borderline-resectable PDAC; documented KRAS G12 mutation (G12D, G12V, or G12R); ECOG performance status 0 or 1; and candidacy for pancreaticoduodenectomy, eligible under the broad pan-KRAS criteria of the RASolute 302 program. An exemplar screening subject, PAT-PDAC-0001, is a 68-year-old with a 3.4 cm head-of-pancreas tumor abutting the superior mesenteric vein at 75 degrees, KRAS G12D, microsatellite-stable disease, CA 19-9 of 412 U/mL, ECOG 1, and four cycles of neoadjuvant modified FOLFIRINOX, illustrating the borderline-resectable phenotype the investigation targets.

The investigational plan rests on a three-part acceleration thesis that is the animating premise of the present repository. A patient already enrolled in a Phase 1 PDAC study has little survival margin to spare on administrative delay; the clinical and operational time of a trial (recruitment, treatment, and waiting for progression-free and overall survival to mature) cannot be compressed by writing faster, and neither can the fixed regulatory review clocks, but the administrative and preparation time spent authoring the large documents between steps can be compressed substantially [9, 14]. Faster, internally consistent paperwork therefore means faster treatment: a complete and consistent package reaches the Institutional Review Board (IRB) and the FDA sooner, the 30-day IND safety clock starts sooner, sites activate sooner, and the next dose-escalation cohort opens sooner. This IND, like the protocol and the supporting publication from which it is assembled, was generated through a single-prompt mermaid to draft to full to final pipeline whose every artifact is committed and reviewable in real time, and the same mechanism that produced this dossier maintains the safety reports, amendments, and study reports that the trial will require throughout its lifecycle.

Figure 1 places this IND within the full Phase 1 oncology document landscape it inhabits. Before submission the repository LLM generates the IND dossier, the Investigator's Brochure, the clinical protocol, and the informed consent form; during the trial it produces the 7-day and 15-day safety reports, the annual Development Safety Update Report (DSUR), protocol amendments, and dose-escalation minutes; and after the trial it assembles the tables, listings, and figures, the ICH E3 clinical study report, and the downstream manuscripts and lay summaries. The IND package is the first and rate-limiting artifact in this relay, which is why the acceleration thesis is most consequential here [9].

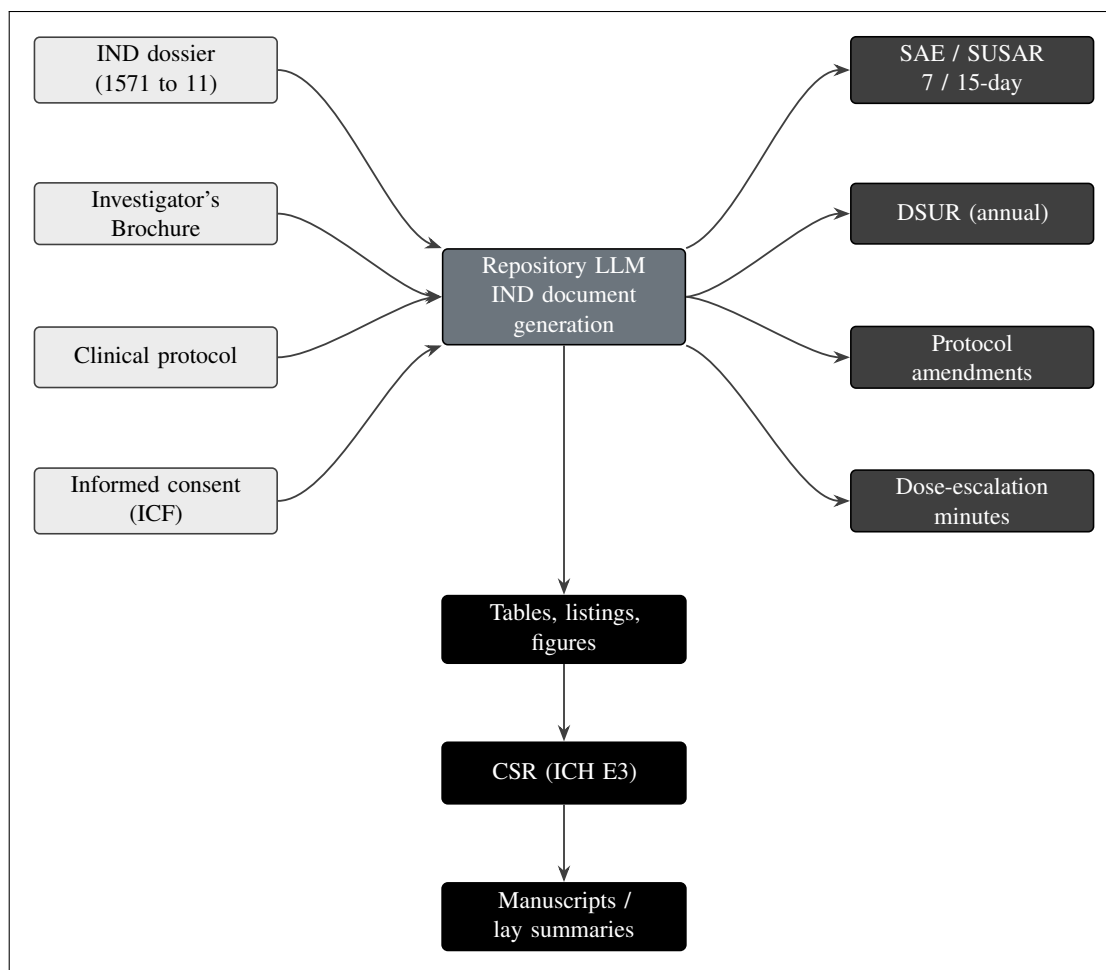


Figure 1. The large Phase 1 oncology document landscape around the IND, grouped into before-submission inputs (the IND dossier, Investigator's Brochure, protocol, and consent), during-trial process documents (7 and 15-day safety reports, the annual DSUR, amendments, and dose-escalation minutes), and after-trial outputs (tables, listings, figures, the ICH E3 clinical study report, and manuscripts and lay summaries), all generated by the repository LLM around the central hub. This figure renders in the Introduction to place the IND in the full document lifecycle the same method maintains.

The remainder of this Introduction follows the ReGARDIND content sequence. It names the drug and its active ingredients (§3.1.1), states the pharmacological class (§3.1.2) and structural description (§3.1.3), summarizes the dosage form (§3.1.4), the route of administration (§3.1.5), and the duration of the proposed clinical investigation (§3.1.6); it then summarizes previous human experience (§3.2), the status of the drug in other countries (§3.3), and the preclinical and computational evidence (§3.4), with the supporting references collected in §3.5. Quantitative and design detail introduced here is developed in full in the General Investigational Plan (§4), the Investigator's Brochure (§5), the proposed clinical research (§6), the Chemistry, Manufacturing, and Control information (§7), and the Pharmacology and Toxicology information (§8).

3.1.1 Name of the Drug and All Active Ingredients

The investigational drug is daraxonrasib, also known by its development code RMC-6236. Daraxonrasib is the single active ingredient of the drug product; there are no additional active pharmaceutical ingredients. It is an orally bioavailable, RAS(ON) multi-selective (pan-RAS) small-molecule inhibitor

that binds the active, guanosine-triphosphate-bound (ON) state of mutant RAS in a tri-complex with the intracellular chaperone cyclophilin A, suppressing downstream effector signaling across the common KRAS oncoprotein variants found in PDAC [1, 2]. The molecular target population for this investigation is defined by the KRAS G12 alleles (G12D, G12V, and G12R) that predominate in pancreatic ductal adenocarcinoma; daraxonrasib provides broad pan-KRAS coverage that spans these alleles, which is the property that makes it suitable for the protocol's molecularly defined eligibility [1].

Figure 2 reproduces the mechanism and pharmacological-class schematic for daraxonrasib that grounds §3.1.1 through §3.1.3. Daraxonrasib forms the tri-complex with cyclophilin A, binds the active GTP-bound state of mutant RAS, intercepts the KRAS G12 variants that define the trial population, suppresses downstream MAPK and PI3K effector signaling, and thereby reduces tumor-cell proliferation and survival; it is dosed orally once daily at the three escalation levels (DL1 160 mg, DL2 220 mg, DL3 300 mg) and carries FDA Breakthrough Therapy designation granted in June 2025 for previously treated metastatic KRAS G12 PDAC [1, 2].

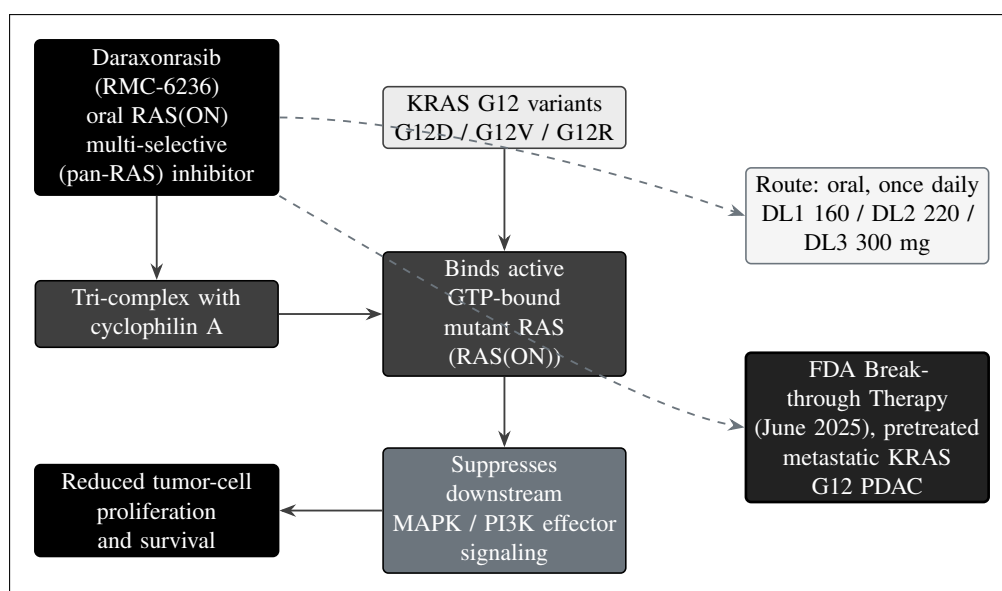


Figure 2. Daraxonrasib (RMC-6236) is an oral RAS(ON) multi-selective (pan-RAS) small-molecule inhibitor. It forms a tri-complex with cyclophilin A and binds the active, guanosine-triphosphate-bound state of mutant RAS, including the KRAS G12 variants (G12D, G12V, G12R) that define the trial population, suppressing downstream MAPK and PI3K effector signaling and reducing tumor-cell proliferation and survival. It is dosed orally once daily at DL1 160 mg, DL2 220 mg, and DL3 300 mg, and holds FDA Breakthrough Therapy designation (June 2025) for previously treated metastatic KRAS G12 PDAC. This figure renders in §3.1.1 (Name of Drug and Active Ingredients), §3.1.2 (Pharmacological Class), and §3.1.3 (Structural Formula).

In the present investigation daraxonrasib is studied in a perioperative, curative-intent surgical context that is distinct from its late-phase metastatic program: it is administered orally once daily in a defined preoperative course, paused across the operative window, and restarted postoperatively on a day chosen by the LLM-bound, human-confirmed advisory described in §3.1.5 and developed in the proposed clinical research. Because the drug is paired with an investigational device in the drug-delivery pathway, the combined IND and IDE submission carries a full Physical AI System Description meeting the eleven fields of 21 CFR §312.23(g) [3], the substance of which is presented in the CMC and device sections of this dossier.

3.1.2 Pharmacological Class of the Drug

Daraxonrasib belongs to the RAS(ON) multi-selective small-molecule inhibitor class. Pharmacologically, the agent is a tri-complex inhibitor: it does not act on the inactive, guanosine-diphosphate-bound (OFF) conformation that earlier-generation allele-specific inhibitors target, but instead engages the active, GTP-bound (ON) conformation of mutant RAS by recruiting the abundant intracellular chaperone cyclophilin A to form a high-affinity ternary complex on the RAS surface [1, 2]. This ON-state, chaperone-mediated mechanism confers multi-selective (pan-RAS) coverage across KRAS, NRAS, and HRAS variants, and in particular across the KRAS G12 alleles (G12D, G12V, G12R) that account for the large majority of RAS-driven PDAC. By occupying the active oncoprotein, the inhibitor blocks engagement of downstream effectors and suppresses the mitogen activated protein kinase (MAPK) and phosphoinositide 3-kinase (PI3K) signaling axes, with the consequence of reduced tumor-cell proliferation and survival, as summarized in Figure 2.

The class assignment is material to the safety profile anticipated in this Phase 1 study and to the dose-finding rationale. As a multi-selective RAS pathway inhibitor, daraxonrasib carries the on-target, mechanism-based toxicities characteristic of broad RAS and MAPK pathway suppression, including gastrointestinal effects (nausea, diarrhea, stomatitis), dermatologic and mucocutaneous events (rash), and hepatic enzyme elevations; these are the toxicity categories that the 3+3 escalation is specifically designed to detect and bound and that inform the dose-limiting toxicity (DLT) definitions in §4. The class also informs the perioperative pause-and-restart logic, because broad pathway inhibition during the immediate postoperative window carries a theoretical risk to wound and anastomotic healing if the restart is mistimed, which is why the restart day is keyed to fistula grade and drug trough rather than fixed.

3.1.3 Structural Formula of the Drug

Daraxonrasib (RMC-6236) is a synthetic, non-peptidic small-molecule organic compound of defined molecular structure designed to bind the RAS-cyclophilin A interface and engage the active GTP-bound state of mutant RAS. It is a single, well-characterized chemical entity rather than a mixture, a biologic, or a nucleic-acid therapeutic, and it is the sole active ingredient of the drug product. The molecule is the macrocyclic tri-complex inhibitor scaffold characteristic of the RAS(ON) class, engineered for oral bioavailability and once-daily exposure, and its structure supports the broad pan-KRAS coverage that defines the molecularly selected trial population.

The complete structural formula of daraxonrasib, including the molecular formula, molecular weight, stereochemistry, the structural diagram of the macrocyclic scaffold, the chemical name, and the characterization data establishing identity and purity, is provided in the Chemistry, Manufacturing, and Control information of this submission at §7.1.1 (Drug Substance), consistent with 21 CFR §312.23(a)(7) [7, 3]. Presenting the full structural characterization in the CMC section avoids duplication and keeps the structural, physicochemical, and manufacturing detail consolidated where the reviewer expects it; this Introduction provides only the qualitative structural description and the cross-reference. The mechanism schematic in Figure 2 summarizes the functional relationship between the structure and its pharmacological target.

3.1.4 Formulation of the Dosage Form(s) to be Used

Daraxonrasib is supplied as oral tablets in the strengths required to compose the three once-daily dose levels evaluated in the 3+3 escalation: DL1 at 160 mg, DL2 at 220 mg, and DL3 at 300 mg. The 300 mg once-daily level (DL3) corresponds to the recommended Phase 2 dose established in the pan-KRAS RASolute 302 program, and the escalation begins one and two steps below it (160 mg and 220 mg) to preserve a conservative first-in-human margin for the novel perioperative surgical context [1, 2]. Tablets

Table 3: Daraxonrasib dose-level structure for the Phase 1 3+3 escalation. DL3 (300 mg once daily) corresponds to the pan-KRAS RASolute 302 recommended Phase 2 dose; DL1 and DL2 sit below it to preserve a conservative first-in-human margin. The DLT window is 28 days, and the recommended Phase 2 dose (RP2D) is selected at or below the maximum tolerated dose (MTD).

Dose level	Daily dose	DLT window	3+3 cohort	Role and rationale
DL1	160 mg QD	28 days	3, expand to 6 if 1 DLT	Conservative starting dose, two steps below the RASolute 302 reference
DL2	220 mg QD	28 days	3, expand to 6 if 1 DLT	Intermediate level, one step below the reference dose
DL3	300 mg QD	28 days	3, expand to 6 if 1 DLT	RASolute 302 recommended Phase 2 dose; highest planned level

are swallowed whole with water and are not crushed, split, or chewed unless a future protocol amendment provides validated instructions. No compounding or reconstitution is required; preparation consists of the site pharmacist verifying participant identity, the assigned dose level, and the expiry, then dispensing the once-daily oral dose with administration instructions.

Tablets are packaged in child-resistant, light-protective containers labeled in accordance with 21 CFR §312.6, bearing the investigational caution statement “Caution: New Drug. Limited by Federal (or United States) law to investigational use,” together with the protocol number, lot number, storage conditions, and a unique container identifier; because the study is open-label, the participant-facing label identifies the product and the assigned daily dose. The product is stored at controlled room temperature in a secured, access-controlled, temperature-monitored location within the investigational-product pharmacy, in its original light-protective packaging, with continuous temperature monitoring and documented excursion handling, and is dispensed only within its labeled expiry or retest date. The dose-level structure that the dosage form must support is summarized in Table 1; the full formulation composition, appearance, packaging, stability, and labeling detail is provided in §7.1.2 (Drug Product) and §7.1.4 (Labeling).

Table 1. Daraxonrasib oral dose levels and the 3+3 escalation structure. At each level three participants are observed across the 28-day DLT window; the cohort escalates on 0 of 3, expands to 6 on 1 of 3, escalates on 1 of 6, and fixes the MTD at the preceding level on 2 or more of 6.

3.1.5 Route of Administration

Daraxonrasib is administered by the oral route, once daily, on a perioperative schedule comprising a defined preoperative course, a protocol-mandated pause spanning the operative window, and a postoperative restart whose timing is governed by an advisory. The preoperative course establishes systemic exposure before surgery; the pause spans the immediate operative and early-recovery window during which broad RAS pathway suppression could impair wound and anastomotic healing; and the restart returns systemic therapy at the earliest safe point to suppress micrometastatic outgrowth. The drug is not delivered by the device, which has no pharmacologic dose of its own; the device conducts the resection and reconstruction and supplies the operative and pathology inputs that the restart advisory consumes [1].

The perioperative pause-and-restart advisory is LLM-bound and human-confirmed, never autonomous. It takes two inputs and produces one of three restart recommendations. The inputs are the pancreaticojejunostomy ISGPS postoperative pancreatic-fistula grade (A, B, or C) and the serum daraxonrasib trough relative to a 0.5 ng/mL threshold. The outputs are a restart on postoperative day T+7d, T+14d, or T+21d, or a continued hold. In the prespecified 32-iteration deterministic simulation sweep, baseline serum was 0.45 ng/mL at the operative timepoint across all iterations (below the 0.5 ng/mL trough), and the advisory recommended T+7d in 29 of 32 iterations (90.6 percent, Grade A fistula with sub-threshold trough), T+14d in 3 of 32 iterations (9.4 percent, Grade B fistula or a delayed serum rise), and T+21d in 0 of 32 iterations; Grade C fistula maps to T+21d or a continued hold and to the drug stop rule. Because the pancreaticojejunostomy is the dominant determinant of postoperative pancreatic-fistula risk, the fistula-grade input to the drug advisory is based on that anastomosis. The advisory is one of only two coupling points between the drug and device components of the integrated intervention, and it keeps the restart decision human-confirmed and fully traceable in the audit trail. The detailed advisory logic and the simulation sweep are presented in the proposed clinical research and Investigator's Brochure sections of this dossier.

3.1.6 Duration of the Proposed Clinical Investigation(s)

The duration of the proposed clinical investigation is defined at two levels: the per-participant timeline and the overall study timeline. For each participant, the investigation comprises up to 28 days of screening (Day -28 to Day -1) during which histologic confirmation, KRAS G12 confirmation, staging, and eligibility are established; a baseline day (Day -1) that carries the Phase 0 simulation sign-off and the Unified Safety Level (USL) and safety-matrix verification; surgery on Day 0; an acute window (Day 1 to Day 7) carrying the ISGPS fistula grading and the daraxonrasib restart advisory; the Day 30 primary safety window during which device-related and procedure-related serious adverse events, including Clavien-Dindo grade III or higher complications, are ascertained; Day 90 pathology and mortality assessment, which carries the R0 resection determination and the 90-day mortality endpoint; and long-term follow-up every 12 weeks to the 24-month overall-survival endpoint, during which progression-free survival (PFS) and overall survival (OS) are assessed by RECIST v1.1 [1].

The daraxonrasib DLT observation window is 28 days from the start of dosing at each dose level, which governs the cadence of the 3+3 escalation. At the study level, the investigation enrolls up to 18 treated participants (approximately 36 screened) at a single academic site, with an anticipated accrual of approximately one to two participants per month under sentinel staggered enrollment; with three dose levels of up to six DLT-evaluable participants each, the maximum of 18 follows directly (three levels times six). Adding the per-participant follow-up to 24-month OS to the accrual period, the total study duration is governed primarily by the time required to mature the overall-survival endpoint in the last-enrolled participant. A mandatory Phase 0 simulation-validation gate (at least 1000 simulated procedures across at least two independent frameworks, with a USL rating of at least 7.0 and a sim-to-real trajectory gap below 2 mm and tip-force gap below 0.5 N) precedes any patient-facing robotic activity and is re-validated after any qualifying change, so the patient-facing portion of the investigation does not begin until that gate is satisfied. The first-year operational plan and the projected enrollment cadence are developed in the General Investigational Plan (§4).

The investigation is statistically descriptive and non-confirmatory, consistent with ICH E9 principles for early-phase trials and with TRIPOD+AI for the AI-enabled components; there is no power calculation and no comparator arm. Estimation uses exact Clopper-Pearson 95 percent confidence intervals, and the maximum sample of 18 participants yields the precision appropriate to a first-in-human dose-finding and early-feasibility study: an observed proportion of 0 of 18 corresponds to 0 percent (95 percent CI 0 to 18 percent), 1 of 18 corresponds to 5.6 percent (95 percent CI 0.1 to 27 percent), 0 of 3 to a CI of

0 to 71 percent, and 0 of 6 to a CI of 0 to 46 percent. The three co-primary endpoints are the incidence of device-related and procedure-related serious adverse events through the Day 30 window (including Clavien-Dindo grade III or higher), the maximum tolerated dose and recommended Phase 2 dose of daraxonrasib determined by the 3+3 escalation across the 28-day DLT window, and technical feasibility defined as the proportion of participants completing the assigned operative tasks without unplanned conversion attributable to device malfunction or unsafe behavior. Binding halt rules govern progression: a device-related or procedure-related serious adverse event in at least one of three participants within a cohort, or two device-related deaths at any time, triggers Data Safety Monitoring Board review and possible halt. Safety reporting follows the 7-calendar-day clock for fatal or life-threatening suspected adverse reactions and the 15-calendar-day clock for other serious and unexpected reactions under 21 CFR §312.32, augmented by six Physical AI reporting triggers under 21 CFR §312.32(g) and a -24 hour to +72 hour audit-trail preservation window around each Physical AI adverse event [7, 3].

3.2 Summary of Previous Human Experience

Daraxonrasib has an active and advancing clinical development program in RAS-mutated solid tumors, with the most mature human experience in pancreatic cancer. The FDA granted daraxonrasib Breakthrough Therapy designation in June 2025 for previously treated metastatic PDAC with KRAS G12 mutations, reflecting preliminary clinical evidence of a meaningful advantage over available therapy in this population [1, 2]. The late-phase program comprises RASolute 302, a Phase 3 study in previously treated metastatic PDAC, and RASolve 301, a first-line program; the broad pan-KRAS eligibility of RASolute 302 defines the molecular eligibility adopted in this protocol, and 300 mg once daily is the established recommended Phase 2 dose carried forward as the highest dose level (DL3) in the present escalation [1]. The previous human experience with daraxonrasib therefore establishes a pan-KRAS antitumor rationale, a tolerability and exposure basis for the dose levels studied here, and a regulatory recognition (Breakthrough Therapy) of unmet need in the same molecular population. The perioperative pause-and-restart logic additionally respects any RASolve 301 checkpoint-inhibitor combination context.

The present investigation differs from the prior daraxonrasib human experience in two material respects that define the scope of the first-in-human determination for this IND. First, it studies the drug in a perioperative, curative-intent surgical setting (resectable and borderline-resectable disease) rather than in the metastatic, palliative setting of the late-phase program, so the wound-healing and anastomotic-healing considerations are new and motivate the pause-and-restart advisory. Second, it couples the drug to an investigational LLM-directed robotic device, which has no prior human experience and which is evaluated here as a significant-risk early-feasibility Class II collaborative platform under the IDE pathway; the device-related and procedure-related safety endpoints in §4 are specific to this novel combination. The full marketed-experience, prior-clinical, and clinical-care experience analysis is presented in the Previous Human Experience section of this dossier (§9).

The investigational design is further grounded by a body of computational and in silico evidence developed by the sponsor-investigator throughout 2025 and 2026, which supplies the simulation scaffolding adapted in this protocol and independently complements the daraxonrasib drug-development program. This evidence comprises a 60-second PDAC robotic Whipple and daraxonrasib simulation; an AI digital-twin pancreatic-cancer simulation for in silico FDA compliance and cost efficiency [15]; a quantitative systems pharmacology (QSP) metastatic-PDAC trial simulation with code, verification, validation, and uncertainty quantification, and a reproducible playbook [16]; a 100,000-patient in silico Phase 3 five-arm PDAC trial triplicate [17]; and end-to-end PDAC digital-twin clinical-trial proposals [18]. The sponsor-investigator's record of LLM evaluation and trust over August 2024 to June 2026 further grounds the device governance, including a comparative code-generation evaluation of

16 proprietary versus open-source LLMs against the FDA Adverse Event Reporting System [19], the on-premises advisory-LLM and verification-before-generation design [6], and the legislative framing aligned with H.R. 9510, the Verification Before Generation in Physical AI Oncology Trials Act of 2026 [5]. The previous human experience and the supporting computational evidence together motivate the favorable benefit-risk determination for this low-survival, limited-alternative population.

3.3 Status of Drug in Other Countries

Daraxonrasib is an investigational agent that has not received marketing approval in the United States or in any other jurisdiction for any indication; it is studied worldwide under investigational authorizations within its sponsor’s RAS(ON) development program, and its U.S. Breakthrough Therapy designation (June 2025) is a development-stage designation rather than a marketing authorization [1, 2]. The drug has therefore not been withdrawn from marketing in any country for reasons of safety or effectiveness, and there is no adverse foreign regulatory action to report under 21 CFR §312.23(a)(5). Within the present investigation, the LLM-directed robotic device component is similarly investigational and is studied under the IDE pathway as a significant-risk Class II collaborative device; it carries no marketing clearance or approval in any jurisdiction.

Although this Phase 1 investigation is conducted at a single United States academic site under the combined IND and IDE, the dossier is authored to align with the principal international device and trial frameworks so that the same documentation can support future multi-jurisdiction extension without re-derivation. The cross-jurisdiction alignment that the device safety architecture and the Physical AI System Description are designed to satisfy is summarized in Table 2. The hard cyber-physical safety envelope (the 10 kHz heartbeat with a 100 microsecond watchdog, the cross-arm emergency stop of 3 ms or less, the per-arm tip-force cap of 3 N or less and cumulative cap of 18 N or less, and the vascular no-fly gating) and the 21 CFR part 11 hash-chained audit trail are constructed to map onto the corresponding requirements of the European Union Medical Device Regulation, the China National Medical Products Administration, the Japan Pharmaceuticals and Medical Devices Agency, the United Kingdom Medicines and Healthcare products Regulatory Agency, Health Canada, and the Australian Therapeutic Goods Administration [10, 3].

Table 2. Cross-jurisdiction status of daraxonrasib and the LLM-directed device. The agent and device are investigational in every jurisdiction; the documentation is authored once to align with the principal international device and trial frameworks so it can support future multi-jurisdiction extension.

3.4 Overview of Preclinical Data

The preclinical and computational basis for this investigation supports both the pharmacologic rationale for daraxonrasib in KRAS G12-mutated PDAC and the safety and feasibility rationale for the LLM-directed robotic device, and is presented in full in the Pharmacology and Toxicology information (§8) and the Investigator’s Brochure (§5). On the pharmacology side, daraxonrasib acts through the mechanism-based, on-target suppression of RAS pathway signaling described in §3.1.2: by engaging the active GTP-bound state of mutant RAS in a tri-complex with cyclophilin A, it suppresses MAPK and PI3K effector signaling and reduces tumor-cell proliferation and survival across the KRAS G12 alleles that define the trial population [1]. The drug-distribution, exposure, and tolerability assumptions that anchor the three dose levels (DL1 160 mg, DL2 220 mg, DL3 300 mg) are carried forward from the pan-KRAS RASolute 302 program, in which 300 mg once daily is the recommended Phase 2 dose, and the anticipated mechanism-based toxicity profile (gastrointestinal, dermatologic and mucocutaneous, and hepatic) informs the DLT definitions and the perioperative pause-and-restart logic [2].

Table 4: Cross-jurisdiction regulatory status and alignment. Daraxonrasib is investigational with no marketing authorization in any country; the device is investigational under the IDE. The dossier is authored to align with the principal international frameworks for future multi-jurisdiction extension.

Jurisdiction	Framework	Status and alignment for this investigation
United States	FDA, 21 CFR §312 / §812	Combined IND / IDE; daraxonrasib investigational with Breakthrough Therapy designation (June 2025); device significant-risk Class II under IDE
European Union	EU MDR 2017/745	No marketing authorization; safety envelope and audit trail authored to map onto MDR essential requirements
China	NMPA	No marketing authorization; device safety architecture aligned for future NMPA submission
Japan	PMDA	No marketing authorization; Physical AI System Description aligned to PMDA expectations
United Kingdom	MHRA	No marketing authorization; trial and device documentation aligned to MHRA framework
Canada	Health Canada	No marketing authorization; dossier structured for future Health Canada extension
Australia	TGA	No marketing authorization; safety and audit-trail design aligned to TGA requirements

On the device and combined-intervention side, the preclinical evidence is dominated by a structured program of high-fidelity simulation and digital-twin work that substitutes for and supplements conventional preclinical testing for the Physical AI system, consistent with the verification, validation, and uncertainty quantification (VVUQ) discipline that the protocol applies. The mandatory Phase 0 simulation-validation gate requires at least 1000 simulated procedures across at least two independent frameworks (for example Isaac, MuJoCo, Gazebo, and PyBullet), a Unified Safety Level rating of at least 7.0, and a sim-to-real gap below 2 mm in trajectory and below 0.5 N in tip force, with re-validation after any qualifying change; this gate is the principal preclinical safety bar that the device must clear before any patient-facing activity. The supporting computational evidence base includes the 60-second PDAC robotic Whipple and daraxonrasib simulation, the AI digital-twin pancreatic-cancer simulation for in silico FDA compliance and cost efficiency [15], the QSP metastatic-PDAC trial simulation with code, VVUQ, and a reproducible playbook [16], the 100,000-patient in silico Phase 3 five-arm PDAC trial triplicate [17], and the end-to-end PDAC digital-twin clinical-trial proposals [18]. Together these establish that the integrated drug-device intervention has been characterized in silico across a large and reproducible body of simulated experience before any human exposure, and that the on-premises advisory LLM operates strictly as a bounded, hash-pinned, second-opinion oracle with full autonomy prohibited consistent with 21 CFR §312.21(e) [6, 5]. The integrated pharmacology and toxicology summary, including the structure of the pharmacology summary, the integrated toxicology summary, and the full data tabulation, is presented in §8.

3.5 References

The references cited throughout this Introduction are drawn from the IND reference library and are collected, formatted, and rendered in the References and Back Matter section of this dossier. The principal sources grounding the clinical subject, the daraxonrasib mechanism and class, the device safety

architecture, the indication, the duration timeline, and the acceleration thesis include the FDA Phase 1 content and format guidance [7], the oncology-trial LLM authoring basis [9], the daraxonrasib robotic Whipple and on-premises device sources [1, 2], the 21 CFR Part 312, Part 50, and ICH Physical AI unification adaptations [3, 11, 12], the PDAC epidemiology and acceleration sources [13, 14, 10], the verification-before-generation and clinical-trust Physical AI sources [5, 6, 4, 8], the comparative LLM evaluation against the FDA Adverse Event Reporting System [19], and the computational digital-twin and QSP evidence base [15, 16, 17, 18]. Each citation key resolves against the IND reference library, and the assembled list renders in the References and Back Matter section.

4 General Investigational Plan

This General Investigational Plan describes, at the level of detail required by 21 CFR §312.23(a)(3)(iv), the rationale for the investigation, the indication to be studied, the general approach to evaluating the investigational articles, the trials to be conducted during the first year, the estimated number of subjects to be exposed, and the foreseeable drug-related and device-related risks together with their mitigations. The investigation is a single, integrated, first-in-human study of two regulated articles administered in combination: perioperative oral daraxonrasib (RMC-6236), an investigational pan-RAS(ON) multi-selective small-molecule inhibitor evaluated under the investigational new drug pathway, and an on-premises large language model (LLM) directed eight-arm robotic pancreaticoduodenectomy (Whipple) system, a significant-risk Class II collaborative device evaluated under the investigational device exemption pathway. The study carries a single protocol, a single informed consent bearing the Physical AI opt-out, and a single three-tier safety governance structure. The plan that follows is intentionally quantitative so that the reviewer can assess the design, the dose schedule, the operating characteristics, and the benefit-risk balance from the numbers themselves [7, 1].

4.1 Rationale

The scientific rationale rests on a population of profound unmet need and a mechanistically matched intervention. Pancreatic ductal adenocarcinoma (PDAC) remains among the most lethal of solid tumors: the five-year overall survival is below 13 percent, and the disease is the third leading cause of cancer death in the United States and the fourth in the European Union [13]. The great majority of PDAC is driven by an activating KRAS mutation at codon 12, most commonly KRAS G12D, G12V, or G12R, and until recently this oncogenic dependency was considered pharmacologically intractable. Daraxonrasib (RMC-6236) is an oral once-daily RAS(ON) multi-selective (pan-RAS) inhibitor that binds the GTP-bound, active conformation of mutant RAS in a tri-complex with cyclophilin A and thereby suppresses downstream effector signaling through the MAPK and PI3K pathways. The agent received FDA Breakthrough Therapy designation in June 2025 for previously treated metastatic PDAC harboring a KRAS G12 mutation, and the 300 mg once-daily dose is the established recommended Phase 2 dose of the RASolute 302 pan-KRAS program [1, 9]. Pairing a genotype-matched systemic agent with a margin-negative curative-intent resection in the resectable and borderline-resectable setting is a scientifically coherent strategy for a disease in which the resection margin is the strongest modifiable determinant of survival.

The second component of the rationale is the surgical platform itself. The Whipple operation is the most technically demanding curative-intent abdominal procedure, and its early morbidity is dominated by pancreatic fistula, intra-abdominal hemorrhage, sepsis, and failure to rescue. The investigational eight-arm robotic system is engineered to perform this operation within hard cyber-physical limits under continuous human oversight: eight arms each with seven degrees of freedom (56 degrees of freedom total), 640 sensor channels (80 per arm) sampled at 100 kHz for force and 10 kHz for other modalities, a positional accuracy of 0.05 mm RMS at 1200 mm/s, per-arm tip-force caps at or below 3 N and a cumulative cross-arm cap at or below 18 N, a 10 kHz safety heartbeat with a 100 microsecond watchdog, arm-park within 50 microseconds, a cross-arm emergency stop within 3 ms to 0.0 N, and a hardware emergency stop within 500 ms consistent with 21 CFR §312.404 [2, 1]. A vascular no-fly gate continuously protects five named vessels (the superior mesenteric vein, portal vein, hepatic artery, celiac axis, and superior mesenteric artery) with a 3.0 mm advisory soft-warning zone, a 1.0 mm no-fly zone for the superior mesenteric and portal veins and a 1.5 mm no-fly zone for the arterial structures, and a 5.0 mm hard-stop zone that triggers an emergency stop. The on-premises LLM operates strictly as a second-opinion advisory oracle, hash-pinned to a deterministic seed and a repository commit, isolated outside the vendor kinematic stack, and prohibited from full autonomy consistent with 21 CFR

§312.21(e); every advisory passes a ten-gate verification-before-generation suite returning ACCEPT, BLOCK, or ESCALATE, aligned with H.R. 9510, the Verification Before Generation in Physical AI Oncology Trials Act of 2026 [5, 6].

The schedule rationale is distinct from, and complementary to, the clinical rationale. This IND is itself the highest-value product of a verified single-prompt authoring method whose purpose is to compress the administrative and preparation time that separates clinical milestones, without ever altering tumor biology or any fixed regulatory clock. Figure 1 ranks the six document targets where faster, validated authoring yields the greatest schedule value. The initial IND and IRB package, the subject of this build, is the highest-value target because it sits on the critical path; one input, faster validated authoring, feeds all six targets, and each contributes to a common compression of administrative and preparation time measured in months to a year. Crucially, the gains are capped by clinical events and by fixed review clocks: faster authoring cannot shorten the 30-day FDA review or manufacture missing toxicology or stability data [14, 10].

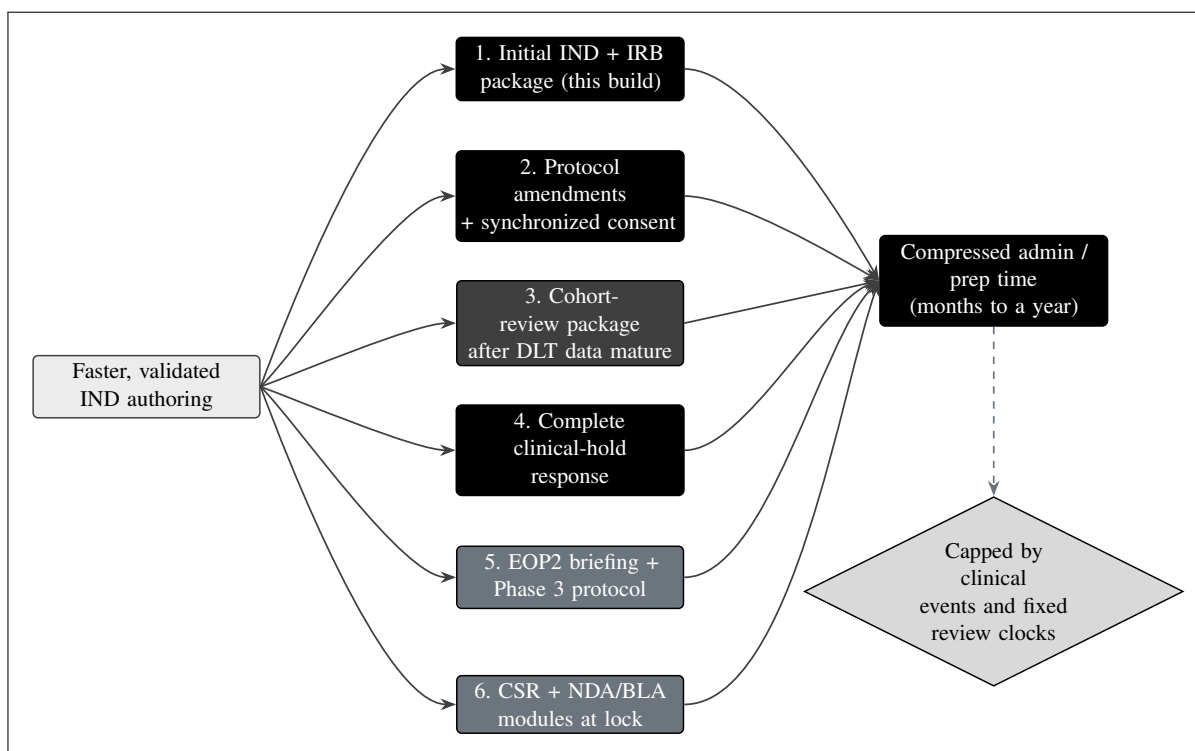


Figure 1. The six greatest-acceleration IND document targets, ranked, with the initial IND and IRB package the highest-value target and the subject of this build. One input, faster validated authoring, feeds all six; each contributes to a common compression of administrative and preparation time of months to a year. The schedule value is highest on the critical path, and the gains are capped by clinical events and fixed review clocks (they cannot shorten the 30-day FDA review or manufacture missing data).

The mechanism of the schedule benefit is specific and is shown in Figure 2. A clinical-trial timeline divides into three buckets: a clinical and operational bucket (recruitment, surgery, the dose-limiting-toxicity window, and progression-free and overall-survival maturation), which is not compressible; an administrative and preparation bucket (data cleaning, analysis, and document authoring), which is compressible; and a fixed regulatory-review bucket (the 30-day IND review and the 30-day clinical-hold review), which is fixed. The repository LLM acts on the middle bucket only. The exact per-document

savings are: the initial IND and IRB package falls from 8 to 12 weeks to 1 to 4 days; a protocol amendment with synchronized consent falls from 2 to 4 weeks to 1 to 2 days; a cohort-review package falls from 1 to 2 weeks to about 1 day; a complete clinical-hold response falls from 3 to 6 weeks to 2 to 4 days; an annual development safety update report (DSUR) falls from 4 to 8 weeks to 2 to 3 days; and a clinical study report under ICH E3 falls from 3 to 6 months to 1 to 2 weeks, summing to months to a year saved cumulatively [14, 8].

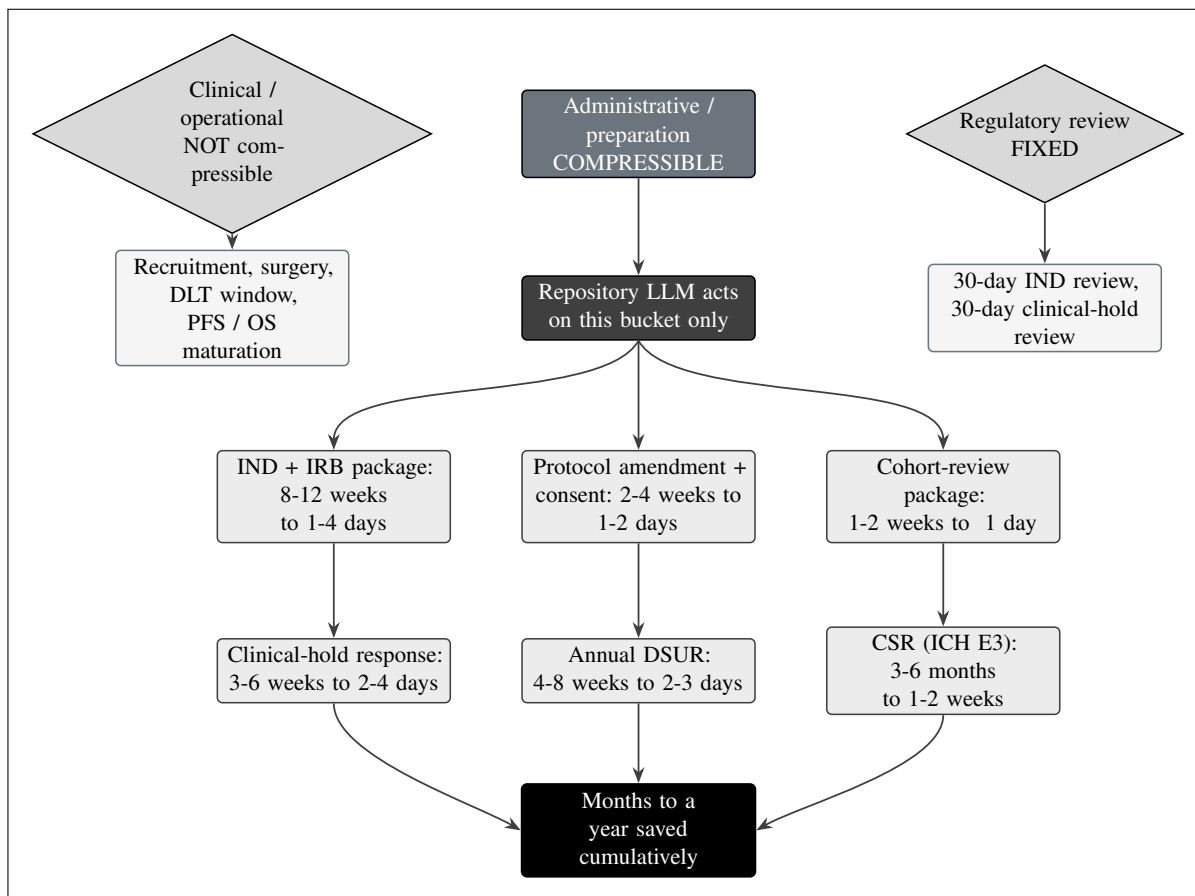


Figure 2. The three timeline buckets. Only the administrative and preparation bucket compresses; the clinical and operational bucket and the fixed regulatory review clocks do not. The repository LLM acts on the middle bucket only, with exact per-document savings: initial IND and IRB package 8 to 12 weeks to 1 to 4 days; protocol amendment with synchronized consent 2 to 4 weeks to 1 to 2 days; cohort-review package 1 to 2 weeks to about 1 day; complete clinical-hold response 3 to 6 weeks to 2 to 4 days; annual DSUR 4 to 8 weeks to 2 to 3 days; clinical study report 3 to 6 months to 1 to 2 weeks, summing to months to a year saved cumulatively.

These two rationales meet at the patient. Figure 3 traces the patient time-saved cascade: faster validated authoring advances each step from IND and IRB submission and site activation, through the elapse of the 30-day clock and the first dose, to faster 3+3 cohort turnover toward the recommended Phase 2 dose, shortening the white space between phases and, for this low-survival population, extending progression-free and overall survival, without changing tumor biology or any fixed review clock. The cascade is the patient-level expression of the schedule benefit and is the bridge between the rationale here and the favorable benefit-risk balance developed in §4.6 [14, 13].

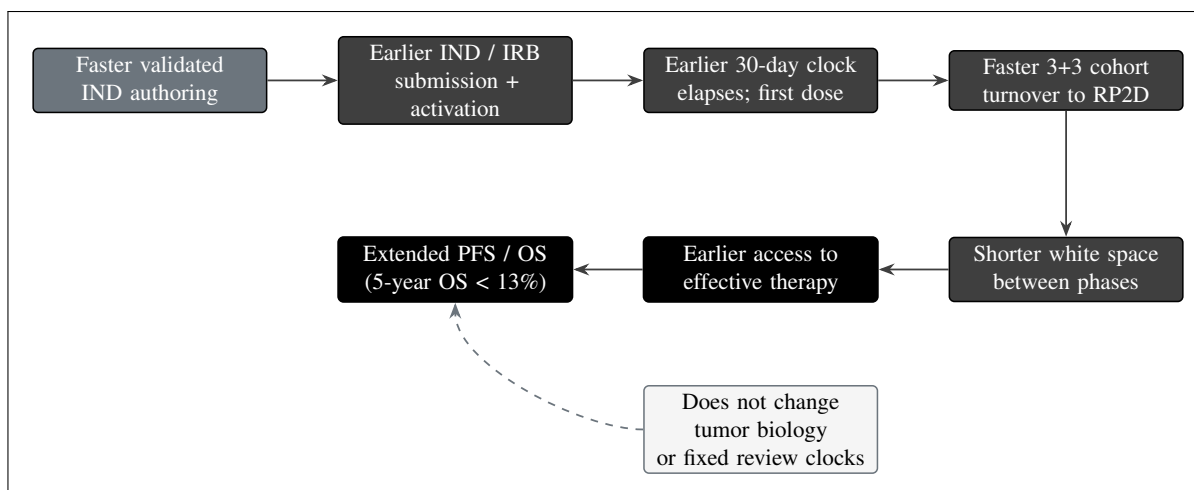


Figure 3. The patient time-saved cascade. Faster validated authoring advances each step from IND submission to first dose, then through faster 3+3 cohort turnover to the recommended Phase 2 dose, shortening the white space between phases and, for this low-survival population (five-year overall survival under 13 percent), extending progression-free and overall survival, without changing tumor biology or the fixed regulatory review clocks.

4.2 Indication to be Studied

The indication to be studied is histologically or cytologically confirmed pancreatic ductal adenocarcinoma that is resectable or borderline-resectable on multidisciplinary radiologic review, harbors a documented KRAS G12 mutation (for example G12D, G12V, or G12R) on a validated tumor genotyping assay, occurs in an adult of at least 18 years with an Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1, and is anatomically compatible with the eight-arm robotic system and with a planned pancreaticoduodenectomy. Eligibility for perioperative daraxonrasib follows the broad pan-KRAS criteria of the RASolute 302 program [1]. The indication is deliberately narrow at this first-in-human stage so that the safety and feasibility readouts are interpretable: distant metastatic disease, vascular involvement precluding a margin-negative resection, prior pancreatic resection precluding the planned reconstruction, uncontrolled intercurrent illness, pregnancy or lactation, and known contraindication to daraxonrasib are exclusionary. Table 1 summarizes the eligibility frame that defines the indication; the full inclusion and exclusion criteria are set out in the protocol and reproduced in the proposed clinical research module of this IND.

Domain	Inclusion (must meet all)	Exclusion (any one disqualifies)
Disease and stage	Histologically or cytologically confirmed PDAC; resectable or borderline-resectable on multidisciplinary review.	Distant metastatic disease on staging; vascular involvement precluding R0 resection or anatomy incompatible with the eight-arm system.
Genotype	Documented KRAS G12 mutation (G12D, G12V, or G12R) on a validated assay; eligible under pan-KRAS RASolute 302 criteria.	Absence of a qualifying KRAS G12 mutation.
Performance and organ function	ECOG performance status 0 or 1; adequate hematologic, hepatic, and renal function for major pancreaticobiliary surgery and daraxonrasib.	Uncontrolled cardiac, pulmonary, hepatic, renal, or active infectious disease making surgery or dosing unsafe.
Surgical candidacy	Candidate for pancreaticoduodenectomy; tumor and vascular anatomy compatible with port placement and the working volume.	Prior pancreatic resection or prior surgery precluding the planned reconstruction.
Age and consent	Age at least 18 years; signed informed consent including explicit election or declination of the Physical AI opt-out.	Inability to consent or to comply with the Schedule of Activities.
Other	Measured baseline CA 19-9 recorded for serial assessment (no threshold imposed).	Pregnancy or lactation; known contraindication or hypersensitivity to daraxonrasib or a required excipient; unmanageable drug interaction.

Table 1. Eligibility summary defining the indication to be studied. Selection is equitable and is determined solely by the scientific and safety criteria; limited English proficiency is not an exclusion, and qualified interpreters and translated consent materials are provided.

A synthetic reference exemplar, designated PAT-PDAC-0001, illustrates a representative eligible participant and is used for design illustration only: a 68-year-old with a 3.4 cm head-of-pancreas tumor abutting the superior mesenteric vein at 75 degrees, KRAS G12D, microsatellite stable, a baseline CA 19-9 of 412 U/mL, ECOG performance status 1, after a completed neoadjuvant modified FOLFIRINOX course of four cycles. This exemplar does not represent an enrolled individual but fixes the anatomic and oncologic context in which the eligibility frame and the device no-fly gating are exercised [1].

4.3 General Approach for Evaluation of Treatment

The general approach is a prospective, open-label, single-arm, first-in-human study that combines a standard Phase 1 3+3 dose-finding evaluation of perioperative daraxonrasib with an early-feasibility evaluation of the LLM-directed eight-arm robotic Whipple system. A single-arm, open-label design is scientifically and ethically appropriate, and a randomized comparator arm is not, because the first-in-human objectives, characterizing early safety, identifying a recommended Phase 2 dose, and demonstrating that a novel surgical system can perform its intended function within hard cyber-physical limits,

are not comparative-efficacy questions and are answered most safely in a small, intensively monitored cohort with sentinel staggered enrollment and case-by-case data and safety monitoring board (DSMB) review. The early-feasibility pathway recognized by FDA for significant-risk devices is explicitly structured around small first-in-human cohorts that generate the device-performance and safety evidence needed before any controlled comparison [7]. The 2025 Dutch nationwide cohort of 1000 robotic pancreaticoduodenectomies provides a robust external human benchmark for the secondary surgical-quality endpoints, so the absence of an internal control does not leave the surgical results uninterpretable [1].

4.3.1 Objectives and Endpoints

The study carries three co-primary objectives, one for each axis of the combined IND/IDE investigation, beneath which sit secondary oncologic-surgical quality endpoints benchmarked against the Dutch 2025 cohort and exploratory non-confirmatory endpoints. The first co-primary objective characterizes the early safety of the combined intervention; its endpoint is the incidence of device-related or procedure-related serious adverse events through 30 days after the procedure, including Clavien-Dindo grade III or higher complications, with device or procedure attribution adjudicated by an independent reviewer blinded to the daraxonrasib dose level. The second co-primary objective determines the maximum tolerated dose (MTD) and recommended Phase 2 dose (RP2D) of perioperative daraxonrasib, defined operationally by the rate of dose-limiting toxicities (DLTs) within the 28-day DLT window of the 3+3 escalation. The third co-primary objective establishes the technical feasibility of the eight-arm robotic Whipple under continuous human oversight; its endpoint is the proportion of procedures that complete all prespecified assigned operative tasks without unplanned conversion caused by device malfunction or unsafe device behavior. Figure 4 renders the objective-to-endpoint hierarchy, and Table 2 sets out the full mapping with justifications and the Dutch 2025 comparators [1, 7].

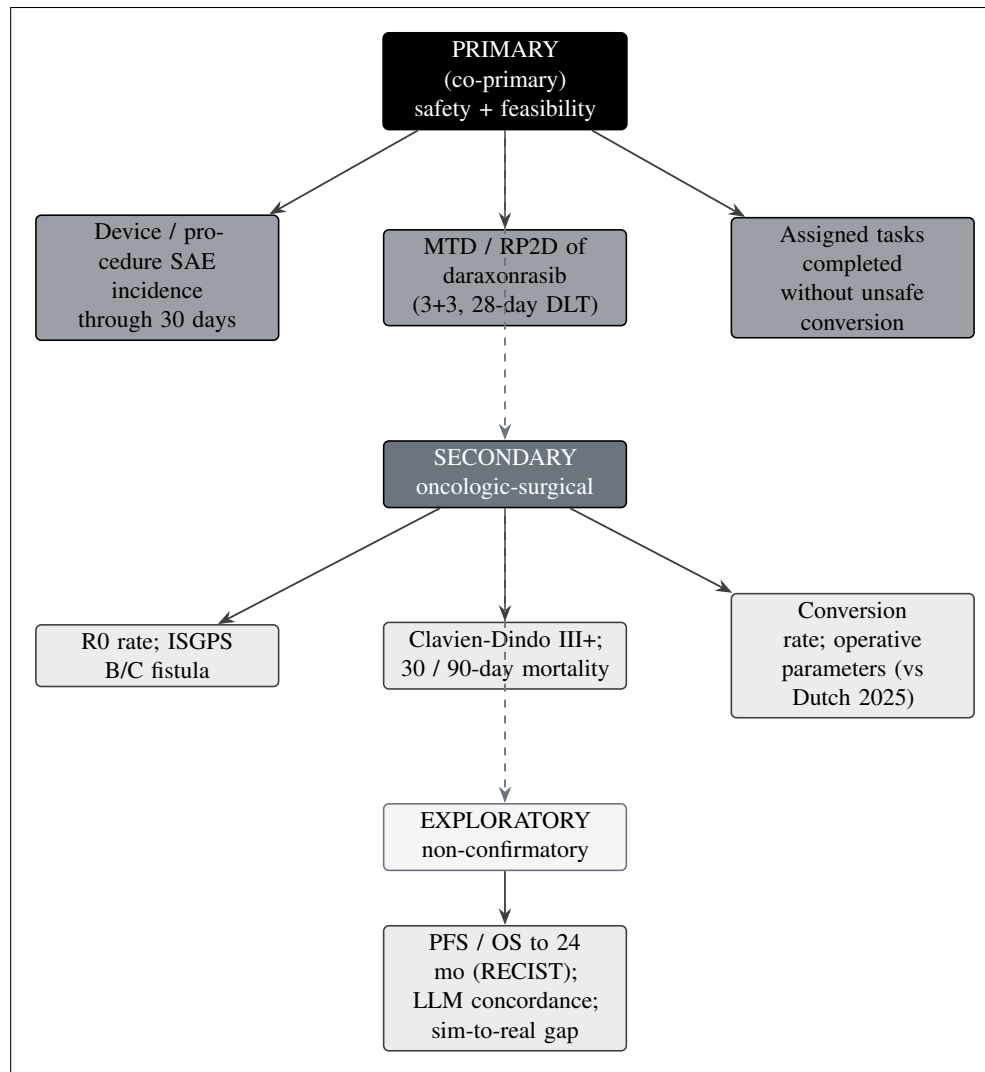


Figure 4. The objective-to-endpoint hierarchy of the general investigational plan. Three co-primary objectives (device or procedure serious-adverse-event incidence through 30 days, the maximum tolerated dose and recommended Phase 2 dose of daraxonrasib by 3+3 over the 28-day window, and completion of assigned operative tasks without unsafe conversion) sit above secondary oncologic-surgical quality endpoints (R0 rate, ISGPS grade B/C fistula, Clavien-Dindo grade III or higher, 30 and 90-day mortality, conversion rate, and operative parameters benchmarked against the Dutch 2025 robotic cohort) and exploratory, non-confirmatory endpoints (progression-free and overall survival to 24 months by RECIST, LLM advisory concordance, and the sim-to-real gap).

Tier	Objective and endpoint	Justification / comparator
<i>Co-primary</i>		
Safety	Device-related or procedure-related SAE incidence through 30 days, including Clavien-Dindo grade III+, dose-blinded attribution.	Standard first-in-human safety anchor; captures the early lethal cascade of fistula, hemorrhage, sepsis, and failure to rescue.
Dose-finding	MTD and RP2D of perioperative daraxonrasib by the DLT rate within the 28-day window of the 3+3 across DL1 to DL3.	Established dose-finding design for targeted oncology agents; anchored to the RASolute 302 pan-KRAS program.
Feasibility	Proportion of procedures completing all assigned operative tasks without unplanned conversion from device malfunction or unsafe behavior.	FDA-recognized early-feasibility readout for a significant-risk device.
<i>Secondary (descriptive, Dutch 2025 comparators)</i>		
Oncologic adequacy	R0 (microscopically margin-negative) resection rate at day-90 pathology.	Strongest modifiable surgical determinant of survival; descriptive comparison with Dutch 2025.
Fistula	ISGPS grade B/C postoperative pancreatic fistula rate.	Leading driver of post-Whipple morbidity; Dutch 2025 comparator 24.4 percent.
Major complications	Clavien-Dindo grade III+ complication rate through 90 days.	Validated severity threshold; Dutch 2025 comparator 41.3 percent.
Mortality	30-day and 90-day all-cause mortality.	Standard short-term survival safety measures; Dutch 2025 30-day comparator 3.9 percent.
Conversion	Unplanned conversion rate from the robotic system to open or conventional approach.	Contextualizes feasibility; Dutch 2025 comparator 10.1 percent.
Operative parameters	Estimated blood loss, operative time, and length of stay.	Mean and median with range; Dutch 2025 learning-curve context.
<i>Exploratory (non-confirmatory)</i>		
Survival	PFS and OS assessed to 24 months by RECIST v1.1, Kaplan-Meier.	Preliminary, hypothesis-generating signal; the small single-arm sample precludes confirmatory inference.
Advisory reliability	Concordance between the LLM advisory output and the independent hard-coded	Characterizes the second-opinion oracle and informs autonomy-graduation decisions reserved for later phases

Table 2. The objective-to-endpoint mapping with justifications and the Dutch 2025 nationwide robotic-pancreatoduodenectomy comparators. Comparisons are descriptive only; the single-arm design supports estimation, not inferential between-group testing.

4.3.2 Dose-Escalation Approach

Daraxonrasib is administered orally once daily. The escalation uses exact doses anchored to the RA-Solute 302 pan-KRAS program, in which the 300 mg once-daily dose is the established RP2D. To preserve a conservative first-in-human margin in the perioperative setting, the escalation begins below that reference dose and steps up to it: DL1 is 160 mg once daily (the starting dose), DL2 is 220 mg once daily, and DL3 is the 300 mg once-daily reference dose. Three participants enroll at each level and are observed over a 28-day DLT window. If no DLT occurs among the first three, the cohort escalates; if one of three experiences a DLT, the cohort expands to six; a single DLT among six permits escalation, whereas two or more DLTs at a level define that level as exceeding the MTD and fix the MTD at the preceding level. The MTD is the highest level at which fewer than two of six DLT-evaluable participants experience a DLT, and the RP2D is selected at or below the MTD with integration of all safety, tolerability, and available pharmacokinetic data. Figure 5 renders the decision automaton and Table 3 states the rule explicitly for each cohort outcome [1, 7].

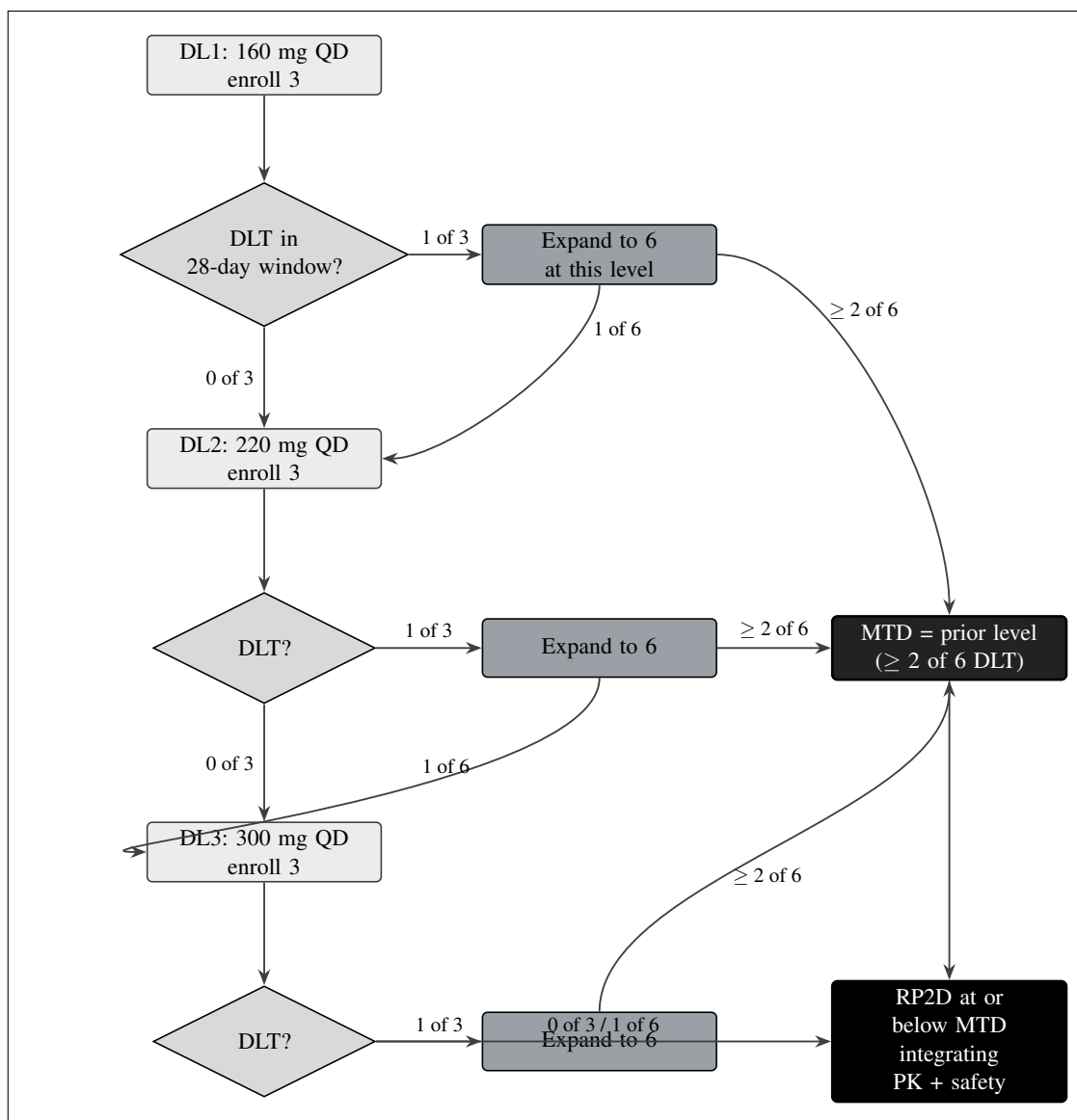


Figure 5. The standard 3+3 dose-escalation automaton across the three daraxonrasib dose levels (DL1 160 mg, DL2 220 mg, DL3 300 mg once daily). Three participants enroll at each level; with no dose-limiting toxicity in the 28-day window the cohort escalates, with one of three it expands to six, and two or more of six fixes the maximum tolerated dose at the prior level. The recommended Phase 2 dose is selected at or below the maximum tolerated dose, integrating pharmacokinetic and safety data. The design maximum is three levels times six, up to 18 treated.

Cohort observation	DLT count	Decision	Consequence
First 3 at a level	0 of 3	Escalate	Open the next higher dose level; current level deemed tolerated at the cohort of 3.
First 3 at a level	1 of 3	Expand to 6	Enroll 3 additional participants at the same level before any escalation decision.
First 3 at a level	≥ 2 of 3	MTD exceeded	Do not enroll further at this level; the MTD is the preceding (lower) level.
Expanded 6 at a level	1 of 6	Escalate	Single DLT in six permits escalation to the next higher level.
Expanded 6 at a level	≥ 2 of 6	MTD exceeded	The level exceeds the MTD; the MTD is the preceding level.
Highest level (DL3)	0 of 3 or 1 of 6	RP2D candidate	DL3 (300 mg) tolerated; RP2D selected at or below DL3 integrating PK and safety.
Any level	≥ 2 of 6	Declare MTD / RP2D	MTD fixed at prior level; RP2D at or below MTD.

Table 3. The 3+3 dose-escalation decision rule. The maximum tolerated dose is the highest level at which fewer than two of six DLT-evaluable participants experience a dose-limiting toxicity within the 28-day window; the recommended Phase 2 dose is selected at or below the MTD. The design maximum is three levels times six, up to 18 treated.

4.3.3 Device Readiness and Statistical Framework

The device has no pharmacologic dose; its dose analogue is the readiness evidence required before any patient-facing robotic activity, escalated only by passing a mandatory Phase 0 simulation-validation gate. Three conditions must be met and documented before the first robotic case: a Unified Safety Level (USL) rating of at least 7.0 on the surgical-readiness scale; completion of at least 1000 simulated procedures across at least two independent frameworks (for example Isaac, MuJoCo, Gazebo, or PyBullet), so the finding is not an artifact of any single simulator; and a sim-to-real fidelity within prespecified limits, a trajectory gap less than 2 mm and a tip-force gap less than 0.5 N between the simulation prediction and the operative record. The gate is re-validated after any change to the system. These limits operate together with the intra-operative hard caps (a cross-arm emergency-stop budget at or below 3 ms, per-arm tip-force caps at or below 3 N, a cumulative cross-arm cap at or below 18 N, and vascular no-fly gating) so the device dose analogue is met only when both the simulation evidence and the real-time safety envelope are demonstrably in place [2, 1].

The statistical framework is that of a small, intensively monitored, single-arm investigation whose primary purpose is to characterize early safety and to identify a recommended dose, not to test a comparative-efficacy hypothesis. The analyses are predominantly descriptive and estimation-based,

follow ICH E9, and are reported in accordance with TRIPOD+AI for the model-assisted advisory component [12]. No power calculation is performed because the study tests no confirmatory efficacy hypothesis. Every proportion is accompanied by an exact (Clopper-Pearson) two-sided 95 percent confidence interval rather than a normal-approximation interval, which is unreliable near the bounds and at small sample sizes; time-to-event variables are summarized by Kaplan-Meier with the number at risk; missing data are not imputed for the primary or safety analyses; and a nominal alpha of 0.05 two-sided applies to supportive estimation only, with no multiplicity adjustment because the analyses are descriptive and non-confirmatory. Four analysis populations are applied consistently: the Safety population (all who received any intervention), the DLT-evaluable population (all dosed participants who completed the 28-day window or had a DLT within it), the Per-Protocol population, and the modified intention-to-treat population [12, 7].

4.4 Description of First Year Trial(s)

During the first year, a single trial is conducted at a single academic medical center: the combined IND/IDE first-in-human study of perioperative daraxonrasib and the LLM-directed eight-arm robotic Whipple. Enrollment is planned for up to 18 treated participants, allocated across the three daraxonrasib dose levels (DL1 through DL3) by the 3+3 rule with staggered sentinel enrollment, from approximately 36 screened. Recruitment proceeds at approximately one to two participants per month through the institutional multidisciplinary pancreatic-cancer program, with eligible individuals identified at tumor-board review and described to them by a member of the research team who is not the treating surgeon, to reduce undue influence. The planned accrual rate, the sentinel staggered enrollment of the 3+3 design, and the Phase 0 readiness gate together give an enrollment period sufficient to treat 18 participants from approximately 36 screened [1, 7].

Sentinel staggering is a binding feature of the first-year conduct: no two participants in a cohort undergo surgery within the same DLT-evaluation interval until the supervising surgeon and the DSMB have reviewed the preceding case. After each participant completes the 28-day DLT-evaluation window and before the next participant in the cohort undergoes surgery, the accumulating safety data are reviewed; after each dose-level cohort is complete, the DSMB convenes for a formal interim safety review and an escalation, expansion, hold, or halt decision. The prespecified halt rules are quantitative and binding: enrollment and escalation halt for a mandatory DSMB review if the device-related or procedure-related serious-AE rate reaches or exceeds one third within a cohort, or if two device-related deaths occur at any point. Additional pause triggers include any unexpected device-related serious AE, any breach of a hard cyber-physical safety limit (emergency-stop latency, positional, or force envelope), and any failure of the pre-procedure safety-matrix verification. The DSMB may halt, pause, modify, or permit continuation, and a halt for an immediate hazard is implemented at once [1].

The device operates within a phase-graduated staged-autonomy model. In this Phase 1 study the system runs only at the clinician-controlled (teleoperation or supervised) level and the supervised semi-autonomous level, both under continuous human oversight with an immediate manual override; full autonomy is prohibited at this phase consistent with 21 CFR §312.21(e). Any future move to higher autonomy (Stage 2 supervised semiautonomous expansion or Stage 3 higher autonomy) is reserved for a later phase and gated by independent safety review, a USL reassessment, and DSMB concurrence. The perioperative daraxonrasib pause-and-restart advisory is exercised in every case: daraxonrasib is paused before surgery and the restart day is set by an LLM-bound, human-confirmed advisory whose inputs are the postoperative pancreatic fistula grade by ISGPS (A, B, or C) and the serum daraxonrasib trough relative to a 0.5 ng/mL threshold, with outputs of T+7 days, T+14 days, T+21 days, or hold. A

32-iteration deterministic sweep returned a baseline serum of 0.45 ng/mL (sub-threshold) in all iterations, a T+7-day restart in 29 of 32 (90.6 percent, Grade A and sub-threshold), a T+14-day restart in 3 of 32 (9.4 percent, Grade B or delayed), and no T+21-day restart [1, 6].

The first-year objectives map onto the endpoint hierarchy of Figure 4: the co-primary safety, dose-finding, and feasibility readouts are accrued as cohorts complete, the secondary surgical-quality endpoints are estimated against the Dutch 2025 benchmark as the per-protocol set matures, and the exploratory survival, advisory-concordance, and sim-to-real measures begin to accumulate through the day-90 pathology review and the 24-month survival follow-up. The realistic first-year deliverable is completion of the DL1 and DL2 cohorts and initiation of the DL3 cohort, with an interim determination of the MTD and a provisional RP2D if DL3 is reached and tolerated, together with the device-feasibility and telemetry summaries that support progression to a later-phase efficacy study [1].

4.5 Number of Subjects to be Evaluated

The planned sample size is up to 18 treated participants, allocated across three daraxonrasib dose levels (DL1 through DL3) by the 3+3 rule, from approximately 36 screened. This number is not derived from a power calculation, because the study tests no confirmatory efficacy hypothesis; it is determined jointly by the dose-finding design and by the early-feasibility precedent for significant-risk devices, and it is justified by its operating characteristics. The 3+3 rule enrolls cohorts of three: a level with no DLT in three escalates, a level with one DLT in three expands to six, and a level with two or more DLTs in up to six is exceeded, defining the dose below it as the MTD. The maximum exposure at any single dose level is therefore six participants, and three escalation levels give a design maximum of three times six, equal to 18 treated participants; many realizations terminate with fewer. This ceiling is consistent with the FDA early-feasibility convention of generally fewer than ten participants per device-feasibility cohort while providing the per-dose exposure the drug-arm 3+3 escalation requires [7].

Rather than power, the design is characterized by its operating characteristics and by the precision of estimation at the planned sample sizes. The probability that the 3+3 rule escalates past a given dose level is a function of the true DLT probability at that level: a dose with a true DLT rate near 10 percent is very likely to be tolerated and escalated, whereas a dose with a true DLT rate of 40 percent or higher is very likely to be identified as excessively toxic and to define the MTD at the level below. The exact-interval precision is summarized in Table 4. Zero events in three yields an exact two-sided 95 percent confidence interval of approximately 0 to 71 percent; zero in six, approximately 0 to 46 percent; zero in 18, approximately 0 to 18 percent; and one event in 18, a point estimate of 5.6 percent with an interval of approximately 0.1 to 27 percent. These intervals make explicit both the value and the limits of a first-in-human cohort: a clean safety record across the full cohort of 18 bounds the true event rate below roughly one in five, while a single event still excludes rates above roughly one quarter [7, 12].

Quantity	Planned value	Operating characteristic / precision
Cohort size	3, expand to 6	3+3 rule: escalate on 0/3; expand on 1/3; exceed dose on $\geq 2/6$.
Per-dose maximum	6	Maximum exposure at any one dose level before the MTD is declared.
Dose levels	3 (DL1 to DL3)	Three-level escalation; design maximum $3 \times 6 = 18$ treated.
Total treated	up to 18 (approx. 36 screened)	At or below the early-feasibility convention of generally fewer than 10 per device cohort, summed across dose levels.
0 events / 3	point estimate 0%	Exact two-sided 95% CI approximately 0% to 71%.
0 events / 6	point estimate 0%	Exact two-sided 95% CI approximately 0% to 46%.
0 events / 18	point estimate 0%	Exact two-sided 95% CI approximately 0% to 18%.
1 event / 18	point estimate 5.6%	Exact two-sided 95% CI approximately 0.1% to 27%.
Significance level	$\alpha = 0.05$ (two-sided)	Applies to supportive estimation only; no α is spent on dose selection.

Table 4. Operating characteristics and exact-interval (Clopper-Pearson) precision at the planned sample sizes. The dose-finding behavior is that of the standard 3+3 rule; no statistical power is computed because the study tests no confirmatory efficacy hypothesis.

The four analysis populations apportion the 18 participants across the analyses without altering the total exposure: the Safety population comprises all who received any intervention and is the primary population for safety and primary-endpoint analyses; the DLT-evaluable population comprises those who completed the 28-day window or had a DLT within it and drives the escalation and MTD determination; the Per-Protocol population supports the secondary surgical-quality analyses; and the modified intention-to-treat population supports the supportive and exploratory efficacy summaries. Because the cohort is small and intensively monitored, individual participant data are tabulated in full, so that every aggregate statistic can be traced to its contributing records [12].

4.6 Drug Related Risks

The foreseeable risks of the investigation arise from three sources, the investigational drug, the investigational device, and the surgical procedure, and each is bounded by a specific control rather than left to clinical judgment alone. The anticipated risks of daraxonrasib are those characteristic of RAS-pathway inhibition and of the RASolute 302 experience, including gastrointestinal effects (nausea, diarrhea, and stomatitis), dermatologic effects (rash), hepatic enzyme elevation, and the class effects of MAPK-pathway suppression; in the perioperative setting the additional concern is impaired wound healing and anastomotic integrity, which is the explicit reason for the pause-and-restart advisory that holds daraxonrasib before surgery and resumes it only on a human-confirmed, fistula-grade-and-trough-informed schedule (T+7, T+14, T+21 days, or hold). Drug toxicity is captured through the 28-day DLT window, the dose-blinded SAE attribution, the exact-interval safety reporting, and the binding cohort-level halt rules [1, 9]. Table 5 summarizes the principal risks, their sources, and their mitigations.

Risk source	Foreseeable risk	Mitigation
Daraxonrasib (drug)	GI effects (nausea, diarrhea, stomatitis), rash, hepatic enzyme elevation, MAPK-pathway class effects; impaired perioperative wound healing.	28-day DLT window; conservative 160 mg starting dose below the 300 mg RP2D; pause-and-restart advisory keyed to ISGPS grade and 0.5 ng/mL trough; cohort-level halt rules.
Robotic system (device)	Device malfunction, unsafe trajectory, excessive tip force, vascular injury near the SMV, PV, HA, CA, or SMA.	Force caps ≤ 3 N per arm and ≤ 18 N cumulative; cross-arm E-stop ≤ 3 ms; hardware E-stop ≤ 500 ms; 10 kHz heartbeat with 100 microsecond watchdog; vascular no-fly gating; Phase 0 gate (USL ≥ 7.0 , ≥ 1000 sims, sim-to-real < 2 mm and < 0.5 N).
LLM advisory layer	Erroneous or unverified advisory output; over-reliance (therapeutic misconception); model degradation.	Second-opinion advisory only; full autonomy prohibited (§312.21(e)); ten-gate verification-before-generation (ACCEPT/BLOCK/ESCALATE); hash-pinned deterministic seed and commit; Physical AI opt-out; advisory-to-gate concordance monitored.
Whipple procedure	Pancreatic fistula, hemorrhage, intra-abdominal infection, sepsis, failure to rescue, unplanned conversion.	Sentinel staggered enrollment with case-by-case DSMB review; three-tier oversight (DSMB, Independent Safety Monitor, Physical AI Safety Review Committee); conversion pathway with continuous human override.
Schedule and population	Delay to effective therapy in a low-survival disease.	Verified single-prompt authoring compresses only the administrative bucket, advancing first dose without altering tumor biology or fixed review clocks (Figure 3).

Table 5. Principal drug-related, device-related, procedural, and schedule-related risks with their controls. Each risk is bounded by a specific, prespecified mitigation; the residual risk after these controls is judged acceptable for the population and the stage of investigation.

The device-related and procedural risks are bounded by the cyber-physical safety envelope and the three-tier oversight structure. The hard caps (per-arm tip force at or below 3 N, cumulative cross-arm force at or below 18 N, cross-arm emergency stop within 3 ms to 0.0 N, and hardware emergency stop within 500 ms), the 10 kHz heartbeat with the 100 microsecond watchdog, and the five-vessel no-fly

gating limit the magnitude of any single excursion, while the Phase 0 gate (USL at least 7.0, at least 1000 simulations across at least two frameworks, and a sim-to-real gap below 2 mm and 0.5 N) ensures the system is demonstrably ready before any patient-facing activity. The LLM advisory risks (an erroneous or unverified advisory and the therapeutic misconception of over-reliance on the system) are bounded by the advisory-only role, the prohibition on full autonomy under 21 CFR §312.21(e), the ten-gate verification-before-generation suite, the hash-pinned deterministic seed and commit, the Physical AI opt-out in the consent, and the monitoring of advisory-to-gate concordance. Safety reporting follows the 21 CFR §312.32 clocks (fatal or life-threatening suspected adverse reactions within 7 calendar days and other serious and unexpected reactions within 15 calendar days), augmented by the six Physical AI reporting triggers under §312.32(g) and an audit trail preserved from 24 hours before to 72 hours after each Physical AI adverse event under 21 CFR part 11 [3, 5].

The benefit-risk balance is favorable for this population and at this stage. The population is vulnerable, with a lethal malignancy whose five-year survival is below 13 percent and for whom the resection margin is the strongest modifiable determinant of survival; the intervention pairs a genotype-matched systemic agent with a margin-negative curative-intent resection delivered within hard cyber-physical limits under continuous human oversight. The schedule benefit reaches the patient through the cascade of Figure 3: faster validated authoring advances IND and IRB submission and site activation, the elapse of the 30-day clock and the first dose, and the 3+3 cohort turnover toward the RP2D, shortening the white space between phases and, for this low-survival population, extending progression-free and overall survival, without changing tumor biology or any fixed review clock. For patients whose disease will not pause, the alternative of slower sequential authoring carries its own, often larger, cost in delayed treatment. With each foreseeable risk bounded by a prespecified control, with the conservative 160 mg starting dose, the sentinel staggered enrollment, the binding halt rules, and the three-tier oversight, the probable benefit of the investigation exceeds its probable risk, and the General Investigational Plan supports the proposed first-year conduct [13, 1, 6].

4.7 References

The clinical, statistical, and acceleration content of this plan is grounded in the trial protocol and the supporting evidence base. The Phase 1 first-in-human guidance and the early-feasibility framework inform the single-arm dose-finding design and the operating characteristics [7]. The daraxonrasib robotic Whipple program and the on-premises eight-arm system specifications supply the drug, device, and advisory facts [1, 2, 9]. The Physical AI adaptations of 21 CFR Part 312 and Part 50 and the ICH guidance govern the safety reporting, consent, and statistical conventions [3, 11, 12]. H.R. 9510 and the clinical-trust framework establish the verification-before-generation discipline [5, 6]. The PDAC epidemiology, the acceleration mechanism, and the supporting computational evidence are drawn from the population and document-acceleration sources [13, 14, 10, 8, 4, 19, 15, 16, 17, 18]. The full bibliography is consolidated in the references and back-matter section of this IND.

5 Investigator Brochure

This section is the investigational new drug (IND) level summary of the Investigator's Brochure (IB) for the LLM-directed eight-arm robotic pancreaticoduodenectomy with perioperative daraxonrasib Phase 1 study. It is provided here as a clearly labeled supporting document under 21 CFR §312.23(a)(5) and §312.55, drawing together, in one place and at IND-summary depth, the physical, chemical, and pharmaceutical properties of the investigational drug daraxonrasib (RMC-6236); the available nonclinical pharmacology, pharmacokinetics, and toxicology that underpin the conservative 160 mg first-in-human starting dose; a full description of the investigational eight-arm collaborative surgical device (56 degrees of freedom, 640 sensor channels, the per-arm and cumulative force caps, the layered emergency-stop chain, and the five-vessel no-fly gate); and a structured summary of the data with explicit guidance for the investigator. The full, standalone Investigator's Brochure is a separate bound document submitted with this IND and incorporated by reference; the present section is the IND-level synopsis and the pointer to that document, consistent with the ReGARDD convention of using a distinct section to indicate and include a supporting document. The clinical-conduct detail that the IB summarizes is carried in the study protocol (§6 of this IND), the chemistry, manufacturing, and control information (§7), and the pharmacology and toxicology information (§8), and the investigator is directed to those sections and to the bound IB for the controlling text [7, 1].

Supporting Document Status and Scope

The Investigator's Brochure assembles, for the clinical investigator and the reviewing institutional review board (IRB), all reference safety and efficacy information needed to conduct the study and to assess the expectedness of adverse events. Because the intervention has two inseparable components, an investigational drug evaluated under 21 CFR part 312 and an investigational device evaluated under 21 CFR part 812 in a single combined IND and IDE submission, the IB has two coordinated parts: the daraxonrasib drug brochure and a Physical AI device supplement that functions as the reference safety information for the device and the on-premises advisory large language model (LLM). An adverse event is judged unexpected, and therefore potentially reportable on an expedited timeline, when it is not listed at the observed specificity or severity in the daraxonrasib IB (for drug events) or in the Physical AI supplement and System Description (for device events), as specified in §8 of this IND and in the protocol assessment section. The IB is reviewed and reissued at least annually and whenever new safety information materially changes the risk profile, and currently enrolled participants are re-consented when such a change occurs. The version of the IB in effect at the time of each procedure is recorded in the case report form so that expectedness determinations are traceable to a fixed reference text [1, 3].

Physical, Chemical, and Pharmaceutical Properties of Daraxonrasib

Daraxonrasib (development code RMC-6236) is an orally bioavailable, small-molecule, RAS(ON) multi-selective (pan-RAS) inhibitor. Its pharmacological mechanism is a tri-complex mode of action: the molecule binds the active, guanosine triphosphate (GTP) bound conformation of mutant RAS in a ternary complex with the abundant intracellular chaperone cyclophilin A, and in doing so it sterically and allosterically blocks the engagement of downstream effectors, suppressing the mitogen-activated protein kinase and phosphoinositide 3-kinase signaling that drives pancreatic ductal adenocarcinoma (PDAC). Because it targets the shared GTP-bound state rather than a single codon substitution, daraxonrasib is active across the spectrum of common oncogenic KRAS variants relevant to PDAC, including the G12D, G12V, and G12R substitutions that define the eligible population of this study. The agent is administered as an oral tablet, once daily, and is supplied in the strengths required to compose the three protocol dose levels of 160 mg, 220 mg, and 300 mg once daily. Tablets are swallowed whole with water and are not crushed, split, or chewed; they are packaged in child-resistant, light-protective

containers labeled under 21 CFR §312.6 with the investigational-use caution statement, the protocol number, the lot number, and the storage conditions, and are stored at controlled room temperature in a secured, temperature-monitored, access-controlled location until dispensed. The detailed chemistry, manufacturing, and control characterization of the drug substance and drug product, including appearance, composition, container-closure system, and the stability program that governs the labeled expiry, is provided in §7 of this IND; the present summary restates only the properties most relevant to the investigator at the point of care [1, 13].

The clinical and regulatory standing of daraxonrasib supports its use as the drug component of this first-in-human perioperative study. The agent received FDA Breakthrough Therapy designation in June 2025 for previously treated metastatic PDAC harboring a KRAS G12 mutation, and the dose and exposure assumptions of this protocol are anchored to the pan-KRAS RASolute 302 program, in which 300 mg once daily is the established recommended Phase 2 dose (RP2D). In the present study the drug is investigational specifically in the perioperative, curative-intent surgical context and for its use in combination with the investigational device; the escalation deliberately begins one and two steps below the established RP2D to preserve a conservative first-in-human margin in this new clinical setting [1, 10].

Nonclinical Pharmacology, Pharmacokinetics, and Toxicology

The pharmacology of daraxonrasib is summarized in §8.1.1 of this IND and rests on the RAS(ON) tri-complex mechanism described above: selective engagement of the GTP-bound mutant RAS in complex with cyclophilin A, with downstream suppression of effector signaling and consequent antiproliferative and pro-apoptotic activity in KRAS-mutant PDAC models. Drug distribution and pharmacodynamics support a once-daily oral schedule with sustained target coverage. The pharmacokinetic behavior most relevant to this protocol is the relationship between the daily dose, the serum trough concentration, and the 0.5 ng/mL trough threshold that governs the perioperative pause-and-restart advisory. A 32-iteration deterministic pharmacokinetic sweep, performed to characterize the perioperative window, returned a baseline serum daraxonrasib concentration of 0.45 ng/mL at the operative timepoint across all 32 iterations, that is, below the 0.5 ng/mL trough threshold in every iteration, consistent with the protocol-mandated pre-operative pause; the advisory then recommended a postoperative restart on day T+7 in 29 of 32 iterations (90.6 percent, corresponding to a Grade A fistula with a sub-threshold trough), on day T+14 in 3 of 32 iterations (9.4 percent, corresponding to a Grade B or delayed serum rise), and on day T+21 in 0 of 32 iterations. This sweep is described in the study protocol (§6 of this IND) and is summarized in §8 and §3.4 [16, 15, 18].

The toxicology integrated summary (§8.1.2) states the safety margins that underpin the conservative DL1 160 mg first-in-human starting dose. The class toxicity profile of RAS(ON) multi-selective inhibition is dominated by gastrointestinal effects (diarrhea, nausea) and dermatologic effects (rash), which are managed symptomatically and through the protocol dose-interruption and dose-reduction rules; rescue management of these class effects, including antidiarrheals and aggressive dermatologic care, is specified in the protocol. The 160 mg starting dose sits a full step below the 220 mg intermediate level and two steps below the 300 mg established RP2D, providing a deliberate margin for the perioperative surgical setting in which the drug is paused across the operative window and restarted only on a fistula-grade-keyed and trough-keyed schedule. The full data tabulation (§8.1.3) tabulates the supporting studies; for this AI-generated submission the supporting nonclinical and in silico evidence base includes the author's quantitative-systems-pharmacology metastatic-PDAC simulation with verification, validation, and uncertainty quantification; the artificial intelligence digital-twin PDAC simulations; the 100,000-patient in silico Phase 3 five-arm PDAC evaluation; and the end-to-end PDAC digital-twin trial work, each grounded in the references identified in the figure below. The pharmacology and toxicology integrated-summary structure that organizes §8 is reproduced in Figure 17 [16, 15, 18, 17].

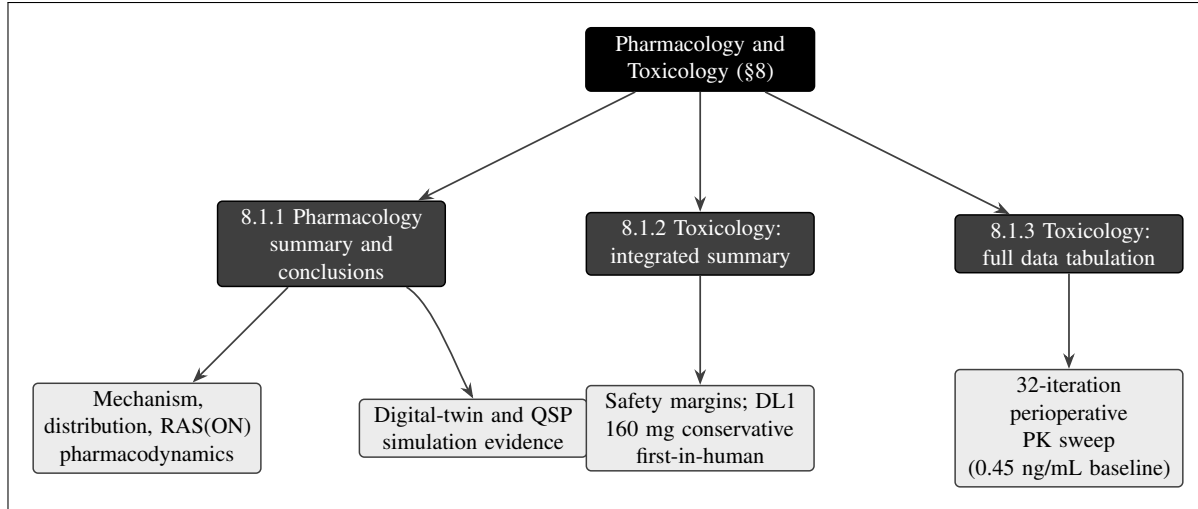


Figure 17. The pharmacology and toxicology integrated-summary structure for §8. The pharmacology summary covers the RAS(ON) mechanism, drug distribution, and pharmacodynamics; the integrated toxicology summary states the safety margins underpinning the conservative DL1 160 mg first-in-human starting dose; and the full data tabulation tabulates the supporting studies, here grounded in the author's digital-twin and quantitative-systems-pharmacology simulation evidence and the 32-iteration perioperative pharmacokinetic sweep (0.45 ng/mL baseline serum trough at the operative timepoint across all iterations).

Investigational Device Description: the Eight-Arm Robotic System

The investigational device is an on-premises, LLM-directed, eight-arm parallel coelomic surgical platform that conducts the pancreaticoduodenectomy and generates the operative and pathology inputs to the perioperative drug-restart advisory. The on-premises, repository-pinned LLM acts strictly as a second-opinion advisory oracle held outside the robot vendor kinematic stack and is never an autonomous controller; full autonomy is prohibited consistent with 21 CFR §312.21(e), and in this Phase 1 study the device operates only at the clinician-controlled teleoperated and supervised semiautonomous autonomy levels under continuous human oversight with immediate manual override. The platform carries 56 mechanical degrees of freedom, composed of eight arms with seven degrees of freedom each (8 multiplied by 7), and a 640-channel mixed-rate sensor stack of 80 channels per arm. Force channels sample at 100 kHz and all remaining channels sample at the 10 kHz command rate. Positioning accuracy is 0.05 mm root mean square at a peak tip velocity of 1,200 mm/s. The cyber-physical safety envelope comprises a per-arm tip-force cap of no more than 3 N and a cumulative cross-arm tip-force cap of no more than 18 N enforced at the heartbeat boundary; a 10 kHz heartbeat coordination bus broadcasting a 64-byte frame on a 100 microsecond per-arm watchdog deadline; an emergency arm park within 50 microseconds of a missed or out-of-parameter frame; a cross-arm emergency stop (E-stop) that halts the full platform within 3 ms at a 0.0 N force clamp; and a software-independent hardware E-stop that meets the 21 CFR §312.404 requirement of no more than 500 ms. The device specifications are summarized in Table 4.1, and the layered safety envelope, including the five-vessel no-fly gate, is summarized in Table 4.2 [1, 2].

The eight cooperating arms are coordinated by the 10 kHz heartbeat bus and carry phase-keyed tool assignments across the eight operative phases. Phases P1 through P4 comprise the dissection and specimen removal: Arms 1 and 2 perform the active dissection under the vessel control of Arm 3 (bipolar plus retractor) and the near-infrared indocyanine-green imaging of Arm 4, with Arm 7 providing suction and irrigation throughout. The specimen is removed at the end of P4 and reconstruction begins.

Table 5: Eight-arm LLM-directed robotic Whipple device specifications (Investigator’s Brochure device supplement; 21 CFR §312.23(g)(iii) hardware fields).

Specification	Value	Notes
Architecture	Eight-arm parallel coelomic platform	On-premises LLM advisory oracle outside vendor kinematic stack
Degrees of freedom	56 total (8 arms x 7 DOF)	Cooperating arms, phase-keyed tool assignment P1 to P8
Sensor channels	640 total (80 per arm)	Force at 100 kHz; all other channels at 10 kHz command rate
Positioning accuracy	0.05 mm RMS at 1,200 mm/s	Verified in the pre-procedure safety matrix each case
Per-arm tip-force cap	3 N maximum	Enforced continuously per arm
Cumulative force cap	18 N maximum	Enforced at the heartbeat boundary across all arms
Heartbeat bus	10 kHz, 64-byte frame	100 microsecond per-arm watchdog deadline
Emergency arm park	50 microseconds maximum	On a missed or out-of-parameter frame
Cross-arm E-stop	3 ms maximum, 0.0 N clamp	Software escalation halting the full platform
Hardware E-stop	500 ms maximum	Software-independent; meets 21 CFR §312.404
Autonomy level (Phase 1)	Teleoperated / supervised	Continuous human oversight; full autonomy prohibited (§312.21(e))

Table 4.1. Device specifications carried in the Investigator’s Brochure device supplement and the Physical AI System Description; the controlling hardware, software, and safety-system text is in §6 and §7 of this IND and in the bound IB.

The three reconstructive anastomoses are then constructed in sequence under closed-loop ring-tension control by the Arm 5 anastomosis probe, which issues a soft warning whenever measured ring tension falls outside its controller band: the duct-to-mucosa pancreaticojejunostomy (PJ) in P5 with a band of 0.30 to 0.60 N, the end-to-side hepaticojejunostomy (HJ) in P6 with a band of 0.20 to 0.50 N, and the antecolic gastrojejunostomy (GJ) in P7 with a band of 0.40 to 0.80 N; P8 covers final imaging, hemostasis, and drain placement. The pancreaticojejunostomy is the dominant determinant of postoperative pancreatic-fistula risk and is therefore the anastomosis whose ISGPS fistula grade supplies the input to the perioperative drug-restart advisory [1, 8].

Safety Envelope: Force Caps, E-Stop Chain, and the No-Fly Gate

The device safety envelope is enforced before and during every procedure rather than reconstructed afterward. The vascular safety-zone gate ticks at 10 kHz and evaluates the instrument-tip proximity to each of five named vessels, the superior mesenteric vein (SMV), the portal vein (PV), the hepatic artery (HA), the celiac axis (CA), and the superior mesenteric artery (SMA), against three concentric thresholds: a soft-warning zone at 3.0 mm that raises an advisory but does not block the command, a no-fly zone that blocks the command (1.0 mm at the SMV and PV; 1.5 mm at the HA, CA, and SMA), and a

Table 6: Device safety envelope: force caps, layered emergency-stop chain, and the five-vessel no-fly gate.

Safety element	Threshold / action	Basis and verdict sample
Per-arm force cap	3 N maximum per arm	Continuous; force channels at 100 kHz
Cumulative force cap	18 N maximum across arms	Enforced at the heartbeat boundary
Heartbeat and watchdog	10 kHz bus, 100 microsecond deadline	On-time frame continues command state
Arm park	50 microseconds maximum	Triggered by a missed or out-of-parameter frame
Cross-arm E-stop	3 ms maximum, 0.0 N clamp	Software escalation; halts full platform
Hardware E-stop	500 ms maximum	Independent of software; 21 CFR §312.404
Soft-warning zone	3.0 mm, all five vessels	Advisory only; 6 of 100 ticks
No-fly zone (block)	1.0 mm SMV/PV; 1.5 mm HA/CA/SMA	Command blocked; 6 of 100 ticks
Hard-stop zone	5.0 mm, all five vessels	Forces E-stop; 5 of 100 ticks
Clear verdict	Outside all zones	Command proceeds; 83 of 100 ticks

Table 4.2. The cyber-physical safety envelope and the five-vessel no-fly gate (SMV, PV, hepatic artery, celiac axis, SMA), sampled at 10 kHz; the representative 100-tick verdict sample partitions into 83 clear, 6 soft warning, 6 no-fly, and 5 hard-stop ticks. The controlling text is in the protocol assessment section (§6 and §8 of this IND).

hard-stop zone at 5.0 mm that forces a cross-arm E-stop within 3 ms. Across a representative 100-tick verdict sample the gate returned clear on 83 ticks, soft warning on 6, no-fly on 6, and hard-stop on 5, exercising all five vessels across all eight operative phases. The halt chain that backs the gate is layered: an on-time heartbeat frame continues the command state, whereas a missed or out-of-parameter frame parks the affected arm within 50 microseconds and escalates to the cross-arm E-stop, and the software-independent hardware E-stop backs the software chain at the 500 ms outer bound. The safety envelope is summarized in Table 4.2 [1, 6].

Every advisory command, every safety-gate verdict, every force and trajectory sample at the recorded rates, and every E-stop event is written to a 21 CFR part 11 hash-chained, append-only audit trail, so that compliance with the cyber-physical envelope is verifiable for each case and any deviation is time-stamped and detectable. The audit trail also satisfies the preservation requirement from 24 hours before to 72 hours after any reportable Physical AI adverse event. Before any patient-facing robotic activity, the platform must clear the Phase 0 simulation validation gate (a Unified Safety Level of at least 7.0; at least 1,000 simulations across at least two simulation frameworks; a sim-to-real trajectory gap below 2 mm; and a tip-force gap below 0.5 N), and the gate is re-cleared after any change. The on-premises LLM is hash-pinned to a deterministic seed and repository commit and is governed by a verification-before-generation gate suite, aligned with H.R. 9510, the Verification Before Generation in Physical AI Oncology Trials Act of 2026 [5, 6, 4].

Summary of Data and Guidance for the Investigator

The investigator should treat the daraxonrasib drug brochure and the Physical AI device supplement as the two controlling reference texts for expectedness and risk. For the drug, the established 300 mg once-daily RP2D and the class toxicity profile (gastrointestinal and dermatologic effects) define the

expected events; the study deliberately starts at 160 mg once daily to preserve a conservative first-in-human margin in the perioperative setting, and the 28-day dose-limiting toxicity window, the 3+3 escalation, and the dose-interruption and dose-reduction rules govern management. For the device, the force caps, the layered E-stop chain, and the five-vessel no-fly gate define the expected safety behavior; any robotic malfunction, AI prediction error, sensor failure, communication loss, unauthorized access, digital-twin divergence, or event requiring activation of the E-stop is a Physical AI adverse event under 21 CFR §312.32(f) and is assessed against this supplement for expectedness. The perioperative pause-and-restart advisory is LLM-bound and human-confirmed and never autonomous; the investigator confirms the restart day keyed to the pancreaticojejunostomy ISGPS fistula grade and the serum trough relative to 0.5 ng/mL, with a Grade C fistula mapping to T+21 or a continued hold and to the drug stopping rule. All adverse events are graded by CTCAE version 5.0 and reported on the regulatory timelines (no later than 7 calendar days for an unexpected fatal or life-threatening suspected adverse reaction and no later than 15 calendar days for any other serious and unexpected suspected adverse reaction), with the six Physical AI reporting triggers of §312.32(g) layered on top [1, 3, 12].

The investigator is reminded that the controlling clinical-conduct text is the study protocol and that the present section is an IND-level summary; the full standalone Investigator's Brochure is a separate bound document submitted with this IND and is the authoritative reference safety information for both the drug and the device. The investigator should hold a current copy of the bound IB, record its version in the case report form for each procedure, and consult §6, §7, and §8 of this IND for the drug-and-device intervention detail, the chemistry, manufacturing, and control information, and the pharmacology and toxicology information that the IB summarizes [7, 1, 9].

6 Proposed clinical research

The proposed clinical research is presented in this section of the investigational new drug (IND) application in the form required by 21 CFR §312.23(a)(6). It consists of the study protocol for the planned Phase 1 first-in-human investigation, the informed consent that governs each participant's enrollment, and the investigator and facilities information that establishes that the study will be conducted by qualified persons at adequate facilities. The investigation described here is a prospective, open-label, single-arm, first-in-human study that combines a standard 3+3 dose-finding evaluation of perioperative daraxonrasib (RMC-6236) with an early-feasibility evaluation of an on-premises large language model (LLM) directed eight-arm robotic pancreaticoduodenectomy (Whipple) system, a significant-risk Class II collaborative device. Because the device participates in the delivery of the investigational drug, the study is conducted under a combined IND (21 CFR §312) and investigational device exemption (IDE, 21 CFR §812) submission with a single integrated protocol, a single informed consent carrying the Physical AI opt-out, and a single integrated safety governance structure. The full clinical protocol, version 1.0.0, is bound to this IND as a supporting document and is archived with the digital object identifier DOI 10.5281/zenodo.20780121 (<https://doi.org/10.5281/zenodo.20780121>); the synopsis and supporting tables in §6.1 summarize that bound document, and every quantitative value reported here is traceable to it [1, 2].

6.1 Study Protocol

Protocol identification and overview

The bound protocol is entitled the LLM-Directed PDAC Robotic Daraxonrasib Phase 1 study (protocol version 1.0.0, DOI 10.5281/zenodo.20780121). It is a prospective, open-label, single-arm, first-in-human dose-finding and early-feasibility study in adults with histologically or cytologically confirmed, resectable or borderline-resectable, KRAS G12-mutated pancreatic ductal adenocarcinoma (PDAC) who are candidates for pancreaticoduodenectomy. The central hypothesis is that the combined drug-device intervention can be delivered with an acceptable early safety profile and can complete the assigned operative tasks under continuous human oversight while a recommended Phase 2 dose (RP2D) of daraxonrasib is identified, supporting progression to a later-phase efficacy study. Enrollment is planned for up to $n = 18$ treated participants (approximately 36 screened) at a single academic medical center, assigned across three daraxonrasib dose levels by the 3+3 rule with staggered sentinel enrollment, at a planned accrual of approximately one to two participants per month. The investigation is structured around the early-feasibility pathway recognized for significant-risk devices, which generates the device-performance and safety evidence required before any controlled comparison is undertaken [7, 1].

Objectives and endpoints

The protocol specifies three co-primary objectives, each evaluated with exact two-sided 95% Clopper-Pearson confidence intervals because the design is descriptive and non-confirmatory under ICH E9 with no formal power calculation and no multiplicity adjustment. The first co-primary endpoint is the incidence of device- or procedure-related serious adverse events through 30 days, including any Clavien-Dindo Grade III or higher complication. The second co-primary endpoint is the maximum tolerated dose (MTD) and the RP2D of perioperative daraxonrasib, determined by the 3+3 rule across a 28-day dose-limiting toxicity (DLT) window. The third co-primary endpoint is technical feasibility, defined as the proportion of participants completing the assigned operative tasks without unplanned conversion attributable to device malfunction or unsafe device behavior. The secondary endpoints are the margin-negative (R0) resection rate on day-90 pathology (per-protocol), the International Study Group of Pancreatic Surgery (ISGPS) Grade B or C postoperative pancreatic-fistula rate, the Clavien-Dindo Grade

III or higher complication rate at 90 days, 30-day and 90-day mortality, the conversion rate, and operative parameters (estimated blood loss, operating-room time, and length of stay), each benchmarked descriptively against the 2025 Dutch nationwide robotic pancreaticoduodenectomy cohort (conversion 10.1%, Clavien-Dindo III+ 41.3%, ISGPS B/C 24.4%, 30-day mortality 3.9%). Exploratory endpoints are progression-free and overall survival to 24 months by RECIST version 1.1 with Kaplan-Meier estimation, the concordance of the LLM advisory with the hard-coded safety gate, and the sim-to-real gap in millimeters and newtons [1, 9].

Table 7 presents the schedule of activities that governs the timing of every assessment, from screening through the operative episode and the 24-month overall-survival follow-up.

Study design and dose escalation

The drug component proceeds under the IND pathway and the device component under the IDE pathway, with a single integrated protocol and a single safety governance structure. The escalation uses three dose levels anchored to the pan-KRAS RASolute 302 program, in which 300 mg once daily is the established RP2D: DL1 is 160 mg once daily (starting dose, chosen below the reference dose to preserve a conservative first-in-human margin), DL2 is 220 mg once daily, and DL3 is the 300 mg once-daily reference dose. Three participants are enrolled at each level and observed across a 28-day DLT window. If no DLT occurs among the first three, the cohort escalates; if one of three experiences a DLT, the cohort expands to six; a single DLT among six permits escalation, whereas two or more DLTs at a level fix the MTD at the preceding level. The MTD is the highest level at which fewer than two of six DLT-evaluable participants experience a DLT, and the RP2D is selected at or below the MTD. With a maximum of three levels of six, the design treats up to 18 participants. The estimation precision that this sample supports is illustrated by the exact Clopper-Pearson intervals: an observed rate of 0 of 18 yields 0% (95% CI 0 to 18%), 1 of 18 yields 5.6% (95% CI 0.1 to 27%), 0 of 3 yields 0% (95% CI 0 to 71%), and 0 of 6 yields 0% (95% CI 0 to 46%) [1]. The analysis populations are the Safety population (all dosed participants), the DLT-evaluable population, the Per-protocol population, and the modified intention-to-treat (mITT) population.

The device has no pharmacologic dose; its dose analogue is the readiness evidence required before any patient-facing activity, escalated only by passing the mandatory Phase 0 simulation-validation gate. Three conditions must be met and documented before the first robotic case: a Unified Safety Level (USL) rating of at least 7.0 on the surgical-readiness scale; at least 1000 simulated procedures completed across at least two independent simulation frameworks (for example Isaac, MuJoCo, Gazebo, or PyBullet) so that the readiness finding is not an artifact of any single simulator; and a sim-to-real fidelity within prespecified limits, a trajectory gap of less than 2 mm and a tip-force gap of less than 0.5 N between the simulation prediction and the operative record. The gate is re-run after any software, model, or configuration change before any further patient-facing use [2, 15].

Population and eligibility

The study population comprises adults with histologically or cytologically confirmed, resectable or borderline-resectable, KRAS G12-mutated PDAC who are eligible for perioperative daraxonrasib under the broad pan-KRAS RASolute 302 criteria and whose tumor and vascular anatomy are compatible with the eight-arm robotic system. Participant selection is equitable; eligibility is determined solely by the scientific and safety criteria, and limited English proficiency is not an exclusion criterion. The synthetic reference exemplar PAT-PDAC-0001 (a 68-year-old with a 3.4 cm head-of-pancreas tumor abutting the superior mesenteric vein at 75°, KRAS G12D, microsatellite stable, CA 19-9 of 412 U/mL,

ECOG performance status 1, after a completed neoadjuvant modified FOLFIRINOX course of four cycles) illustrates a representative eligible participant and does not represent an enrolled individual. The full eligibility criteria are summarized in Table 8.

Intervention: drug and device

Daraxonrasib is an orally bioavailable RAS(ON) multi-selective (pan-RAS) small-molecule inhibitor that binds the active, guanosine-triphosphate-bound state of mutant RAS in complex with cyclophilin A, suppressing downstream effector signaling across the common KRAS variants in PDAC; it received FDA Breakthrough Therapy designation in June 2025 for previously treated metastatic KRAS G12-mutant PDAC. In this study it is administered orally once daily in a perioperative schedule comprising a defined pre-operative course, a protocol-mandated pause across the operative window, and a postoperative restart governed by the advisory in §6.1 below. The device component is an eight-arm parallel coelomic surgical platform carrying 56 mechanical degrees of freedom (8 arms $\times$ 7 per-arm degrees of freedom) and a 640-channel mixed-rate sensor stack (80 channels per arm), with force channels sampling at 100 kHz and all remaining channels at the 10 kHz command rate, and a positioning accuracy of 0.05 mm root mean square at a peak tip velocity of 1,200 mm/s. An on-premises repository-pinned, open-weights LLM acts as a second-opinion advisory oracle held strictly outside the robot vendor kinematic stack and hash-pinned to a deterministic seed and a repository commit; it is never an autonomous controller, and full autonomy is prohibited at this phase consistent with 21 CFR §312.21(e). The advisory is gated by a verification-before-generation ten-gate suite returning ACCEPT, BLOCK, or ESCA-LATE, aligned with H.R. 9510, the Verification Before Generation in Physical AI Oncology Trials Act of 2026 [5, 2].

The hard cyber-physical safety envelope comprises a per-arm tip-force cap of ≤ 3 N, a cumulative cross-arm tip-force cap of ≤ 18 N enforced at the heartbeat boundary, a 10 kHz heartbeat coordination bus broadcasting a 64-byte frame on a 100 μ s per-arm watchdog deadline, an emergency arm park within 50 μ s of a missed or out-of-parameter frame, a cross-arm emergency stop (E-stop) that halts the full platform in ≤ 3 ms at a 0.0 N force clamp, and a software-independent hardware E-stop meeting the 21 CFR §312.404 requirement of ≤ 500 ms. A vascular no-fly gate sampled at 10 kHz evaluates instrument-tip proximity to five named vessels (superior mesenteric vein, portal vein, hepatic artery, celiac axis, and superior mesenteric artery) against three concentric thresholds: a soft-warning zone at 3.0 mm raises an LLM advisory, a no-fly zone (1.0 mm at the superior mesenteric and portal veins; 1.5 mm at the hepatic artery, celiac axis, and superior mesenteric artery) blocks the command, and a hard-stop zone at 5.0 mm forces the ≤ 3 ms E-stop; across a representative 100-tick verdict sample the gate returned clear on 83 ticks, soft-warning on 6, no-fly on 6, and hard-stop on 5. The operation proceeds across eight operative phases: phases P1 through P4 comprise dissection and specimen removal, phase P5 the duct-to-mucosa pancreaticojejunostomy under a controller ring-tension band of 0.30 to 0.60 N, phase P6 the end-to-side hepaticojejunostomy at 0.20 to 0.50 N, phase P7 the antecolic gastrojejunostomy at 0.40 to 0.80 N, and phase P8 final imaging, hemostasis, and drain placement [2].

Perioperative daraxonrasib pause-and-restart advisory

The two components are coupled at one decision point only: the perioperative pause-and-restart advisory for the drug is informed by the operative and pathology results of the device procedure. Daraxonrasib is paused before surgery and restarted postoperatively on a day chosen by an advisory that is LLM-bound and human-confirmed, never autonomous. The advisory takes two inputs, the pancreaticojejunostomy ISGPS fistula grade (A, B, or C) and the serum daraxonrasib trough relative to a 0.5 ng/mL threshold, and recommends a restart at postoperative day T+7d, T+14d, or T+21d (or a continued hold). In the 32-iteration deterministic simulation sweep, baseline serum was 0.45 ng/mL at the

operative timepoint across all iterations (below the 0.5 ng/mL trough threshold), and the advisory recommended T+7d in 29 of 32 iterations (90.6%, Grade A fistula with sub-threshold trough), T+14d in 3 of 32 (9.4%, Grade B or delayed serum rise), and T+21d or continued hold in 0 of 32 (reserved for Grade C, which also maps to the drug stopping rule). Figure 18 reproduces the advisory logic and the sweep distribution.

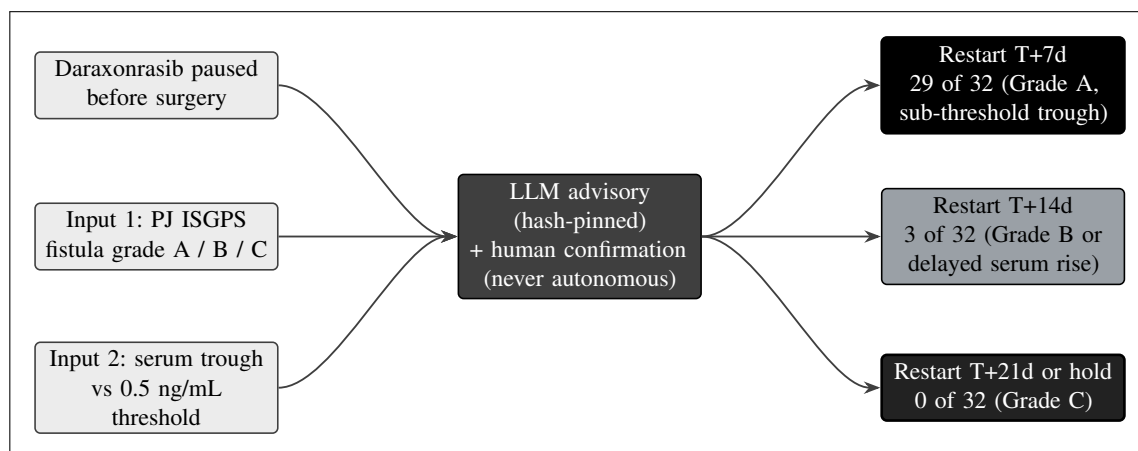


Figure 18. The perioperative daraxonrasib pause-and-restart advisory. The drug is paused before surgery and the postoperative restart day is set by a hash-pinned, human-confirmed LLM advisory (never autonomous) from two inputs: the pancreaticojejunostomy ISGPS fistula grade (A, B, or C) and the serum daraxonrasib trough relative to a 0.5 ng/mL threshold. In the 32-iteration deterministic sweep the advisory recommended restart at T+7 days in 29 of 32 iterations (Grade A with sub-threshold trough), T+14 days in 3 of 32 (Grade B or delayed serum rise), and T+21 days or continued hold in 0 of 32 (reserved for Grade C).

Assessments and the vascular and halt safety chain

Clinical safety assessments comprise vital signs, a directed physical examination, hematology, serum chemistry including hepatic and pancreatic indices, coagulation studies, the serum daraxonrasib trough assay, and protocol-specified imaging, at the frequencies in the schedule of activities. In addition, the device generates a continuous safety telemetry stream that is a source of protocol observations under 21 CFR §312.23(g): per-arm and cumulative force profiles, trajectory accuracy against the 0.05 mm specification, every vascular safety-zone verdict, every heartbeat and watchdog event, and every E-stop activation, each captured at its recorded rate in the hash-chained audit trail. The halt chain is layered: the 10 kHz heartbeat bus broadcasts a 64-byte frame on a 100 μ s per-arm watchdog deadline; a frame that is on time continues the command state, whereas a missed or out-of-parameter frame parks the affected arm within 50 μ s and escalates to the cross-arm E-stop that halts the platform in ≤ 3 ms at a 0.0 N force clamp, backed by the software-independent hardware E-stop. Postoperative pancreatic fistula is graded by the ISGPS definition as a biochemical leak, Grade B (requiring a change in management), or Grade C (requiring reoperation or causing organ failure or death), and the pancreaticojejunostomy grade is the dominant determinant and the input to the restart advisory and the drug stopping rule [1, 9].

Safety reporting clocks and the six Physical AI triggers

Adverse events are captured through two parallel and equally binding streams, a conventional pharmacovigilance stream for clinical events graded by CTCAE version 5.0, and a Physical AI safety stream for device- and AI-related events under 21 CFR §312.32(f). Serious adverse events are reported by the investigator to the sponsor within 24 hours of awareness, and the sponsor submits IND/IDE safety reports on the regulatory clocks: any unexpected fatal or life-threatening suspected adverse reaction is

reported as soon as possible but no later than 7 calendar days after initial receipt of the information, and any other serious and unexpected suspected adverse reaction is reported no later than 15 calendar days after the determination that the event is reportable (21 CFR §312.32). In addition, six Physical AI reporting triggers are reported under §312.32(g): a serious Physical AI adverse event on the 7- or 15-day timeline as applicable; a system-drug interaction causing incorrect drug administration, regardless of seriousness, within 15 days; a cybersecurity incident within 15 days (or 7 days if it has the potential to cause patient harm); sustained AI model degradation below the pre-specified acceptance criteria for three or more consecutive procedures or a cumulative 24 hours; a sim-to-real divergence beyond twice the validated gap tolerance (trajectory deviation greater than 4 mm or force discrepancy greater than 1.0 N); and a digital-twin discrepancy beyond the pre-specified accuracy thresholds. For every Physical AI adverse event, the complete hash-chained audit trail from 24 hours before the event to 72 hours after the event is preserved under 21 CFR part 11 and made available to the FDA on request. Figure 19 reproduces the clocks and the six triggers [7, 3].

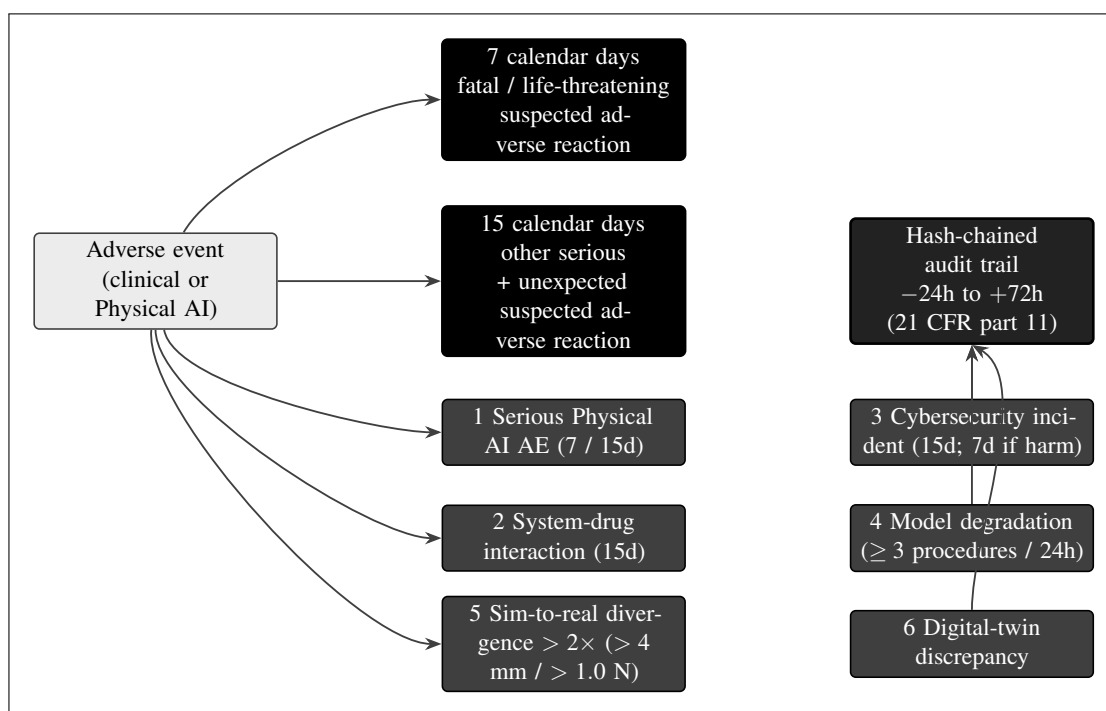


Figure 19. The safety-reporting clocks and the six Physical AI reporting triggers. A fatal or life-threatening suspected adverse reaction is reported within 7 calendar days and any other serious and unexpected suspected adverse reaction within 15 calendar days (21 CFR §312.32). The six Physical AI triggers of §312.32(g) are a serious Physical AI adverse event (7- or 15-day), a system-drug interaction (15-day), a cybersecurity incident (15-day, or 7-day if patient harm is possible), sustained model degradation (at least 3 consecutive procedures or 24 cumulative hours), a sim-to-real divergence exceeding twice the validated tolerance (trajectory over 4 mm or force over 1.0 N), and a digital-twin discrepancy. Each preserves the hash-chained audit trail from minus 24 to plus 72 hours under 21 CFR part 11.

Discontinuation and stopping rules

The protocol distinguishes discontinuation of an intervention from withdrawal from the study; a participant who stops one or both components for a protocol-defined reason remains in the study for safety and survival follow-up unless consent for that follow-up is withdrawn. The investigational device is discontinued for the index procedure, with conversion to the appropriate fallback technique, whenever any

of five prespecified conditions occurs: a conversion to open or manual technique judged by the supervising surgeon to better serve patient safety or oncologic adequacy (conversion is always available and is never itself a protocol violation); repeated excursions against the ≤ 3 N per-arm or ≤ 18 N cumulative caps; any breach of a vascular no-fly zone (1.0 mm at the superior mesenteric and portal veins; 1.5 mm at the hepatic artery, celiac axis, and superior mesenteric artery) or a recurring pattern of no-fly approaches; any sim-to-real divergence beyond twice the validated tolerance (trajectory deviation greater than 4 mm or force discrepancy greater than 1.0 N), which also triggers a §312.32(g) report; and any hard-stop cross-arm E-stop event, after which device resumption requires that the cause be identified and the system re-pass the pre-procedure safety matrix. Daraxonrasib is discontinued in an individual participant for a drug-attributable DLT during the 28-day window, for any other unacceptable toxicity not resolving with protocol dose interruption and reduction, for a confirmed ISGPS Grade C fistula (which precludes restart and maps to the T+21d or continued-hold arm of the advisory), for pregnancy, for contraindicating intercurrent illness, or at the participant's request [2, 1].

Two binding halt rules govern the study as a whole and are applied by the data safety monitoring board (DSMB): a device- or procedure-related serious adverse event in at least one of three participants within a cohort, or two device-related deaths at any time, triggers DSMB review and a halt. On any hold or closure, enrollment and patient-facing robotic activity stop immediately, affected participants are transitioned to appropriate clinical care, and a controlled decommissioning and data-preservation procedure seals the complete hash-chained command and telemetry record for the regulatory retention period.

Statistical considerations and oversight

The statistical approach is descriptive and estimation-based, non-confirmatory, and follows ICH E9 and the TRIPOD+AI reporting standard for the model-assisted advisory component. No formal power calculation is performed; exact two-sided 95% Clopper-Pearson confidence intervals are reported for the binomial endpoints, and a two-sided alpha of 0.05 is used only in a supportive sense with no multiplicity adjustment. Independent safety oversight operates on three tiers: the DSMB reviews accumulating safety data at each cohort boundary and at any triggered review and holds halt authority; the Independent Safety Monitor (ISM) is designated for each procedure and provides real-time case-level oversight with the authority to call a stop; and the Physical AI Safety Review Committee meets on a cadence not exceeding 90 days (and ad hoc on any triggering event) to review the autonomy-bearing behavior of the system, the verification-before-generation gate-surface decisions, the USL rating, and any software change, and to recommend USL reassessment or oversight changes. Seven Physical AI record types (the LLM advisory record, the robot command record, the sensor and telemetry record, the safety-limit-event record, the verification-before-generation gate-surface record, the human-override and intervention record, and the binding hash-chained audit trail) are retained for the period required by 21 CFR §312.57, at least two years after a marketing application is approved or, if none is filed, two years after the investigation is discontinued and the FDA is notified [1, 6].

The full protocol, version 1.0.0, with its complete schedule of activities, statistical analysis plan, and Physical AI System Description meeting the eleven fields of 21 CFR §312.23(g), is the bound supporting document for this section and is archived at DOI 10.5281/zenodo.20780121 (<https://doi.org/10.5281/zenodo.20780121>).

6.2 Informed Consent

Informed consent is obtained by a qualified member of the research team who is not the treating surgeon, before any study-specific procedure, in language the participant can understand and with adequate time and opportunity to ask questions, consistent with 21 CFR part 50 and ICH E6(R3) [11, 12].

The consent process incorporates each of the basic and additional elements of informed consent required by 21 CFR §50.25, and a copy of the form and any participant information sheet are submitted with the protocol. The consent discussion covers the diagnosis, the alternatives (including standard open or conventional robotic pancreaticoduodenectomy without the investigational advisory layer), and the prognosis of a malignancy with a total five-year survival below 13%; the perioperative darax-onrasib regimen, its once-daily oral administration, the protocol-mandated operative pause, and the toxicities characteristic of the RAS(ON) inhibitor class; and, in addition to the standard elements, an explicit and distinct disclosure of the on-premises LLM-directed eight-arm robotic system, its role as a second-opinion advisory oracle held outside the vendor kinematic stack, its continuous human oversight with immediate manual override, and its system-specific risks. The voluntary nature of participation, the right to withdraw at any time without penalty and without loss of any benefit to which the participant is otherwise entitled, and the maintenance of confidentiality under the HIPAA Safe Harbor de-identification method are stated at consent and reaffirmed at each contact.

The consent form addresses the therapeutic misconception directly. Because the population is vulnerable, facing the most technically demanding curative-intent abdominal operation for a lethal cancer, the form and the consent conversation make explicit that the study is a first-in-human, dose-finding, and early-feasibility investigation whose primary purpose is to characterize safety and feasibility rather than to confer individual therapeutic benefit, and that the LLM directs no autonomous action. The staged-autonomy model is explained: in this Phase 1 the system operates only at the clinician-controlled (teleoperation or supervised) and supervised semiautonomous levels under continuous human oversight, full autonomy is prohibited consistent with 21 CFR §312.21(e), and any future move to higher autonomy is reserved for a later phase and gated by independent review. An independent consent monitor is used when indicated, and qualified interpreters and translated materials are provided so that consent is meaningful for participants with limited English proficiency [6, 9].

A defining feature of this trial class is the documented Physical AI opt-out. Under 21 CFR §312.60(f), as adapted for Physical AI investigations, the participant is offered, and may exercise, a documented right to request administration of the operative intervention without the Physical AI advisory layer where clinically feasible, without penalty and without affecting eligibility for the remainder of the study or for clinical care [3]. The participant's election or declination of the opt-out is recorded in the source document and the case report form, and the schedule of activities (Table 7) captures that election at the consent visit. Where an unanticipated problem changes the risk profile of the combined intervention, including a relevant Physical AI safety event, the consent document is updated and currently enrolled participants are re-consented before any further study procedure, consistent with the original opt-out.

6.3 Investigator and Facilities Data

The investigation is conducted at a single academic medical center through its institutional multidisciplinary pancreatic-cancer program. Eligible individuals are identified at tumor-board review and at surgical and medical-oncology clinics, and the study is described to them by a member of the research team who is not the treating surgeon, to reduce the potential for undue influence. The site possesses an established high-volume pancreaticobiliary surgical service, a robotic surgical program, an investigational-product pharmacy with controlled, temperature-monitored storage and a drug-accountability log, an anatomic-pathology service applying a standardized synoptic margin protocol with masked independent adjudication, a central imaging capability for RECIST version 1.1 reads by a blinded reviewer, and the clinical laboratory capacity for the serum darax-onrasib trough assay and the hematologic, hepatic, renal, and pancreatic safety panels. The device facilities include the qualified clinical environment in which the eight-arm platform is maintained and calibrated between procedures, the

operator control stations, the on-premises compute environment that hosts the repository-pinned advisory LLM in network isolation from the vendor kinematic stack under a deny-by-default authentication and authorization model, and the 21 CFR part 11 records environment that maintains the hash-chained, append-only audit trail. Day-to-day monitoring is delegated to a robotics- and artificial-intelligence-competent clinical research organization (CRO) over the site team, which comprises the operating surgeon, a primary operator, and a qualified backup operator ready to assume control or invoke the manual override at any time [2, 8].

The governance structure that supervises these investigators and facilities is the three-tier oversight architecture described in §6.1: the Sponsor-Investigator answers to a reviewing IRB and to the independent DSMB, convenes the Physical AI Safety Review Committee on a cadence not exceeding 90 days, designates the Independent Safety Monitor for each procedure, and delegates monitoring to the CRO. Figure 20 presents the governance and safety-oversight structure that binds the investigators, the facilities, and the independent reviewers together.

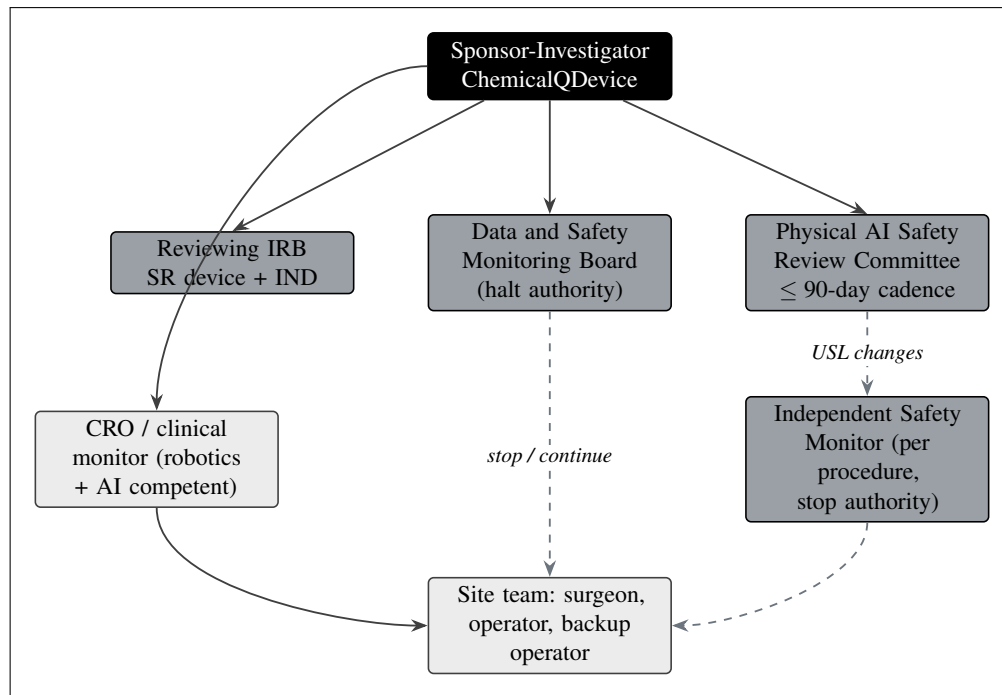


Figure 20. Study governance and safety oversight. The Sponsor-Investigator answers to the reviewing IRB and the DSMB (halt authority), convenes a Physical AI Safety Review Committee on a cadence not exceeding 90 days, designates an Independent Safety Monitor (per-procedure, stop authority), and delegates monitoring to a robotics- and AI-competent CRO over the site team (surgeon, operator, and a qualified backup operator).

6.3.1 FDA Form 1572

A Statement of Investigator (FDA Form 1572) is executed by each investigator responsible for the conduct of the study at the clinical site, in satisfaction of 21 CFR §312.53(c), and is assembled with the submission together with the investigator's current curriculum vitae and the documentation of relevant qualifications. The executed Form 1572 attests that the investigator has the training and experience to assume responsibility for the proper conduct of the investigation, identifies the name and address of the clinical site and of the clinical laboratory and other facilities to be used, names the reviewing IRB responsible for review and approval of the study, lists the sub-investigators (the operating surgeon, the

primary and backup device operators, the medical oncologist, the pathologist, and the research coordinators) who will assist in the conduct of the investigation, and identifies the protocol by its title and version. By signing, the investigator commits to conduct the study in accordance with the protocol, to personally conduct or supervise the investigation, to inform participants that the drug and device are being used for investigational purposes and to ensure that consent and IRB requirements are met, to report adverse events to the sponsor in accordance with 21 CFR §312.64, to read and understand the investigator's brochure and the Physical AI System Description, to ensure that all associates and staff are informed of their obligations, to maintain adequate and accurate records available for inspection under 21 CFR §312.68, and to comply with all other requirements of 21 CFR part 312. The executed forms, with attachments, are bound into the submission and are updated whenever a sub-investigator or facility changes [7].

6.3.2 Disclosure of Financial Interests

Financial interests are disclosed and managed in accordance with 21 CFR part 54, Financial Disclosure by Clinical Investigators, and the certification and disclosure statements (FDA Forms 3454 and 3455 as applicable) are collected before participation and updated through one year after completion of the study. The disclosure makes explicit the principal conflict of interest in this investigation: the Sponsor-Investigator, Kevin Kawchak, is the chief executive officer of ChemicalQDevice, the entity that develops the on-premises LLM-directed eight-arm robotic Physical AI system under study and holds the combined IND and IDE. This relationship is disclosed in full to the FDA and to the reviewing IRB, and it is managed by the independent oversight structure described in §6.3 so that safety and escalation decisions are not made by a party with a financial interest in the outcome alone [3, 6].

The mitigation is structural and operates at every decision point that could be influenced by the conflict. The DSMB, composed of members independent of the conduct of the study with surgical, oncologic, biostatistical, and device-safety expertise, holds the binding halt authority and applies the prespecified halt rules (a device- or procedure-related serious adverse event in at least one of three participants within a cohort, or two device-related deaths at any time). The Independent Safety Monitor, designated for each procedure and independent of the operating team, holds real-time stop authority at the case level. The Physical AI Safety Review Committee reviews the autonomy-bearing behavior of the system, the verification-before-generation gate-surface decisions, the USL rating, and any software change on a cadence not exceeding 90 days, and recommends USL reassessment or oversight changes. Routine monitoring is delegated to a CRO independent of the device manufacturer, and audits may be conducted by a designee independent of the conduct of the study. Because the binding safety, escalation, and halt decisions rest with these independent bodies rather than with the Sponsor-Investigator, the financial conflict is disclosed and mitigated to a degree commensurate with the first-in-human, autonomy-bearing nature of the investigation, and any identified conflict is disclosed and managed before it can affect the conduct, analysis, or reporting of the study [1].

Table 7: Schedule of activities (summary) for the bound Phase 1 protocol. The intensive safety-collection window spans consent through the 30-day postoperative visit and encompasses the 28-day daraxonrasib dose-limiting toxicity window.

Assessment	Screen	Pre-op / day 0	Post-op to day 30	Long-term follow-up
Informed consent and Physical AI opt-out election	X	-	-	re-consent on amendment
Eligibility, ECOG, KRAS G12 genotype, staging imaging	X	-	-	-
Multidisciplinary resectability review	X	X	-	-
Daraxonrasib pre-operative course (oral once daily)	-	X	-	-
Protocol-mandated operative pause of daraxonrasib	-	X	-	-
Robotic Whipple (8 operative phases P1 to P8)	-	X	-	-
Pre-procedure safety matrix and USL confirmation	-	X	-	re-check after any change
Perioperative pause-and-restart advisory (T+7/14/21d)	-	-	X	-
Vitals, hematology, chemistry, hepatic/pancreatic indices	X	X	X	per follow-up schedule
Serum daraxonrasib trough assay	-	X	X	as indicated
ISGPS fistula grading (PJ dominant)	-	-	X	-
Adverse-event and Physical AI AE capture	X	X	X	SAE/PAI AE to 24 months
RECIST v1.1 imaging (central read)	X	-	X	per follow-up schedule
Pathology, R0 margin (day-90 read, per-protocol)	-	-	X	day 90
Progression-free and overall survival	-	-	-	to 24 months (q12 weeks)

Table 8: Key inclusion and exclusion criteria for the bound Phase 1 protocol.

Inclusion criteria (all required)	Exclusion criteria (any one excludes)
Signed informed consent before any study-specific procedure, including explicit Physical AI opt-out election or declination	Distant metastatic disease on staging
Age 18 years or older at consent	Vascular involvement precluding a margin-negative (R0) resection, or anatomy incompatible with the eight-arm robotic approach
Histologically or cytologically confirmed resectable or borderline-resectable PDAC on multidisciplinary review	Prior pancreatic resection or prior surgery precluding the planned reconstruction
Documented KRAS G12 mutation (for example G12D, G12V, or G12R) on a validated assay	Uncontrolled intercurrent illness or comorbidity making major surgery or daraxonrasib unsafe
ECOG performance status 0 or 1	Pregnancy or lactation, or unwillingness to use adequate contraception where applicable
Adequate organ and marrow function for major pancreaticobiliary surgery and daraxonrasib	Known contraindication or hypersensitivity to daraxonrasib or a required excipient, or an unmanageable drug interaction
Measured baseline CA 19-9 recorded for serial assessment; eligible for daraxonrasib under RASolute 302 pan-KRAS criteria	Inability to provide informed consent or to comply with the Schedule of Activities
Tumor and vascular anatomy compatible with the eight-arm robotic system on preoperative imaging	Anatomy or comorbidity otherwise incompatible with safe conduct of the combined intervention

7 Chemistry, Manufacturing and Control Information

7.1 Chemistry, Manufacturing and Control

This section presents the Chemistry, Manufacturing and Control (CMC) information for the investigational drug daraxonrasib (RMC-6236) supporting the Phase 1, first-in-human, open-label, single-arm investigation described elsewhere in this Investigational New Drug application (IND v1.0, repository v4.3.0). It is organized and submitted in accordance with 21 CFR §312.23(a)(7), which requires sufficient information about the composition, manufacture, and control of the drug substance and the drug product to assure the proper identification, quality, purity, and strength of the investigational drug, scaled to the early phase of investigation, the limited number of participants, and the short duration of exposure characteristic of a Phase 1 study [7]. Consistent with the FDA guidance on the content and format of CMC information for a Phase 1 study, the emphasis here is on the information needed to assure participant safety, including identity, strength, quality, purity, and stability, rather than on the detailed validation data appropriate to later phases [7]. As the investigation proceeds and the manufacturing process and analytical methods mature, the CMC information will be updated through information amendments under 21 CFR §312.31 as required.

The investigational regimen couples this oral drug with an on-premises large language model (LLM) directed eight-arm robotic pancreaticoduodenectomy (Whipple) platform that is the subject of the parallel investigational device exemption content of this submission. The device is a durable instrument and is not a drug, a biologic, or a consumable manufactured lot; its design, specification, calibration, and quality controls are addressed in the device description and the Physical AI System Description fields of 21 CFR §312.23(g), not in this CMC section [1, 2]. Accordingly, the four CMC subsections below address only the orally administered drug substance and drug product, the non-applicable placebo, and labeling, followed by the environmental assessment. The CMC information architecture and the mapping of each subsection to its content is shown in Figure 16.

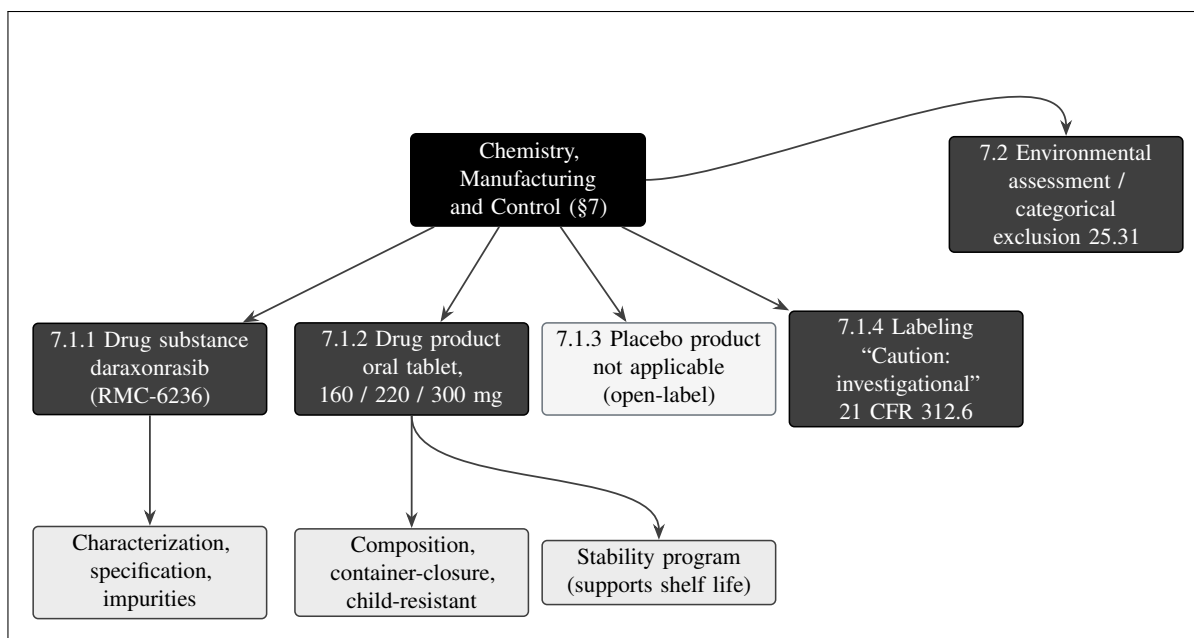


Figure 16. The Chemistry, Manufacturing and Control information architecture for §7. The drug substance (daraxonrasib, RMC-6236) is described by characterization, specification, and impurity controls; the drug product is the oral tablet at the 160, 220, and 300 mg strengths with its composition and child-resistant, light-protective container-closure under 21 CFR §312.6, supported by a stability program; the placebo product is not applicable for this open-label study; labeling carries the investigational caution statement; and an environmental assessment or categorical exclusion under 21 CFR §25.31 is provided.

7.1.1 Drug Substance

Nomenclature, description, and pharmacological class. The drug substance is daraxonrasib, also designated by the developmental code RMC-6236, an orally bioavailable small molecule of the RAS(ON) multi-selective (pan-RAS) inhibitor class. Daraxonrasib acts by binding the active, guanosine triphosphate (GTP) bound state of mutant RAS in a tri-complex with the endogenous chaperone cyclophilin A, thereby occluding the RAS to effector interface and suppressing downstream mitogen-activated protein kinase and phosphoinositide 3-kinase signaling across the common KRAS oncoprotein variants found in pancreatic ductal adenocarcinoma (PDAC), including the G12D, G12V, and G12R substitutions targeted by this protocol [1, 13]. The agent received FDA Breakthrough Therapy designation in June 2025 for previously treated metastatic PDAC bearing a KRAS G12 mutation, and 300 mg administered orally once daily is the recommended Phase 2 dose established in the pan-KRAS RASolute 302 program to which the dose and exposure assumptions of this protocol are anchored [1, 13]. The drug substance is a non-biologic, chirally defined organic small molecule; it is not a radioactive drug, is not a controlled substance, and is not a peptide, protein, or nucleic acid. The drug substance is presented as a solid, and in the finished product it is formulated as an immediate-release oral tablet.

Manufacturer and method of preparation. The drug substance is manufactured by, or under contract to, the sponsor's designated active pharmaceutical ingredient manufacturer under current good manufacturing practice (cGMP) appropriate to the investigational stage, and is released against the drug-substance specification summarized in Table 7. The synthetic route is a defined, multi-step convergent organic synthesis with isolated, characterized intermediates and controlled reagents, catalysts,

and solvents; in-process controls govern the critical steps, and the final isolation and drying steps establish the solid-state form, particle size distribution, and residual-solvent profile of the substance. The name, address, and establishment information of the drug-substance manufacturer, the detailed reaction scheme, the list of starting materials and reagents, the in-process controls, and the batch records are provided in the supporting CMC documentation on file with the sponsor and are cross-referenced here consistent with the Phase 1 level of CMC detail [7]. A statement that the drug substance is manufactured in compliance with cGMP, and the certificate of analysis for the clinical batch, accompany each shipment.

Characterization. The structure and identity of daraxonrasib are established by a panel of orthogonal physicochemical methods. Structural elucidation and confirmation use proton and carbon nuclear magnetic resonance spectroscopy, high-resolution mass spectrometry, infrared spectroscopy, and ultraviolet to visible spectroscopy; elemental analysis confirms empirical composition. Chiral purity is confirmed by chiral high-performance liquid chromatography (HPLC) to assure the correct stereochemical configuration of the defined stereocenters. The solid-state form (polymorph) is characterized by X-ray powder diffraction and differential scanning calorimetry, and the manufacturing process is controlled to deliver the intended crystalline form, because solid-state form governs dissolution and hence the oral exposure on which the pharmacokinetically gated restart advisory of the protocol depends. Hygroscopicity, aqueous and physiological-pH solubility, the partition coefficient, and the pKa are determined to support the biopharmaceutical understanding of the tablet. These characterization data constitute the reference standard against which clinical batches are compared.

Specification and control of impurities. The drug-substance specification, with tests, analytical procedures, and acceptance criteria appropriate to a first-in-human Phase 1 study, is given in Table 7. Organic impurities (process-related and degradation-related), residual solvents, and elemental impurities are controlled to limits consistent with the applicable International Council for Harmonisation (ICH) quality guidelines (ICH Q3A for impurities in new drug substances, ICH Q3C for residual solvents, and ICH Q3D for elemental impurities), as adapted to the limited duration and participant exposure of a Phase 1 investigation [12, 7]. Any individual or total impurity exceeding its qualification threshold is qualified by reference to the nonclinical toxicology program summarized in the pharmacology and toxicology section of this IND. Genotoxic-impurity risk is assessed and controlled under the ICH M7 framework, with structural alerts evaluated and relevant potential mutagenic impurities controlled to the threshold of toxicological concern. The acceptance criteria below are interim Phase 1 specifications and will be tightened as process knowledge and additional batch data accumulate.

7.1.2 Drug Product

Description and composition. The drug product is an immediate-release, film-coated oral tablet supplied in the three strengths required to compose the once-daily doses of the 3+3 escalation: 160 mg for dose level 1 (DL1), 220 mg for dose level 2 (DL2), and 300 mg for dose level 3 (DL3), the last of which corresponds to the RASolute 302 recommended Phase 2 dose [1, 13]. The three strengths are dose-proportional, sharing a common qualitative composition and a common blend so that the only quantitative difference among strengths is the content of drug substance and the corresponding fill weight, with the tablets differentiated by size, debossing, or both to prevent dose confusion in this open-label study. The qualitative and representative quantitative composition is given in Table 8. The excipients are standard, compendial pharmaceutical excipients used at conventional levels; the formulation contains a filler or diluent, a binder, a disintegrant, a glidant, a lubricant, and a non-functional aqueous

film coat. No excipient of human or animal origin presenting a transmissible spongiform encephalopathy risk is used, and no novel excipient is included.

Manufacture. The drug product is manufactured under cGMP appropriate to the investigational stage by the sponsor's designated drug-product manufacturer. The process is a conventional solid-oral-dosage process: dispensing and screening of the drug substance and excipients, blending (with wet or dry granulation as defined in the batch record), lubrication, compression of the final blend into core tablets on a rotary tablet press under in-process controls for tablet weight, hardness, thickness, and friability, and aqueous film coating of the cores in a perforated coating pan to a target weight gain, followed by drying. In-process controls at compression include tablet weight and weight uniformity, hardness, thickness, disintegration, and visual inspection; the coating step is controlled for weight gain and appearance. The batch formula, the master and executed batch records, the in-process control limits, and the name and address of the drug-product manufacturer are provided in the supporting CMC documentation and are cross-referenced here at the Phase 1 level of detail [7]. A certificate of analysis accompanies each clinical batch, and each batch is released against the drug-product specification before use.

Drug-product specification. Each clinical batch is tested and released against a drug-product specification that includes appearance, identity of the drug substance (by HPLC retention time and a spectroscopic method), assay (95.0 to 105.0 percent of label claim by HPLC), content uniformity by the USP uniformity-of-dosage-units test, degradation products controlled to interim Phase 1 limits consistent with ICH Q3B (not more than 0.5 percent per specified or unspecified degradant and not more than 2.0 percent total), dissolution by the USP apparatus and medium validated for the product (for example, not less than 80 percent dissolved in 30 minutes as a provisional immediate-release criterion to be confirmed by the developing dissolution method), water content, and microbial limits within USP acceptance criteria for an oral non-sterile product. The dissolution method and acceptance criterion are provisional and will be finalized as biopharmaceutic and stability data accrue.

Container-closure system. The drug product is packaged in a child-resistant, light-protective, moisture-protective container-closure system, namely high-density polyethylene bottles with a child-resistant, induction-sealed closure and an appropriate desiccant, or equivalent child-resistant unit-dose blister packaging. The container-closure components contacting the product are of compendial pharmaceutical grade and are suitable for an oral solid dosage form; the child-resistant feature satisfies the special-packaging expectation for an investigational oral oncology agent, and the light-protective and moisture-protective features support the stability claim. Packaging and labeling operations are performed under cGMP, and the investigational caution statement and the label content described in Table 9 and in §7.1.4 appear on the immediate and any secondary container in accordance with 21 CFR §312.6 [7].

Stability program. Stability of the drug product is supported by an ongoing stability program conducted under ICH Q1A conditions on representative batches in the proposed container-closure system, using stability-indicating analytical methods. The program comprises long-term, intermediate, and accelerated conditions, with attributes monitored at each station including appearance, assay, degradation products, dissolution, and water content; a photostability evaluation under ICH Q1B and an in-use stability evaluation for the multi-dose bottle presentation are included. The program and its current status are summarized in Table 10. The provisional shelf life (expiry or retest period) assigned to the clinical drug product is supported by the available long-term and accelerated data and is applied through expiry

dating on the label; product is dispensed only within the labeled expiry or retest date. Stability is monitored throughout the investigation, and the shelf life will be extended through information amendments as additional long-term data become available [7]. The drug substance is also placed on a parallel ICH Q1A stability program to establish its retest period.

7.1.3 Placebo Product

This is an open-label, single-arm, first-in-human study; there is no comparator arm, no masking of treatment assignment, and consequently no placebo product. The absence of a placebo is consistent with the dose-finding and early-feasibility objectives of a Phase 1 investigation and with ICH E9 principles for early-phase trials [12, 7]. In accordance with the ReGARDD IND content map, this subsection is retained for completeness and to preserve the table-of-contents numbering, and the placebo product is recorded here as not applicable. The bias-control measures appropriate to an open-label design, including the masking of the pathologist who determines R0 resection status and the central blinded radiographic reader, are described in the study protocol and in §6.1 of this IND rather than through a placebo. Should a future amendment introduce a randomized or placebo-controlled component, a corresponding placebo product description, composition, and matching strategy would be submitted by information amendment at that time.

7.1.4 Labeling

The investigational drug product is labeled in accordance with 21 CFR §312.6, which provides that the immediate package of an investigational drug intended for human use bear a label with the statement “Caution: New Drug. Limited by Federal (or United States) law to investigational use,” and that the label and labeling bear no false or misleading statement and make no claim that the drug is safe or effective for the purposes under investigation [7]. Because the study is open-label, the participant-facing label identifies the product and the assigned once-daily dose level; no information that would unblind a masked outcome assessor (for example, the masked pathologist or the central radiographic reader) appears on any label or in any labeling provided to those assessors. The required content of the immediate-container label and of the supplied labeling is summarized in Table 9. The label is affixed during cGMP packaging operations and is verified at the investigational-product pharmacy against the dispensing record before any unit is released to a participant. The investigational device and its single-use accessories carry the investigational-device caution statement under 21 CFR part 812 and are not relabeled or repackaged at the site; their labeling is addressed in the device content of this submission.

7.2 Environmental Assessment

The sponsor claims a categorical exclusion from the requirement to prepare an environmental assessment or an environmental impact statement under 21 CFR §25.31(e), which categorically excludes from environmental review the action of issuing an investigational exemption for a new drug intended to be administered to human subjects, provided that the conditions of 21 CFR §25.15(c) and §25.15(d) are met and provided that no extraordinary circumstances exist that would result in a significant environmental effect. The investigation described in this IND meets these conditions. Daraxonrasib will be administered orally to a small investigational population of up to 18 treated participants at a single academic site, at clinical doses of 160 to 300 mg once daily over a limited perioperative schedule; the quantities of drug substance and drug product manufactured, distributed, and used are correspondingly small and are confined to the controlled clinical setting and its regulated pharmaceutical-waste stream. The estimated concentration of the drug substance entering the environment at the expected use level is below the threshold at which an environmental assessment is ordinarily required, and no extraordinary circumstances are present: the action does not significantly affect the quality of the human environment,

does not involve a substance that is persistent, bioaccumulative, or otherwise of recognized environmental concern at the quantities used, and does not affect any species or habitat protected under federal environmental law [7]. Investigational product that is unused, returned, expired, or otherwise unsuitable is destroyed only after sponsor authorization and in accordance with applicable federal, state, and local regulations governing the disposal of pharmaceutical waste, and the destruction is documented in the drug-accountability record. To the sponsor's knowledge, no data or information exist that would indicate that the proposed investigation may significantly affect the quality of the human environment. Accordingly, neither an environmental assessment nor an environmental impact statement is required for this submission, and the categorical exclusion under 21 CFR §25.31(e) is claimed. Should the scope of the investigation expand in a manner that would exceed the conditions of the categorical exclusion, an environmental assessment under 21 CFR §25.40 would be submitted by information amendment before implementing the change.

Table 9: Drug-substance specification for daraxonrasib (RMC-6236): interim Phase 1 release and stability tests, analytical procedures, and acceptance criteria.

Attribute / test	Analytical procedure	Acceptance criterion (interim Phase 1)
Appearance	Visual	White to off-white solid; free of foreign matter
Identity (spectroscopic)	IR; UV to visible	Conforms to reference standard
Identity (chromatographic)	HPLC retention time	Retention time matches reference standard
Identity (structural)	¹ H and ¹³ C NMR; HRMS	Consistent with assigned structure
Stereochemical purity	Chiral HPLC	Single defined stereoisomer; undesired enantiomer not more than 0.5 percent
Assay (content)	HPLC with UV detection	98.0 to 102.0 percent on the anhydrous, solvent-free basis
Organic impurities (each)	HPLC, related substances	Not more than 0.5 percent per specified or unspecified impurity (ICH Q3A)
Organic impurities (total)	HPLC, related substances	Not more than 2.0 percent total
Genotoxic impurities	LC-MS or GC-MS, ICH M7	Below the threshold of toxicological concern for specified mutagenic impurities
Residual solvents	Headspace GC	Within ICH Q3C Class limits for solvents used in synthesis
Elemental impurities	ICP-MS	Within ICH Q3D oral permitted daily exposure limits
Water content	Karl Fischer titration	Not more than 1.0 percent w/w
Residue on ignition / sulfated ash	USP general chapter	Not more than 0.2 percent
Solid-state form	X-ray powder diffraction	Conforms to the designated reference polymorph
Particle size distribution	Laser diffraction	Within the validated process range supporting dissolution
Microbial limits	USP total aerobic / yeast and mold	Within USP limits for an oral non-sterile substance

Table 7. Interim Phase 1 drug-substance specification. Acceptance criteria are scaled to the early phase of investigation and will be tightened as batch and process data accumulate; impurity limits reference ICH Q3A, Q3C, Q3D, and M7 as adapted for Phase 1.

Table 10: Representative qualitative and quantitative composition of the daraxonrasib immediate-release oral tablet (per the 160 mg strength; the 220 mg and 300 mg strengths are dose-proportional from a common blend).

Component	Function	Quantity (160 mg strength) and notes
Daraxonrasib (RMC-6236)	Active pharmaceutical ingredient	160 mg per tablet (220 mg / 300 mg in higher strengths)
Microcrystalline cellulose	Filler / diluent	Quantity sufficient to target fill weight
Lactose monohydrate or mannitol	Filler / diluent	Compendial grade, conventional level
Povidone or hypromellose	Binder	Conventional level
Croscarmellose sodium	Disintegrant	Conventional level for immediate release
Colloidal silicon dioxide	Glidant	Conventional level
Magnesium stearate	Lubricant	Conventional level (typically not more than 1.0 percent)
Film coat (hypromellose-based)	Aqueous non-functional coat	Applied to target weight gain; removed as water on drying
Purified water	Coating vehicle	Removed during processing (not present in finished tablet)

Table 8. Drug-product composition. All excipients are compendial and used at conventional levels; the film coat is non-functional and cosmetic. The 220 mg and 300 mg strengths are manufactured dose-proportionally from a common blend, differing only in drug-substance content and fill weight and in tablet size or debossing for identification.

Table 11: Drug-product stability program for the daraxonrasib oral tablet: storage conditions, time points, attributes monitored, and purpose.

Condition	Storage	Time points (months)	Attributes monitored / purpose
Long-term (ICH Q1A)	25 C / 60 percent RH	0, 3, 6, 9, 12, 18, 24	Appearance, assay, degradants, dissolution, water; supports assigned shelf life
Intermediate (ICH Q1A)	30 C / 65 percent RH	0, 6, 12	Same panel; triggered if a significant change occurs at accelerated
Accelerated (ICH Q1A)	40 C / 75 percent RH	0, 1, 2, 3, 6	Same panel; supports provisional dating and excursion assessment
Photostability (ICH Q1B)	ICH option 1 light exposure	Single challenge	Confirms need for light-protective packaging
In-use stability	25 C / 60 percent RH, bottle opened	Per protocol after first opening	Supports the labeled in-use period for the multi-dose bottle
Temperature excursion	Per documented excursion	As incurred	Supports continued-use decisions for monitored excursions

Table 10. Drug-product stability program. Stability-indicating methods are used at every station; the provisional shelf life is supported by the long-term and accelerated data and is extended by information amendment as additional long-term data accrue.

Table 12: Investigational-product label and labeling content for the daraxonrasib oral tablet under 21 CFR §312.6.

Label element	Content and basis
Investigational caution statement	“Caution: New Drug. Limited by Federal (or United States) law to investigational use” (21 CFR §312.6)
Product identity and strength	Daraxonrasib (RMC-6236) oral tablet; 160 mg, 220 mg, or 300 mg as assigned
Protocol and IND reference	Protocol number and IND number; sponsor protocol identifier
Lot and batch identification	Manufacturing lot number and a unique container identifier for accountability
Storage conditions	“Store at controlled room temperature; protect from light and moisture; keep in original container”
Expiry or retest date	Date supported by the stability program; not dispensed beyond this date
Quantity and directions	Net contents and the once-daily directions for use; “Swallow whole; do not crush, split, or chew”
Child-resistant statement	“Keep out of reach of children”; child-resistant closure per special packaging
Sponsor identification	Name and address of the sponsor-investigator (ChemicalQDevice)
Unblinding control	No information appearing on labeling provided to masked assessors that would reveal dose level or outcome

Table 9. Label and labeling content. The immediate package bears the 21 CFR §312.6 investigational caution statement and makes no claim of safety or effectiveness; the label supports drug accountability and protects the masked outcome assessors of this open-label study.

8 Pharmacology and Toxicology Information

8.1 Pharmacology and Drug Distribution

This section presents the pharmacology and toxicology information supporting the Phase 1, first-in-human, open-label, single-arm investigation of perioperative daraxonrasib (RMC-6236) in resectable and borderline-resectable pancreatic ductal adenocarcinoma (PDAC) described elsewhere in this Investigational New Drug application (IND v1.0, repository v4.3.0). It is organized and submitted in accordance with 21 CFR §312.23(a)(8), which requires adequate information about the pharmacological and toxicological studies of the drug, on the basis of which the sponsor has concluded that it is reasonably safe to conduct the proposed clinical investigation, including an integrated summary of the toxicological effects of the drug in animals and in vitro and, where applicable, a full data tabulation and the names and qualifications of the individuals who evaluated the studies [7]. Consistent with the FDA guidance on the content and format of an IND for a Phase 1 study, the depth of the pharmacology and toxicology information presented here is scaled to the early phase of investigation, to the small number of participants (up to eighteen treated), and to the short, bounded duration of drug exposure characteristic of a perioperative dose-finding study [7].

Daraxonrasib is a clinically advanced investigational agent with an established nonclinical and clinical pharmacology package generated within its late-phase sponsor program, and it carries FDA Breakthrough Therapy designation granted in June 2025 for previously treated metastatic PDAC harboring KRAS G12 mutations [1]. The present IND does not seek to recapitulate that established package; rather, it cross-references the existing nonclinical pharmacology and toxicology of daraxonrasib and supplements it with the study-specific computational pharmacology, drug-distribution, and toxicology evidence that is unique to the perioperative, curative-intent surgical context and to the combined drug-device use evaluated here [1, 2]. That supplementary evidence is the principal subject of this section and consists of three classes of in silico work: an AI digital-twin PDAC simulation [15], a quantitative systems pharmacology (QSP) metastatic-PDAC trial simulation with verification, validation, and uncertainty quantification (VVUQ) [16], and a 100,000-patient in silico Phase 3 five-arm PDAC trial conducted in triplicate [17, 18]. The combined regimen also pairs the oral drug with an on-premises large language model (LLM) directed eight-arm robotic pancreaticoduodenectomy (Whipple) platform that is the subject of the parallel investigational device exemption content of this submission; the device has no pharmacologic dose and is not addressed as a pharmacology or toxicology article here, but the operative and pathology outputs of the device procedure are inputs to the perioperative drug pause-and-restart advisory and therefore bear on the drug-distribution discussion that follows [1].

The remainder of this section is organized into the three subsections required by the integrated-summary structure of 21 CFR §312.23(a)(8): a pharmacology summary and conclusions, an integrated toxicology summary, and a full data tabulation. The pharmacology summary states the RAS(ON) mechanism of action, the pharmacodynamics, and the drug-distribution basis for the perioperative pause-and-restart schedule, including the 32-iteration deterministic pharmacokinetic sweep anchored to a 0.45 ng/mL operative-window baseline serum concentration. The integrated toxicology summary states the safety margins supporting the conservative DL1 160 mg once-daily first-in-human starting dose and the dose-limiting toxicity (DLT) framework that bounds the escalation. The full data tabulation summarizes the tabulated computational studies that are provided as supporting documents to this IND. The overall structure of §8 and its relationship to the computational-evidence base is described in Figure 17, which is rendered with the Investigator Brochure content of §5 and is not reproduced here; the present section relies on tables and prose.

8.1.1 Pharmacology Summary and Conclusions

Mechanism of action and pharmacological class. Daraxonrasib (RMC-6236) is an orally bioavailable, RAS(ON) multi-selective (pan-RAS) small-molecule inhibitor. It acts by binding the active, guanosine triphosphate (GTP) bound state of mutant RAS in a tri-complex with the ubiquitous chaperone protein cyclophilin A, and the resulting daraxonrasib to cyclophilin A to RAS(ON) complex sterically occludes the effector-binding interface, suppressing downstream mitogen-activated protein kinase and phosphoinositide 3-kinase effector signaling [1]. Because the agent targets the shared active conformation rather than a single mutant side chain, it provides broad coverage across the common KRAS oncoprotein variants found in PDAC, including the G12D, G12V, and G12R alleles that define the molecular eligibility for this study [1, 2]. The pharmacological class is therefore a RAS(ON) multi-selective inhibitor, and the mechanistic rationale for evaluating the agent in resectable and borderline-resectable PDAC is the suppression of micrometastatic outgrowth during the perioperative window when systemic therapy must otherwise be interrupted for surgery. The dose and exposure assumptions used throughout this IND are anchored to the pan-KRAS RASolute 302 program, in which 300 mg once daily is the established recommended Phase 2 dose, and to the broader late-phase program comprising RASolute 302 and the first-line RASolve 301 study [1]. Because daraxonrasib engages the shared GTP-bound conformation common to the principal mutant alleles rather than discriminating among individual side chains, the molecular eligibility used in this study, namely documented KRAS G12 status (G12D, G12V, or G12R), is satisfied by the same broad pan-KRAS criteria that define eligibility in RASolute 302, and no allele-specific dose adjustment is anticipated within the studied range [1, 2]. This multi-selectivity is the pharmacologic property that distinguishes the agent from allele-locked inhibitors and that motivates its evaluation as the systemic backbone of the perioperative, curative-intent regimen described in this IND.

Pharmacodynamics. The pharmacodynamic profile of daraxonrasib follows directly from its mechanism: target engagement is a function of the fraction of GTP-bound mutant RAS that is captured in the inhibitory tri-complex, which in turn tracks systemic drug exposure. The pharmacodynamic markers relevant to this perioperative study, the direction of the on-treatment change, and the way each marker is used in the study are summarized in Table 8. The dominant on-target, class-characteristic toxicities, principally gastrointestinal events (nausea, diarrhea, stomatitis) and dermatologic and mucocutaneous events, are themselves pharmacodynamic consequences of pan-RAS pathway suppression and are managed by the supportive-care and dose-modification machinery of the protocol rather than by withholding the mechanism [2].

Table 8. Pharmacodynamic markers of daraxonrasib and their use in the perioperative PDAC study. Target engagement and effector suppression are mechanistic; the serum trough is the operative quantitative input to the restart advisory.

Drug distribution and the perioperative pharmacokinetic basis. Daraxonrasib is administered orally, once daily, in a perioperative schedule consisting of a defined pre-operative course, a protocol-mandated pause spanning the operative window, and a postoperative restart whose timing is governed by an LLM-bound, human-confirmed advisory [2]. The drug-distribution question that this schedule poses is when, after surgery, systemic exposure should be re-established so as to suppress micrometastatic outgrowth at the earliest point that does not jeopardize anastomotic healing. The advisory takes two inputs, the pancreaticojejunostomy (PJ) International Study Group of Pancreatic Surgery (ISGPS) postoperative pancreatic-fistula grade (A, B, or C) and the serum daraxonrasib trough relative to a 0.5 ng/mL threshold, and it recommends a restart on postoperative day T+7d, T+14d, or T+21d, or a

Table 13: Pharmacodynamic markers of daraxonrasib relevant to the perioperative PDAC study and their use in the protocol.

Pharmacodynamic marker	On-treatment direction	Use in this study
RAS(ON) tri-complex target engagement	Increases with exposure	Mechanistic basis for exposure-anchored dose escalation; not measured directly in participants but modeled in the QSP and digital-twin evidence [16].
Downstream MAPK / PI3K effector signaling	Suppressed	Mechanistic surrogate for antitumor effect; informs the micrometastatic-outgrowth rationale for restarting drug early postoperatively.
Serum daraxonrasib trough concentration	Rises to steady state on continuous dosing; falls during the protocol-mandated operative pause	Objective adherence confirmation and the quantitative input (relative to the 0.5 ng/mL threshold) to the perioperative restart advisory (§5, §6).
Gastrointestinal and dermatologic class effects	Increase with exposure	On-target pharmacodynamic toxicities detected and bounded by the 3+3 DLT framework; managed by protocol supportive care.
CA 19-9 tumor marker (exploratory)	Expected to fall with antitumor response	Exploratory biochemical correlate of systemic disease control across the perioperative course.

continued hold [2]. The advisory is bound to a deterministic seed and a specific repository commit and is always human-confirmed; it is never an autonomous controller, consistent with the prohibition on full autonomy at 21 CFR §312.21(e) [3]. The drug-distribution logic of the schedule is therefore explicit and re-traceable: the pre-operative pause is required to clear systemic exposure ahead of the operation, the operative-window trough is measured against a single fixed threshold rather than estimated, and the postoperative restart is keyed jointly to that trough and to the dominant determinant of fistula risk, the pancreaticojejunostomy, so that systemic exposure is re-established neither before the highest-risk anastomosis has declared its healing trajectory nor so late that the therapeutic window for micrometastatic suppression is lost [2, 16].

The pharmacokinetic basis for this schedule was characterized in a 32-iteration deterministic simulation sweep, the results of which are summarized in Table 9. Across all 32 iterations the baseline serum daraxonrasib concentration at the operative timepoint was 0.45 ng/mL, that is, below the 0.5 ng/mL trough threshold, reflecting the washout produced by the protocol-mandated pre-operative pause [2]. Under these sub-threshold operative-window conditions the advisory recommended a restart on postoperative day T+7d in 29 of 32 iterations (90.6 percent), corresponding to an ISGPS Grade A fistula with a sub-threshold trough; it recommended T+14d in 3 of 32 iterations (9.4 percent), corresponding to an ISGPS Grade B fistula or a delayed serum rise; and it recommended T+21d in 0 of 32 iterations (0.0 percent) in this sweep. An ISGPS Grade C fistula maps to a T+21d restart or a continued hold and to

Table 14: Perioperative pharmacokinetic sweep: 32-iteration deterministic restart-advisory distribution at a uniform 0.45 ng/mL operative-window baseline serum daraxonrasib concentration.

Advisory output	Iterations (of 32)	Proportion	Determining condition
Restart T+7d	29	90.6%	ISGPS Grade A fistula with sub-threshold trough (serum 0.45 ng/mL < 0.5 ng/mL); earliest safe re-establishment of systemic exposure.
Restart T+14d	3	9.4%	ISGPS Grade B fistula or a delayed serum rise; restart deferred one week to protect anastomotic healing.
Restart T+21d	0	0.0%	ISGPS Grade C fistula maps here or to a continued hold; not triggered in this sweep given the uniform Grade A / B inputs.
Continued hold	0	0.0%	Grade C fistula with persistent risk; defers to the protocol drug stop rule.
Baseline (all 32)	32	100%	Operative-window serum daraxonrasib 0.45 ng/mL (sub-threshold) in every iteration, reflecting pre-operative washout.

the drug stop rule of the protocol. The restart-day distribution and the uniform 0.45 ng/mL baseline are the quantitative drug-distribution conclusions that support the perioperative schedule and the advisory thresholds.

Table 9. The 32-iteration deterministic perioperative pharmacokinetic sweep. The operative-window baseline serum daraxonrasib concentration was 0.45 ng/mL (below the 0.5 ng/mL trough threshold) in all iterations; the advisory recommended T+7d in 29 of 32, T+14d in 3 of 32, and T+21d in 0 of 32.

Pharmacology conclusions. On the basis of the established RAS(ON) multi-selective mechanism, the broad pan-KRAS coverage encompassing the G12 alleles that predominate in PDAC, the exposure-anchored pharmacodynamics summarized in Table 8, and the deterministic perioperative pharmacokinetic sweep summarized in Table 9, the sponsor concludes that the pharmacology of daraxonrasib supports the proposed perioperative, once-daily oral schedule and the exposure thresholds used by the restart advisory. The 0.45 ng/mL sub-threshold operative-window baseline and the resulting predominant T+7d restart recommendation provide a quantitative basis for re-establishing systemic exposure early, while the deferral logic for higher fistula grades preserves anastomotic healing, so that the schedule is reasonably expected to balance micrometastatic suppression against surgical risk in the study population [2, 16].

8.1.2 Toxicology: Integrated Summary

Conservative first-in-human starting dose and its safety margin. The first-in-human starting dose in this study is DL1, 160 mg once daily, which is deliberately set below the RASolute 302 reference recommended Phase 2 dose of 300 mg once daily [1]. The escalation then steps up through DL2 at 220 mg once daily to DL3 at the 300 mg once-daily reference dose, so that the entire dosing range evaluated in this study (160 mg to 300 mg) lies at or below an exposure that has already been characterized in the late-phase sponsor program [1, 2]. The starting dose therefore carries a built-in safety margin in two

Table 15: Integrated toxicology summary: dose levels, safety margin relative to the reference dose, and the 3+3 dose-limiting toxicity framework.

Level	Dose (oral QD)	Fraction of 300 mg reference	Toxicology basis and escalation rule
DL1	160 mg	53%	Conservative first-in-human starting dose; below the RASolute 302 reference; 3 enrolled, 28-day DLT window; 0/3 escalate, 1/3 expand to 6.
DL2	220 mg	73%	Intermediate level; entered only after DL1 clears the DLT window; same 3+3 expansion and escalation logic.
DL3	300 mg	100%	Equal to the established RASolute 302 recommended Phase 2 dose; highest level studied; MTD if fewer than 2/6 DLTs.

DLT rule. MTD is the highest level with fewer than 2 of 6 DLT-evaluable DLTs; RP2D is selected at or below the MTD. Two device-or-procedure related deaths at any time, or one device-or-procedure serious adverse event within a cohort, trigger DSMB review and a binding halt.

senses: it is approximately 53 percent of the established reference dose by milligram amount (160 of 300 mg), and it sits within, rather than above, the exposure range for which the established nonclinical and clinical toxicology of daraxonrasib already supports administration. Because the agent is clinically advanced and its established toxicology package supports dosing at and above the doses studied here, the conservative DL1 starting dose, combined with the bounded perioperative exposure duration, provides the integrated-summary basis for the sponsor's conclusion that it is reasonably safe to begin the proposed clinical investigation [7, 1].

Dose-limiting toxicity framework. The toxicology of the escalation is bounded prospectively by a standard 3+3 dose-limiting toxicity (DLT) framework with sentinel staggered enrollment, rather than retrospectively. Three participants are enrolled at each dose level and observed across a 28-day DLT window. If no DLT occurs among the first three, the cohort escalates; if one of three experiences a DLT, the cohort expands to six; a single DLT among six permits escalation, whereas two or more DLTs at a level fix the maximum tolerated dose (MTD) at the preceding level. The MTD is the highest level at which fewer than two of six DLT-evaluable participants experience a DLT, and the recommended Phase 2 dose is selected at or below the MTD [2]. The dose levels and their relationship to the reference dose and the DLT framework are summarized in Table 10. The class-characteristic on-target toxicities of pan-RAS inhibition, principally gastrointestinal events (nausea, diarrhea, stomatitis) and dermatologic and mucocutaneous events, are the toxicities most likely to manifest as DLTs and are precisely the events the 28-day window is designed to detect and bound; perioperative exposure additionally raises a theoretical risk to wound and anastomotic healing if restart timing is suboptimal, which is the specific hazard the pause-and-restart advisory (Table 9) is designed to mitigate [2].

Table 10. Integrated toxicology summary. The 160 mg starting dose is 53 percent of the 300 mg reference dose, and the entire 160 mg to 300 mg range lies at or below an exposure characterized in the established daraxonrasib program; the 3+3 framework bounds the escalation prospectively.

Computational toxicology and QSP evidence. The integrated toxicology conclusion is reinforced by study-specific computational toxicology evidence. A quantitative systems pharmacology (QSP) metastatic-PDAC trial simulation, conducted with code, verification, validation, and uncertainty quantification (VVUQ), and an accompanying playbook, models the exposure to response and exposure to toxicity relationships of daraxonrasib in a mechanistic systems framework and supports the plausibility of the exposure thresholds and the dose range studied here [16]. An AI digital-twin pancreatic-cancer simulation, developed for in silico regulatory-compliance and cost-efficiency purposes, provides a complementary patient-level toxicodynamic representation [15]. These computational works do not substitute for the established nonclinical toxicology of daraxonrasib; rather, they supplement it with study-specific, in silico characterization of the perioperative context, and they are tabulated as supporting documents in the full data tabulation below [16, 15].

Integrated toxicology conclusion. Taking together the conservative 160 mg first-in-human starting dose set at 53 percent of an already-characterized reference exposure, the prospective 3+3 DLT framework with a 28-day window and sentinel staggered enrollment, the binding halt rules, the perioperative pause-and-restart advisory that mitigates the theoretical anastomotic-healing hazard, and the supporting QSP and digital-twin computational toxicology, the sponsor concludes that the toxicological profile of daraxonrasib supports the conclusion that it is reasonably safe to conduct the proposed Phase 1 investigation at the doses and schedule specified, within the meaning of 21 CFR §312.23(a)(8) [7, 3].

8.1.3 Toxicology: Full Data Tabulation

Scope and form of the tabulation. For a clinically advanced agent such as daraxonrasib, the full data tabulation required by 21 CFR §312.23(a)(8)(ii) for the established nonclinical toxicology studies is provided by cross-reference to the sponsor's existing nonclinical package, and the qualifications of the individuals who evaluated those studies are stated in that package [7, 1]. What is tabulated here, and provided as supporting documents to this IND, is the study-specific computational-evidence base that characterizes daraxonrasib and the perioperative drug-device regimen in silico. These works were generated by the sponsor-investigator over 2025 and 2026 and comprise the digital-twin, QSP, and large-scale in silico Phase 3 simulations described above. Their identity, the evidence each contributes, and the location of each as a supporting document are tabulated in Table 11.

Table 11. Full data tabulation of the study-specific computational evidence (digital-twin, QSP with VVUQ, and the 100,000-patient in silico Phase 3 PDAC triplicate) provided as supporting documents, with the established nonclinical toxicology incorporated by cross-reference.

Evaluators and incorporation by reference. The study-specific computational studies tabulated in Table 11 were designed, executed, and evaluated by the sponsor-investigator and the study computational team, whose qualifications are stated in the Investigator and Facilities data of this submission; the verification, validation, and uncertainty-quantification methodology applied to the QSP work is documented in that study's accompanying VVUQ record [16]. The established nonclinical pharmacology and toxicology of daraxonrasib, including the full data tabulations and evaluator qualifications for those studies, is incorporated by reference to the sponsor's existing nonclinical package consistent with the Phase 1 IND content guidance [7, 1]. As the investigation proceeds, any new nonclinical or computational toxicology information bearing on participant safety will be submitted through information amendments under 21 CFR §312.31 and, where a safety signal meets the reporting thresholds, through IND safety reports under 21 CFR §312.32 [3].

Table 16: Full data tabulation: study-specific computational-evidence supporting documents for the pharmacology and toxicology of the perioperative daraxonrasib PDAC regimen.

Computational study	Form	Evidence contributed to §8
AI digital-twin PDAC simulation [15]	Patient-level digital twin	In silico, regulatory-compliance-oriented toxicodynamic and disease-progression representation of PDAC; supports the perioperative drug-distribution and toxicology reasoning at the individual-patient level.
QSP metastatic-PDAC trial simulation with VVUQ [16]	Mechanistic QSP model + VVUQ + playbook	Exposure-to-response and exposure-to-toxicity relationships in a verified, validated, uncertainty-quantified systems-pharmacology framework; underpins the dose range and exposure thresholds.
100,000-patient in silico Phase 3, 5-arm PDAC, triplicate [17, 18]	Large-scale virtual trial (triplicate)	Population-scale in silico characterization of a five-arm PDAC Phase 3 across 100,000 virtual patients, run in triplicate; supports the population-level plausibility of the regimen and the trial design.
End-to-end PDAC digital-twin trial proposals [18]	Trial-design proposals	End-to-end digital-twin clinical-trial scaffolding adapted for the perioperative drug-device design described in this IND.

Cross-reference. The established nonclinical toxicology full data tabulation for daraxonrasib (RMC-6236) is provided by reference to the sponsor's existing nonclinical package; the qualifications of the evaluators are stated there. The 32-iteration perioperative pharmacokinetic sweep (Table 9) is tabulated above.

Conclusion of the pharmacology and toxicology information. The pharmacology summary (§8.1.1), the integrated toxicology summary (§8.1.2), and the full data tabulation (§8.1.3) together provide adequate information about the pharmacological and toxicological properties of daraxonrasib, scaled to a Phase 1 study, on the basis of which the sponsor has concluded that it is reasonably safe to conduct the proposed perioperative, once-daily oral investigation in resectable and borderline-resectable PDAC at the 160 mg, 220 mg, and 300 mg dose levels under the 3+3 dose-limiting toxicity framework, in satisfaction of 21 CFR §312.23(a)(8) [7, 3, 1].

9 Previous Human Experience

This section assembles the previous human experience that supports the proposed Phase 1 investigation, organized under 21 CFR §312.23(a)(9). It states the marketed status of daraxonrasib (RMC-6236), summarizes the prior and ongoing clinical research experience with the drug, situates the proposed combination within the contemporary clinical care experience for robotic pancreaticoduodenectomy, and identifies the safety-reporting framework under which post-submission human experience will continue to be characterized. Because this is a first-in-human investigation of the drug in a perioperative, curative-intent, device-directed surgical context, the evidence base that substitutes for direct prior experience in that exact setting is drawn from three sources: the late-phase metastatic-disease program for daraxonrasib, the established human experience with robotic pancreaticoduodenectomy benchmarked by the 2025 Dutch nationwide cohort, and the author's 2025 to 2026 computational program that supplies the in silico scaffolding for the device-directed component [1, 2]. Figure 22 maps this evidence base, and the narrative below develops each contributing strand in turn, consistent with the phased-development expectations for an initial Phase 1 submission [7].

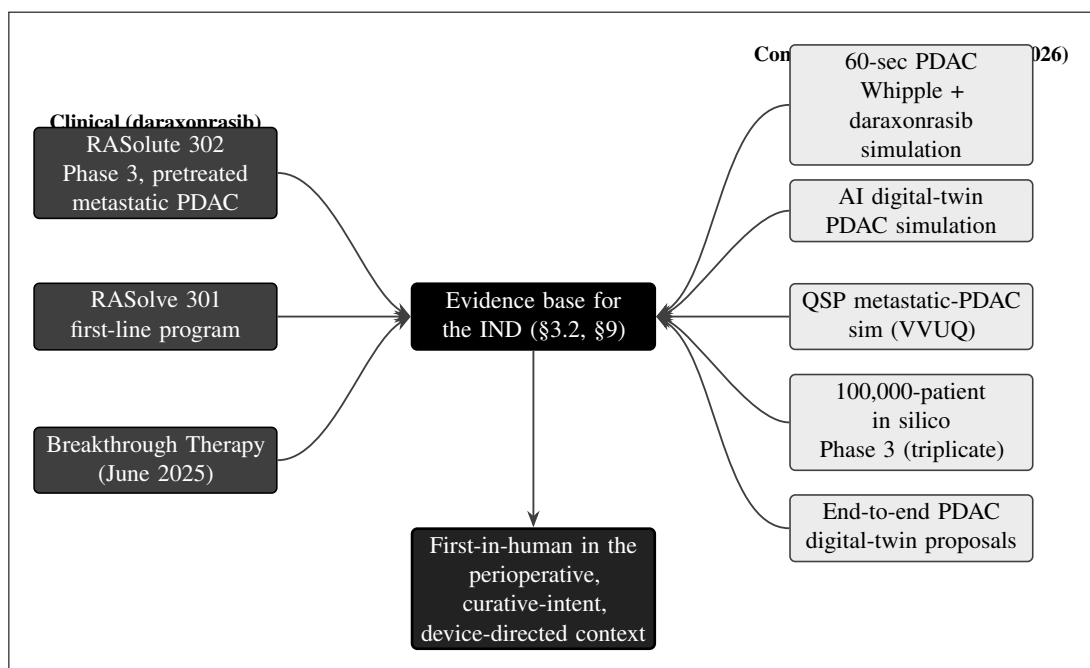


Figure 22. The previous human experience and evidence base. Clinically, daraxonrasib is supported by the RASolute 302 Phase 3 program in previously treated metastatic PDAC, the RASolve 301 first-line program, and its June 2025 Breakthrough Therapy designation; computationally, the author's 2025 to 2026 works supply the simulation scaffolding (the 60-second PDAC Whipple and daraxonrasib simulation, the AI digital-twin PDAC simulation, the quantitative-systems-pharmacology metastatic PDAC simulation with verification, validation, and uncertainty quantification, the 100,000-patient in silico Phase 3 triplicate, and the end-to-end PDAC digital-twin proposals). Daraxonrasib remains first-in-human in this perioperative, curative-intent, device-directed surgical context.

9.1 Marketed experience

Daraxonrasib (RMC-6236) is an investigational new drug and is not approved for marketing in the United States or in any other country for any indication. It is not available commercially, is not the subject of any New Drug Application approval, and therefore has no marketed experience, no postmarketing surveillance history, and no labeling derived from marketing authorization. All human exposure to

date has occurred within controlled clinical investigations conducted under investigational status, and the safety and efficacy information available for the drug is accordingly derived entirely from that investigational program rather than from any marketed use [1]. Because there is no marketed experience, this subsection records that fact and points instead to the investigational program characterized in §9.2; no withdrawal, suspension, or restriction of any marketing authorization has occurred in any country, because none exists.

Daraxonrasib is an orally administered, once-daily RAS(ON) multi-selective (pan-RAS) small-molecule inhibitor. Mechanistically, it binds the active, GTP-bound (ON) state of multiple RAS variants in a tri-complex with the intracellular chaperone cyclophilin A and thereby suppresses downstream effector signaling, providing broad pan-KRAS coverage that includes the G12 alleles (G12D, G12V, and G12R) that predominate in pancreatic ductal adenocarcinoma (PDAC). This mechanism and pharmacological class are developed in full in the Introduction (§3.1) of this IND and are summarized here only insofar as they frame the human experience: the drug acts on a validated oncogenic driver in a disease whose five-year overall survival remains below 13 percent, the third leading cause of cancer death in the United States and the fourth in the European Union [13].

The most significant regulatory milestone in the drug's investigational history to date is the grant of Breakthrough Therapy designation by the FDA in June 2025 for previously treated metastatic PDAC harboring KRAS G12 mutations. Breakthrough Therapy designation is reserved for an investigational drug intended to treat a serious or life-threatening condition for which preliminary clinical evidence indicates that the drug may demonstrate substantial improvement over available therapy on one or more clinically significant endpoints; the designation does not constitute marketing approval and does not alter the investigational status of the drug, but it does signal that the FDA has reviewed preliminary clinical data and found a credible signal of substantial improvement [1, 7]. For the purposes of this IND, the designation is relevant as a qualitative indicator of the maturity and promise of the human experience accumulated in the metastatic setting, not as a basis for any efficacy claim in the perioperative, resectable population studied here. Table 9.1 records the marketed and designation status concisely.

Attribute	Status for daraxonrasib (RMC-6236)
Marketing authorization	None in any country; investigational new drug only.
Drug class	Oral RAS(ON) multi-selective (pan-RAS) small-molecule inhibitor; binds GTP-bound active mutant RAS in a tri-complex with cyclophilin A.
Route and schedule	Oral, once daily.
Molecular target population	PDAC with documented KRAS G12 mutations (G12D, G12V, G12R); broad pan-KRAS coverage.
FDA Breakthrough Therapy	Granted June 2025 for previously treated metastatic PDAC with KRAS G12 mutations.
Marketed safety experience	None; all human safety data derive from the investigational program (§9.2).
Postmarketing surveillance	Not applicable; no marketed use exists.

Table 9.1. Marketed and regulatory-designation status of daraxonrasib. The drug is investigational only; the June 2025 Breakthrough Therapy designation is a development milestone, not a marketing authorization.

9.2 Prior Clinical Research Experience

The prior clinical research experience with daraxonrasib is concentrated in the late-phase metastatic-disease program, which comprises two pivotal studies and supplies the molecular eligibility, dosing, and safety reference framework adapted for this Phase 1 investigation. The two studies are RASolute 302, a Phase 3 study in previously treated metastatic PDAC, and RASolve 301, a first-line program; each is summarized below, followed by the way their experience is translated into the proposed protocol [1, 2].

RASolute 302 (Phase 3, previously treated metastatic PDAC)

RASolute 302 is the pivotal Phase 3 study of daraxonrasib in patients with previously treated metastatic PDAC. It establishes the human experience base for the drug at the dose level that the present protocol adopts as its highest escalation step: the RASolute 302 recommended Phase 2 dose corresponds to daraxonrasib 300 mg administered orally once daily, which this IND designates as dose level 3 (DL3) of its 3+3 escalation, with DL1 at 160 mg and DL2 at 220 mg. The molecular eligibility of RASolute 302 is broad pan-KRAS, encompassing the G12 alleles, and that breadth is carried forward directly into the present protocol: eligible participants must have histologically or cytologically confirmed PDAC with a documented KRAS G12 mutation (G12D, G12V, or G12R), anchoring the present study's molecular criterion to the validated RASolute 302 eligibility rather than introducing a narrower or novel mutational requirement [1]. The safety experience accumulated in RASolute 302, including the characterization of gastrointestinal effects (nausea, diarrhea, stomatitis), dermatologic and mucocutaneous events, and hepatic enzyme elevations, provides the reference safety information against which expectedness is judged for drug-attributed adverse events in this Phase 1 study, as detailed in the safety framework below.

RASolve 301 (first-line program)

RASolve 301 is the first-line program for daraxonrasib in PDAC and extends the human experience to the previously untreated metastatic population, including combination contexts. Its relevance to the present IND is twofold. First, it broadens the dose and exposure experience that informs the conservative, sub-RASolute-302 starting dose chosen for this first-in-human-in-context study (DL1 at 160 mg, well below the 300 mg recommended Phase 2 dose). Second, the perioperative pause-and-restart advisory specified in this protocol is designed to respect any first-line combination context that RASolve 301 may establish: the restart logic keys the resumption day to the postoperative pancreatic fistula grade and the serum daraxonrasib trough relative to the 0.5 ng/mL threshold, and the advisory framework is constructed so that it remains valid whether the drug is given as monotherapy or within a combination regimen [1, 2].

Translation of prior experience into the proposed protocol

The prior clinical research experience is translated into the present protocol through three explicit linkages, summarized in Table 9.2. The molecular eligibility is anchored to the broad pan-KRAS RASolute 302 criteria; the dosing ladder is anchored to the RASolute 302 recommended Phase 2 dose as its ceiling, with two lower steps below it; and the reference safety information for drug causality and expectedness is taken from the metastatic program. The present study does not seek to reproduce or confirm the efficacy demonstrated in the metastatic setting; it is a descriptive, non-confirmatory, first-in-human investigation of the drug in a resectable or borderline-resectable, curative-intent population undergoing a device-directed pancreaticoduodenectomy, with co-primary endpoints directed at device and procedure safety, the maximum tolerated dose (MTD) and recommended Phase 2 dose (RP2D) via 3+3 escalation over a 28-day dose-limiting toxicity (DLT) window, and technical feasibility [7, 1].

Prior study	Population and status	Contribution to the present Phase 1 IND
RASolute 302	Phase 3, previously treated metastatic PDAC; broad pan-KRAS eligibility	Defines the molecular eligibility (KRAS G12) used here; its recommended Phase 2 dose (300 mg once daily) is adopted as DL3, the ceiling of the 3+3 ladder; supplies drug reference safety information.
RASolve 301	First-line metastatic PDAC program, including combination contexts	Broadens the dose and exposure experience supporting the conservative 160 mg starting dose; the perioperative pause-and-restart advisory is built to remain valid in any first-line combination context.
Breakthrough Therapy (June 2025)	Regulatory designation for previously treated metastatic PDAC with KRAS G12	Qualitative indicator of the maturity of the metastatic human experience; not a basis for any efficacy claim in the resectable population studied here.

Table 9.2. How the prior clinical research experience with daraxonrasib is translated into the eligibility, dosing, and safety reference framework of the present Phase 1 IND.

The dose ladder and its 3+3 escalation rule are stated in full in the General Investigational Plan (§4) and are reproduced here only to fix the relationship between prior experience and the proposed exposures: three participants are enrolled at each level; if zero of three experience a DLT the cohort escalates, if one of three does the cohort expands to six, and if one of six does the cohort escalates, whereas two or more DLTs in six define an exceeded MTD with the MTD declared at the prior level. The maximum treated sample is 18 participants (three levels times six), with approximately 36 screened, at a single academic site enrolling approximately one to two participants per month under staggered sentinel enrollment. Because the highest planned exposure (DL3, 300 mg once daily) does not exceed the already-characterized RASolute 302 recommended Phase 2 dose, the prior human experience directly bounds the upper end of the exposure range proposed here [1, 7].

9.3 Clinical Care Experience

The clinical care experience relevant to this IND is the established human experience with robotic pancreaticoduodenectomy (the Whipple procedure), which provides the contemporary benchmark against which the device-directed component of this study will be evaluated, together with the safety-reporting framework that governs how new human experience is captured once the IND is in effect. The Whipple procedure is the most technically demanding curative-intent abdominal oncology operation: it requires resection and reconstruction of four luminal structures (the pancreas, the duodenum, the common bile duct, and the stomach or pylorus) and dissection around named vessels (notably the superior mesenteric vein and the portal vein), and its leading lethal complication is the fistula-sepsis-hemorrhage cascade graded under the International Study Group of Pancreatic Surgery (ISGPS) classification of postoperative pancreatic fistula [2].

The 2025 Dutch nationwide robotic-Whipple benchmark cohort

The contemporary human baseline for robotic pancreaticoduodenectomy is the 2025 Dutch nationwide cohort of 1000 robotic procedures, which anchors the secondary endpoint comparisons in this study. That cohort reported a conversion rate of 10.1 percent, Clavien-Dindo grade III or higher complications of 41.3 percent, ISGPS grade B or C (clinically relevant) postoperative pancreatic fistula of 24.4

percent, and 30-day mortality of 3.9 percent [2]. These four figures are used descriptively, not as a comparator arm: the present single-arm study is non-confirmatory, with no power calculation and no formal hypothesis test against the Dutch cohort, and all comparisons are framed as exact 95 percent Clopper-Pearson interval estimates around the small-sample proportions observed in the trial. The benchmark serves to demonstrate that even expert, high-volume robotic surgery leaves a substantial residual of conversions, high-grade complications, and clinically relevant fistulae, which are precisely the failure modes the device-directed component is designed to mitigate through its layered hard safety limits (the 10 kHz heartbeat bus, the cross-arm emergency stop of 3 ms or less, per-arm and cumulative tip-force caps, and the five-vessel vascular no-fly gate) [1]. Table 9.3 places the benchmark alongside the corresponding study secondary endpoints and the small-sample precision the study can achieve at its maximum size.

Outcome	Dutch 2025 benchmark (n = 1000)	Role in the present study
Conversion to open surgery	10.1%	Secondary endpoint; device-directed conversion (from malfunction or unsafe behavior) is also a co-primary feasibility input.
Clavien-Dindo grade III+ complications (90-day)	41.3%	Secondary endpoint; device- or procedure-related serious adverse events through 30 days, including Clavien-Dindo III+, are a co-primary safety endpoint.
ISGPS grade B or C pancreatic fistula	24.4%	Secondary endpoint and event of special interest; the pancreaticojejunostomy fistula grade is the input to the perioperative drug-restart advisory.
30-day mortality	3.9%	Secondary endpoint; 30-day and 90-day mortality are both followed.

Table 9.3. The 2025 Dutch nationwide robotic-Whipple benchmark cohort (n = 1000) and the corresponding secondary endpoints in the present study. The benchmark is used descriptively; all study estimates are reported with exact 95 percent Clopper-Pearson confidence intervals.

Because the study is small by design, its endpoint estimates carry wide exact confidence intervals, and the benchmark is therefore a contextual reference rather than a statistical comparator. At the maximum treated sample of 18, an observed rate of zero of 18 corresponds to 0 percent with a 95 percent Clopper-Pearson interval of 0 to 18 percent, and one of 18 corresponds to 5.6 percent with an interval of 0.1 to 27 percent; within a single dose cohort, zero of three corresponds to an interval of 0 to 71 percent and zero of six to an interval of 0 to 46 percent. These precision figures make explicit that the present study is designed to characterize the safety and feasibility of the combined intervention, not to establish superiority or non-inferiority against the Dutch benchmark [7]. The surgical and oncologic care context, including the duct-to-mucosa pancreaticojejunostomy under controller ring-tension monitoring (0.30 to 0.60 N), the end-to-side hepaticojejunostomy (0.20 to 0.50 N), and the antecolic gastrojejunostomy (0.40 to 0.80 N), is specified in full in the protocol and is referenced here as the established clinical care experience the device is designed to reproduce within validated envelopes [2].

The safety-reporting framework for ongoing human experience

The human experience generated by this study will be captured and reported under a two-stream safety framework that pairs a conventional pharmacovigilance stream for clinical events with a Physical AI

safety stream for device- and AI-related events, each equally binding. On the clinical stream, any unexpected fatal or life-threatening suspected adverse reaction is reported to the FDA as soon as possible but no later than 7 calendar days after initial receipt of the information, and any other serious and unexpected suspected adverse reaction is reported no later than 15 calendar days after the determination that the event is reportable, consistent with 21 CFR §312.32 [3]. A Physical AI adverse event, per 21 CFR §312.32(f), is any untoward occurrence associated with the operation of the device during the investigation, whether or not caused by it, including robotic malfunction, AI prediction error, sensor failure, communication loss, unauthorized system access, digital-twin divergence, and any event requiring activation of the emergency stop. Six Physical AI reporting triggers are reported under §312.32(g): a serious Physical AI adverse event on the applicable 7-day or 15-day timeline; any system-drug interaction causing incorrect drug administration regardless of seriousness; any cybersecurity incident within 15 calendar days, or 7 days if it has the potential to cause patient harm; sustained AI model performance degradation below the pre-specified acceptance criteria for three or more consecutive procedures or a cumulative 24 hours; any sim-to-real divergence beyond twice the validated gap tolerance (trajectory deviation greater than 4 mm or force discrepancy greater than 1.0 N); and any digital-twin discrepancy beyond the pre-specified accuracy thresholds. For every Physical AI adverse event reported, the complete hash-chained audit trail covering the period from 24 hours before the event to 72 hours after the event is preserved under 21 CFR part 11 and made available to the FDA on request [3, 11].

Once the IND is in effect and the 30-day clock has elapsed, post-submission authoring is driven by this safety data rather than by the initial submission, and the same generation method that produced the IND maintains the document suite in internal consistency. Figure 13 shows the post-submission maintenance flow: serious adverse events and suspected adverse reactions feed the 7-day and 15-day IND safety reports; emerging pharmacokinetic and pharmacodynamic data and cohort completion feed the annual Development Safety Update Report (DSUR) and amendments; data safety monitoring board (DSMB) cohort reviews feed the dose-escalation minutes and the RP2D memorandum; the principal investigator assesses causality and scientific justification across all of these; and the LLM then synchronizes the protocol, consent, and site documents so the whole suite remains internally consistent [2, 12].

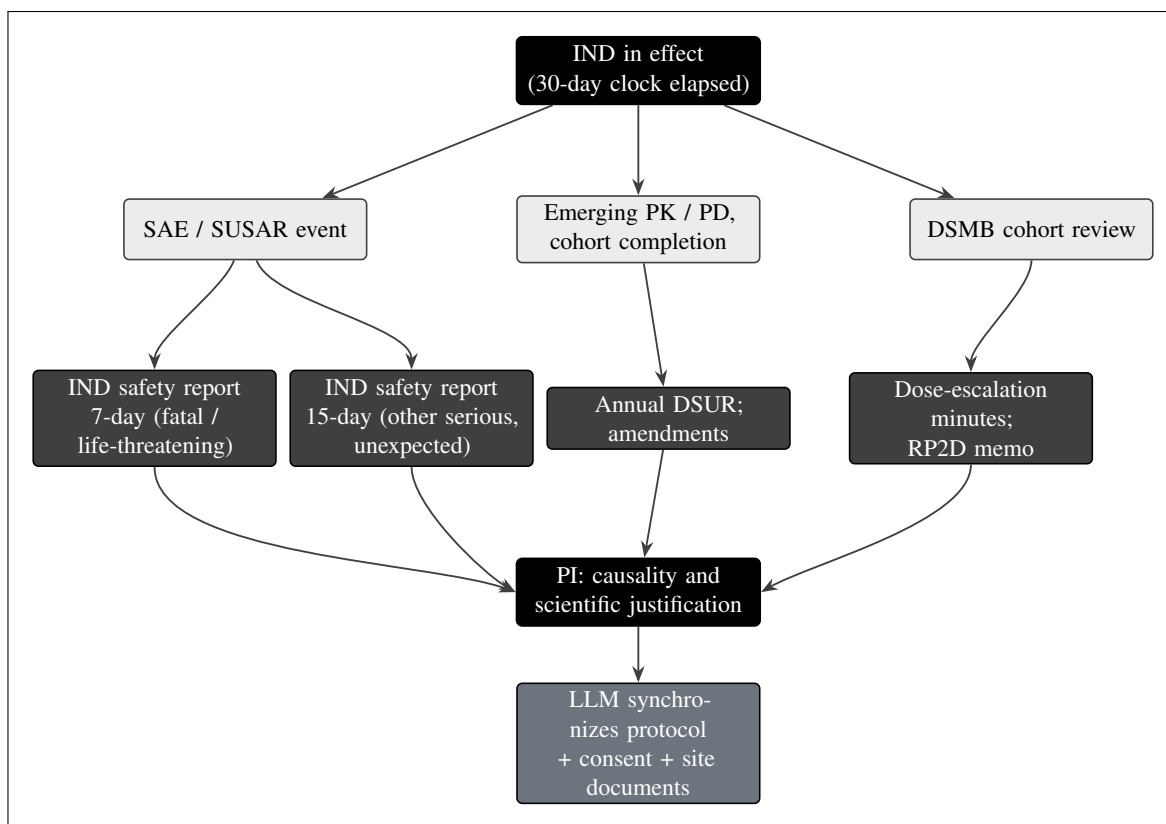


Figure 13. Post-submission IND maintenance authoring once the IND is in effect (the 30-day clock elapsed). Safety data drive the authoring: IND safety reports on the 7-day (fatal or life-threatening) and 15-day (other serious and unexpected) clocks, the annual DSUR, amendments, and dose-escalation minutes, with the principal investigator assessing causality and scientific justification and the LLM synchronizing the protocol, consent, and site documents so the whole suite stays internally consistent.

The three-tier oversight that reviews this accumulating human experience comprises the DSMB (acting at cohort boundaries and on triggered review, with halt authority), an Independent Safety Monitor (per-procedure, real-time, with stop authority), and a Physical AI Safety Review Committee (convening at intervals not exceeding 90 days during active enrollment). The binding halt rules are that one or more device- or procedure-related serious adverse events within a cohort, or two device-related deaths at any time, trigger DSMB review and possible halt [2, 5]. This governance ensures that the previous human experience characterized in this section is continuously updated and acted upon as the study generates the first direct human experience with the drug in this device-directed, curative-intent context, the gap that Figure 22 identifies as first-in-human [6].

9.4 References

The references supporting this section are listed in full in the back matter and are cited inline above. The clinical and computational evidence base is documented in the daraxonrasib robotic-Whipple and on-premises Whipple sources [1, 2]; the phased-development and Phase 1 expectations in the Phase 1 guidance [7]; the PDAC epidemiology and metastatic-program context in the disease and program references [13]; the safety-reporting and Part 11 framework in the 21 CFR adaptations [3, 11, 12]; the verification-before-generation and clinical-trust framing in the legislative and trust references [5, 6]; and the author's 2025 to 2026 computational program in the digital-twin, quantitative-systems-pharmacology, in silico Phase 3, and digital-twin-proposal works [15, 16, 17, 18, 19].

10 Additional Information

This section addresses the categories of additional information enumerated for an original Investigational New Drug (IND) submission under 21 CFR §312.23(a)(10), namely drug dependence and abuse potential, radioactive drugs, pediatric studies, and any other information bearing on the safe and ethical conduct of the proposed clinical investigation. Because this is a combined IND and Investigational Device Exemption (IDE) first-in-human study of orally administered daraxonrasib (RMC-6236) delivered alongside an on-premises large language model (LLM) directed eight-arm robotic pancreaticoduodenectomy, the Other Information subsection additionally consolidates the Physical AI governance, cybersecurity, record retention, and three-tier oversight content that the autonomy-bearing investigational device requires for Phase 1 acceptance. Each statement below is grounded in the protocol (v1.0.0, DOI 10.5281/zenodo.20780121) and is consistent with FDA Phase 1 guidance [7] and with the regulatory adaptations developed for Physical AI investigations [3, 11, 12].

10.1 Drug Dependence and Abuse Potential

Daraxonrasib has no known dependence or abuse potential and is not a controlled substance under the Controlled Substances Act. The investigational drug is an oral, once-daily RAS(ON) multi-selective (pan-RAS) small-molecule inhibitor that binds the guanosine triphosphate (GTP) bound active form of mutant RAS in a complex with cyclophilin A and thereby suppresses downstream effector signaling [1, 2]. Its pharmacological mechanism is targeted oncogenic-signaling inhibition; it has no activity at the central nervous system receptors, transporters, or ion channels that mediate reinforcement, euphoria, sedation, or physical dependence, and it does not belong to a pharmacological class associated with recreational use, tolerance with escalating self-administration, or a withdrawal syndrome. The drug carries an FDA Breakthrough Therapy designation (June 2025) for previously treated metastatic pancreatic ductal adenocarcinoma (PDAC) with KRAS G12 mutations, and the prior and ongoing human experience summarized in the Previous Human Experience section has not identified any signal of abuse, dependence, drug-seeking behavior, or diversion.

Accordingly, no abuse-liability program is proposed for this IND. No dedicated nonclinical abuse-potential studies (for example, self-administration, drug discrimination, or physical-dependence and withdrawal studies under the FDA abuse-potential assessment framework) are warranted, and no controlled-substance scheduling recommendation applies. The investigational product is dispensed under the accountability controls described in the Chemistry, Manufacturing, and Control Information and Proposed Clinical Research sections, with drug accountability logs maintained at the single academic site, so that the limited quantities used for the planned cohorts (up to eighteen treated participants across three dose levels) are reconciled from receipt through administration, return, and destruction. Should any unanticipated central-nervous-system finding suggestive of abuse potential emerge during the conduct of the study, it would be captured through the routine adverse event and serious adverse event reporting pathways under 21 CFR §312.32 and reviewed by the Data Safety Monitoring Board (DSMB).

10.2 Radioactive Drugs

This subsection is not applicable. Daraxonrasib is not a radioactive drug and contains no radionuclide. The investigational product is a non-radioactive oral small-molecule tablet administered once daily; it is neither a positron-emitting nor a single-photon-emitting radiopharmaceutical, and it is not the subject of a Radioactive Drug Research Committee (RDRC) determination under 21 CFR §361.1. No radiation dosimetry, no radiation safety committee review, and no radioactive drug-specific manufacturing, handling, storage, or waste-disposal controls are required for the investigational agent.

Imaging performed during the study (for example, diagnostic computed tomography and any standard-of-care nuclear medicine staging that may precede enrollment) is part of routine clinical care and is not an investigational radioactive drug administered under this IND; such imaging is performed and interpreted according to institutional standards and is outside the scope of this subsection. The intraoperative imaging, hemostasis, and drain placement of operative phase P8 are optical and instrumental and involve no administered radioactivity. No information under 21 CFR §312.23(a)(10)(ii) is therefore required.

10.3 Pediatric Studies

The study population is restricted to adults aged eighteen years or older. Eligible participants have histologically or cytologically confirmed PDAC that is resectable or borderline-resectable, a documented KRAS G12 mutation (G12D, G12V, or G12R), an Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 1, and are candidates for pancreaticoduodenectomy (Whipple resection), consistent with the broad pan-KRAS eligibility framework of the RASolute 302 program [1]. No participant under the age of eighteen will be enrolled, consented, screened, dosed, or operated upon under this protocol, and pediatric assent provisions are therefore inapplicable in practice even though the informed-consent architecture (21 CFR part 50 with ICH E6(R3)) provides for assent where it would otherwise apply [11].

A first-in-human Phase 1 evaluation in a pediatric population is neither warranted nor proposed at this stage, and a pediatric study plan is deferred. The deferral rationale rests on three considerations. First, the disease under study, PDAC, is overwhelmingly a disease of older adults; pancreatic ductal adenocarcinoma in children is exceedingly rare, the five-year overall survival of PDAC is below thirteen percent, and the condition is the third leading cause of cancer death in the United States and the fourth in the European Union [13], so the target indication does not arise materially in the pediatric population. Second, this is a first-in-human study of an autonomy-bearing investigational device combined with an investigational dosing regimen; the prudent and ethical sequence is to establish the device and procedure serious adverse event profile, the maximum tolerated dose and recommended Phase 2 dose, and the technical feasibility of the LLM-directed robotic Whipple in adults before any consideration of younger populations, and full autonomy is in any case prohibited at this stage under 21 CFR §312.21(e) [3]. Third, the staged-autonomy model of this trial class confines the present study to clinician-controlled teleoperation and supervised operation (Stage 1), reserving higher-autonomy operation for future study; extension to a pediatric population, if ever pursued, would follow successful completion of the adult program and would be the subject of a separate pediatric study plan with its own age-appropriate formulation, dosing, device sizing, and assent and parental-permission provisions.

Because this is an investigation of a serious and life-threatening oncology indication that does not occur meaningfully in children, the study qualifies for the conditions under which a pediatric assessment is not required for an adult-only first-in-human Phase 1 program; any subsequent pediatric commitments would be addressed at the appropriate later stage of development in consultation with the FDA. No pediatric data are submitted with this original IND.

10.4 Other Information

The autonomy-bearing investigational device requires governance, cybersecurity, record-retention, and oversight controls beyond those of a conventional small-molecule Phase 1 study. This subsection consolidates that Physical AI overlay. The on-premises LLM operates strictly as a second-opinion advisory oracle: it is hash-pinned to a deterministic seed and a repository commit, isolated outside the vendor kinematic stack, and never granted control authority over the robotic arms; full autonomy is prohibited

under 21 CFR §312.21(e) [3], and every proposed robot-patient command passes a verification-before-generation ten-gate suite returning accept, block, or escalate, aligned with the Verification Before Generation in Physical AI Oncology Trials Act of 2026 (H.R. 9510) [5]. The governance design follows the trust and accountability principles developed for clinical Physical AI [6, 4].

10.4.1 Physical AI Governance and the Three-Tier Oversight Structure

Independent safety oversight operates on three coordinated tiers layered over the reviewing Institutional Review Board (IRB) and the Sponsor-Investigator. The DSMB, composed of members independent of the conduct of the study with surgical, oncologic, biostatistical, and device-safety expertise, reviews accumulating safety data at each cohort boundary and at any triggered review and holds binding halt authority. The Independent Safety Monitor (ISM) is designated for each procedure and provides real-time, case-level safety oversight independent of the operating team, with the authority to call an immediate stop. The Physical AI Safety Review Committee meets on a cadence of ninety days or less (and ad hoc on any triggering event) to review the autonomy-bearing behavior of the system, including the telemetry and audit record, the verification-before-generation gate-surface decisions, the Unified Safety Level (USL) readiness rating, and any software change, and recommends USL reassessment or oversight changes to the Sponsor-Investigator. The three tiers are coordinated so that no autonomy-bearing change and no escalation decision proceeds without appropriate independent review. The Sponsor-Investigator (ChemicalQDevice, Kevin Kawchak, Chief Executive Officer) holds the combined IND and IDE; the conflict of interest arising from the Sponsor-Investigator's relationship to the developer of the Physical AI system is disclosed in full under 21 CFR part 54 and is mitigated structurally by this independent oversight, so that safety and escalation decisions are not made by a financially interested party alone. Table 17 summarizes the oversight bodies, their cadence, and their authority, and Figure 1 renders the governance structure.

Table 17: Three-tier independent safety oversight and the human-subjects-protection structure for the combined IND and IDE study. Each body's review cadence and the authority it exercises are stated; the binding halt rules drive the continue, modify, pause, or halt decision.

Body	Review cadence	Role and authority
Sponsor-Investigator (ChemicalQDevice, K. Kawchak)	Continuous	Holds combined IND and IDE; protocol custody, safety reporting under 21 CFR parts 312 and 812, and compliance assurance; conflict of interest disclosed and managed under 21 CFR part 54.
Reviewing IRB	Per submission and continuing review	Human-subjects protection; reviews protocol, consent, and the significant-risk device determination; may suspend or terminate approval.
Data Safety Monitoring Board (DSMB)	Each cohort boundary and any triggered review	Independent surgical, oncologic, biostatistical, and device-safety members; applies the prespecified halt rules; recommends continue, modify, pause, or halt; binding halt authority.
Independent Safety Monitor (ISM)	Per procedure, real-time	Case-level safety oversight independent of the operating team; authority to call an immediate stop.
Physical AI Safety Review Committee	Cadence of 90 days or less, plus ad hoc	Reviews telemetry and audit records, verification-before-generation gate-surface decisions, USL readiness, and software changes; recommends USL reassessment or oversight changes.

Table 1. The oversight bodies, their cadence, and their authority for the combined IND and IDE first-in-human study.

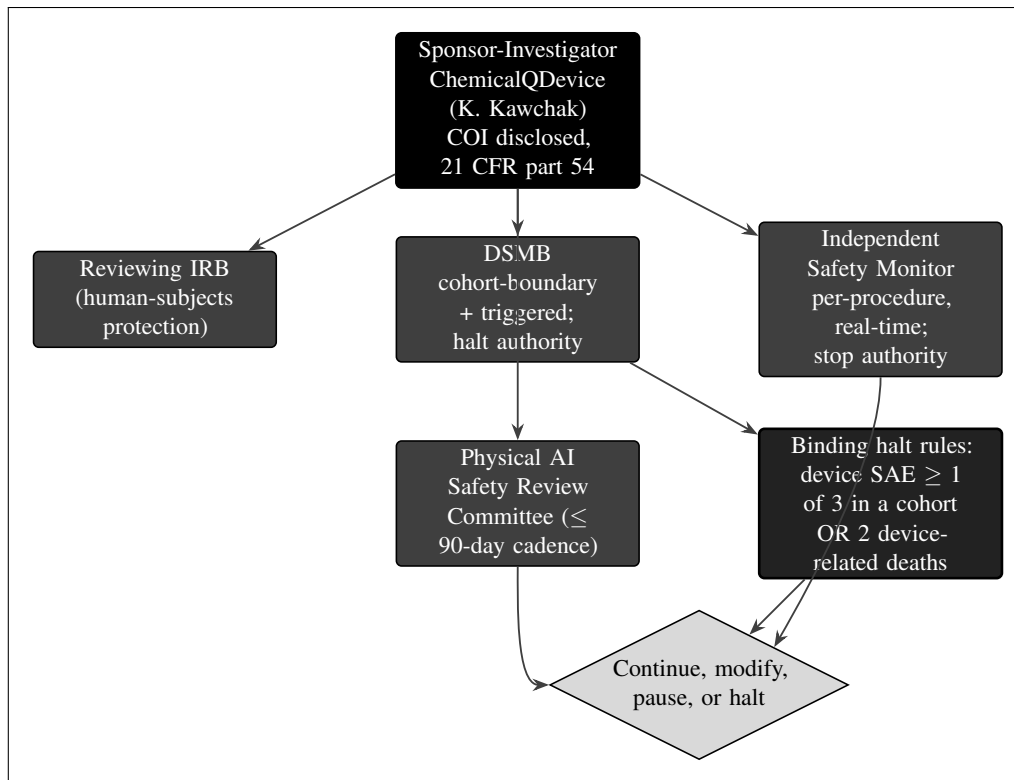


Figure 1. Study governance and three-tier safety oversight. The Sponsor-Investigator (ChemicalQDevice, Kevin Kawchak, with the conflict of interest disclosed under 21 CFR part 54) operates under a reviewing IRB and three independent bodies: the Data Safety Monitoring Board (cohort-boundary and triggered reviews, with halt authority), the Independent Safety Monitor (per-procedure, real-time, with stop authority), and the Physical AI Safety Review Committee (a cadence of 90 days or less). The binding halt rules, a device-related serious adverse event in at least one of three within a cohort or two device-related deaths at any point, drive the continue, modify, pause, or halt decision.

10.4.2 Binding Halt Rules and Stopping Authority

Two prespecified, binding halt rules govern the study and are applied by the DSMB. First, a device-related or procedure-related serious adverse event in at least one of three participants within a dose cohort triggers a mandatory DSMB review with authority to halt. Second, two device-related deaths at any point in the study trigger a mandatory DSMB review with authority to halt. On invocation of either rule, enrollment and any patient-facing robotic activity stop pending review. These rules operate independently of, and in addition to, the dose-finding rules of the 3+3 escalation (dose levels 160, 220, and 300 mg once daily, with a twenty-eight-day dose-limiting-toxicity window) and the per-procedure stop authority of the Independent Safety Monitor. The cyber-physical safety envelope provides further hard stops that are not discretionary: force caps of no more than 3 N per arm and no more than 18 N cumulative across the eight arms, a 10 kHz heartbeat with a sixty-four-byte frame and a one-hundred-microsecond watchdog, arm-park within fifty microseconds, a cross-arm emergency stop within three milliseconds to a commanded 0.0 N, and a hardware emergency stop within five hundred milliseconds consistent with 21 CFR §312.404. The vascular no-fly gate samples at 10 kHz and enforces a soft-warning advisory zone at 3.0 mm, blocking no-fly margins of 1.0 mm at the superior mesenteric vein and portal vein and 1.5 mm at the hepatic artery, celiac artery, and superior mesenteric artery, and a hard-stop emergency-stop zone at 5.0 mm. Any breach of a hard cyber-physical limit, a failure of the verification-before-generation gate surface, or a loss of audit integrity constitutes a clinical-hold ground

for the autonomy-bearing system, on which all patient-facing robotic activity ceases and the decommissioning and data-preservation procedure is executed.

10.4.3 Cybersecurity and System Integrity

Cybersecurity is treated as a patient-safety control rather than an administrative adjunct. The LLM-directed system is deployed on premises and isolated outside the vendor kinematic stack, with no requirement for external network connectivity during a procedure; the advisory model is hash-pinned to a deterministic seed and a repository commit so that the validated configuration in use is exactly the configuration that was tested and recorded. Access to identifiable records and to the cyber-physical control surface is governed by a deny-by-default authorization model, under which no role can read, write, or export a record except where an access right has been explicitly granted, and electronic signatures are bound to the records they authenticate. Every access, change, and export is written to a hash-chained audit trail in which each entry cryptographically incorporates the hash of the prior entry, so that the record is tamper-evident and any alteration is detectable. A cybersecurity incident is one of the six Physical AI reporting triggers under 21 CFR §312.32(g) and is reportable within fifteen calendar days, or within seven calendar days where there is potential for patient harm; the audit trail is preserved from twenty-four hours before to seventy-two hours after each Physical AI adverse event under 21 CFR part 11. The other Physical AI triggers, namely a serious Physical AI adverse event (seven or fifteen days), a system-drug interaction (fifteen days), sustained AI model degradation across at least three consecutive procedures or twenty-four cumulative hours, a sim-to-real divergence exceeding twice the tolerance (a trajectory error above 4 mm or a force error above 1.0 N), and a digital-twin discrepancy, are reported on the same Subpart J pathway.

10.4.4 The Seven Physical AI Record Types and 21 CFR Part 11

Seven Physical AI record types are captured and retained for this trial class, in addition to the conventional clinical-trial records. Table 18 enumerates each record type, its content, and its primary regulatory anchor. All seven are bound to a deterministic seed and a repository commit by the hash-chained audit trail (record type seven), so that any reported behavior can be replayed deterministically and any requested record reconstructed. All electronic records, signatures, and the cyber-physical telemetry stream are maintained under 21 CFR part 11, with validated systems, secure time-stamped audit trails that do not obscure prior entries, role-based access controls, and electronic signatures bound to the records they authenticate. Records are retained for the period required by 21 CFR §312.57, namely at least two years after a marketing application is approved for the indication or, if no application is filed or approved, two years after the investigation is discontinued and the FDA is notified; the raw, losslessly reconstructable command and sensor tier is preserved so that a requested record can be reconstructed on FDA request under 21 CFR §312.57. Data released for analysis, deposition, or publication are de-identified by the HIPAA Safe Harbor method, with removal of all eighteen categories of protected health information identifiers.

Table 2. The seven Physical AI record types, their content, and their regulatory anchors. Record type seven binds the preceding six to a deterministic seed and repository commit so that any reported behavior can be replayed deterministically.

10.4.5 Quality Assurance, Re-Validation, and Decommissioning

Quality assurance spans the clinical and the cyber-physical domains. Before every procedure, the pre-procedure safety matrix is verified, the USL readiness rating is confirmed at or above the deployment threshold, and the emergency-stop, positional, and force envelopes are checked. The Phase 0 simulation-validation gate, which must be satisfied before any patient-facing robotic activity, requires

Table 18: The seven Physical AI record types captured and retained for the LLM-directed robotic Whipple, in addition to the conventional clinical-trial records. All records are retained under 21 CFR part 11 for the period required by 21 CFR §312.57.

No.	Record type	Content and regulatory anchor
1	LLM advisory record	Each prompt, the supplied context, and the advisory output of the second-opinion oracle, hash-pinned to seed and commit; advisory only, never control authority (21 CFR §312.21(e)).
2	Robot command record	Each command issued to the eight arms (56 degrees of freedom), with timestamps for deterministic replay.
3	Sensor and telemetry record	The cyber-physical stream from 640 sensor channels (80 per arm; 100 kHz force, 10 kHz other), tiered from compact summaries to the raw losslessly reconstructable level.
4	Safety-limit-event record	Emergency-stop activations and latencies, positional and force-envelope excursions (≤ 3 N per arm, ≤ 18 N cumulative), and vascular no-fly gating events (soft 3.0 mm, no-fly 1.0 to 1.5 mm, hard-stop 5.0 mm).
5	Verification-before-generation gate-surface record	Each accept, block, or escalate decision of the ten-gate suite, aligned with H.R. 9510 (Verification Before Generation in Physical AI Oncology Trials Act of 2026).
6	Human-override and intervention record	Each manual override, takeover by the backup operator, and stop invoked by the Independent Safety Monitor.
7	Hash-chained audit trail	Cryptographically chained binding of records 1 to 6 to a deterministic seed and repository commit; tamper-evident; preserved from -24 h to $+72$ h around each Physical AI adverse event (21 CFR part 11).

a USL of at least 7.0, at least one thousand simulations across at least two independent frameworks (for example Isaac, MuJoCo, Gazebo, or PyBullet), and a sim-to-real agreement of less than 2 mm in trajectory and less than 0.5 N in tip force. After any software update, model change, or configuration change to the LLM-directed system, a Phase 0 re-validation is performed and the USL is reassessed before any further patient-facing use; no update reaches a participant without re-validation. On any clinical hold or closure, enrollment and patient-facing robotic activity stop immediately, affected participants are transitioned to appropriate clinical care, and a controlled decommissioning and data-preservation procedure is executed under which the robotic system is locked out of patient contact, the deterministic configuration and repository commit are recorded, and the complete hash-chained command and telemetry record is sealed and retained for the regulatory retention period. The computational evidence supporting this assurance approach includes prior in silico and digital-twin work by the author, namely a sixty-second PDAC Whipple with daraxonrasib simulation, an AI digital-twin PDAC simulation, a quantitative systems pharmacology metastatic-PDAC simulation with verification, validation, and uncertainty quantification, a one-hundred-thousand-patient in silico five-arm Phase 3 PDAC study in triplicate, and end-to-end PDAC digital-twin trial proposals [16, 15, 17, 18], together with a sixteen-LLM code-generation evaluation against FDA adverse-event reporting data [19].

10.5 Selected References

The following selected references support the additional information in this section and are cited in full in the back matter. The FDA Phase 1 guidance frames the content and graded rigor appropriate to an original first-in-human IND [7], and the daraxonrasib and on-premises robotic Whipple sources establish the drug mechanism and the device context [1, 2]. The Physical AI regulatory adaptations of 21 CFR parts 312 and 50 and of ICH guidance govern the autonomy-bearing overlay, the opt-out, and the Subpart J reporting and record-retention controls [3, 11, 12]. The Verification Before Generation in

Physical AI Oncology Trials Act of 2026 (H.R. 9510) anchors the verification-before-generation gate surface [5], and the trust, accountability, and site-readiness literature for clinical Physical AI informs the governance and oversight design [6, 4, 10, 8]. The PDAC epidemiology and the prior computational evidence supporting the deferral and assurance rationale are drawn from the indication and digital-twin sources [13, 16, 15, 17, 18, 19, 14]. A complete reference list appears in the back matter.

11 Relevant Information

This section provides the additional relevant information that the U.S. Food and Drug Administration may request under 21 CFR §312.23(a)(11), together with the method-level grounding and verification disclosures that apply because this Investigational New Drug application (IND v1.0) was assembled by an artificial intelligence build operating over the repository at version v4.3.0. The disclosures are placed here, at the close of the package, so that a reviewer can read the entire submission first and then evaluate exactly how the document was constructed, how each figure and number was grounded, how the build is monitored in real time, and where the residual limitations lie. Nothing in this section substitutes for the clinical, statistical, and regulatory substance of the preceding sections; rather, it states the controls that give a reviewer confidence that the substance is internally consistent, traceable to named source files, and reproducible. The verification posture described here is aligned with the verification-before-generation discipline that the program has advanced for physical artificial intelligence in oncology trials [5, 6], and it is consistent with the phased-development expectations for an initial Phase 1 submission [7].

11.1 Method-Level Grounding and Verification

The IND was produced by a single-prompt, multi-file build that proceeds through four stages, mermaid to draft to full to final, with the present full stage expanding each draft scaffold into regulator-grade prose, tables, and figures. To keep that build verifiable rather than merely fluent, the method applies five name-matching and grounding verifications to every artifact it emits. The five verifications are summarized in Figure 23 and detailed, with a concrete IND example for each, in Table 17. They are not abstract quality slogans; each one is a check that a reviewer can independently reproduce by inspecting the public repository tree, the rendered document, and the commit and pull-request history. The five together yield a grounded, monitorable IND in which every figure names its source, every figure is cited where its section discusses it, every file is visible on GitHub as it is written, the repository and directory names mirror the document structure, and the section file names map one to one onto the regulatory sections they contain.

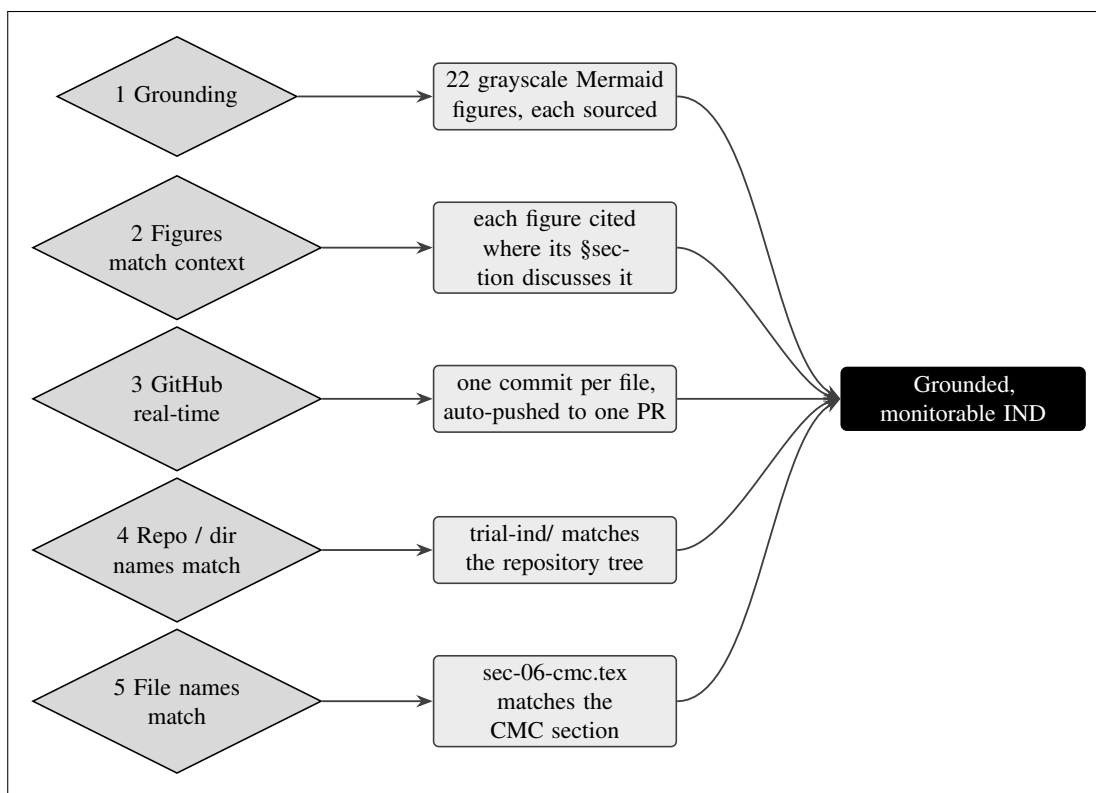


Figure 23. Five name-matching and grounding verifications, each with a concrete IND example: grounding (every one of the 22 grayscale figures names its source files); figures match context (each is cited where its section discusses it); GitHub real-time (one commit per file, auto-pushed to one continuously updated pull request); repository and directory names match (the trial-ind tree mirrors the document structure); and file names match (sec-06-cmc.tex maps to the Chemistry, Manufacturing and Control section). Together they yield a grounded, monitorable IND.

11.1.1 The Five Verifications in Detail

Verification 1, grounding. Every quantitative claim that a figure carries is traceable to a named source file rather than invented at generation time. The 22 grayscale figures of this IND each declare, in their Mermaid source under `trial-ind/mermaid/`, the source files from which their nodes, edge labels, and numbers were drawn, and those numbers are the same document counts, dose levels, force caps, sampling rates, and review clocks reused in the surrounding prose. For example, the dose levels of 160 mg, 220 mg, and 300 mg once daily, the 28 day dose-limiting-toxicity window, and the maximum enrollment of 18 treated subjects appear identically in the dose-escalation figure, in the General Investigational Plan, and in the protocol from which they originate. This binds the figures to the trial facts and prevents a figure from drifting away from the text it illustrates [9, 18].

Verification 2, figures match context. Each figure is cited in the sentence or paragraph that discusses it, in the section where that discussion belongs, so that no figure floats unreferenced and no claim in the prose lacks its illustrating figure. The mechanism-of-action figure is cited in the introductory material that names the drug and its pharmacological class, the 3+3 automaton is cited in the General Investigational Plan, the chemistry information architecture is cited in the Chemistry, Manufacturing and Control section, and the two figures of the present section are cited here. A reviewer can confirm this by following any figure number from its caption back to the citing sentence.

Verification 3, GitHub real time. Each distinguishable file is committed and pushed at the moment it is written, and those commits accumulate on a single continuously updated pull request, so that the IND is observable on GitHub as it is constructed rather than only after completion. This gives the sponsor-investigator, and in principle an auditor, a real-time view of provenance: who or what wrote each file, in what order, and against which commit. The auto-commit and auto-pull-request behavior is the documentary analogue of the audit-trail and monitorability expectations that govern the device-directed component of the trial [4, 8].

Verification 4, repository and directory names match. The directory tree named in the prose is the directory tree on disk. References to `trial-ind/mermaid/`, `trial-ind/full-ind/sections/`, and `trial-ind/inputs/` denote the actual directories of the repository at version v4.3.0, so a reader can navigate from any path mentioned in the document straight to the corresponding location in the public tree without translation.

Verification 5, file names match. The section file names map one to one onto the regulatory sections they contain. `sec-06-cmc.tex` is the Chemistry, Manufacturing and Control section, `sec-07-pharmacology-toxicology.tex` is the Pharmacology and Toxicology section, `sec-10-relevant-information.tex` is the present section, and `fig-09-five-grounding-verifications.md` is the Mermaid source for Figure 23. The naming convention makes the package self-describing and removes ambiguity about which file holds which content.

#	Verification check	Concrete IND example
1	Grounding via Mermaid figures	The 22 grayscale figures in <code>trial-ind/mermaid/</code> each name their source files and carry the dose levels (160, 220, 300 mg), the 28 day DLT window, and the 30 day FDA clock reused verbatim in the prose
2	Figures match IND context	Each figure is cited in the sentence that discusses it, in the section where that discussion belongs (mechanism in the Introduction, 3+3 in the Plan, CMC architecture in the CMC section, these two figures here)
3	AI files viewable on GitHub in real time	One commit per file is pushed at generation, accumulating on a single continuously updated pull request, so each artifact appears on the branch as it is made
4	Repository and directory names match	The prose refers to <code>trial-ind/mermaid</code> and <code>full-ind/sections</code> , which are the directories on disk in repository v4.3.0
5	File names match repository file names	A cited <code>sec-06-cmc.tex</code> is the CMC section; <code>fig-09-five-grounding-verifications.md</code> is the Mermaid source for Figure 23

Table 17. The five name-matching and grounding verifications, each with a concrete example drawn from this IND. Every check is independently reproducible from the public repository tree, the rendered document, and the commit and pull-request history.

11.2 Figure Grounding: Mermaid to TikZ

A second, finer-grained discipline governs the figures themselves: every figure in this IND is authored once as a grayscale Mermaid source in `trial-ind/mermaid/` and then reproduced as a TikZ figure

of identical complexity in the LaTeX document, with no simplification of nodes, edges, edge labels, or quantitative content. The translation pathway and its verify-twice checkpoint are shown in Figure 24. The Mermaid source defines the nodes, edges, grayscale palette, and quantitative data; the conversion preserves all of these; the TikZ `mermaidfig` environment renders them using the `mm*` node styles defined in `indstyle.sty` (the same eight-tone ramp that maps one to one from the Mermaid classDef classes); and the result is verified twice, for text-box and arrow overlaps, for the specified arrow looseness, and for proper box spacing, before it appears identically in the draft, full, and final stages. A figure that fails the verify-twice checkpoint is returned to the TikZ step and corrected; only a figure that passes is allowed to render in the document.

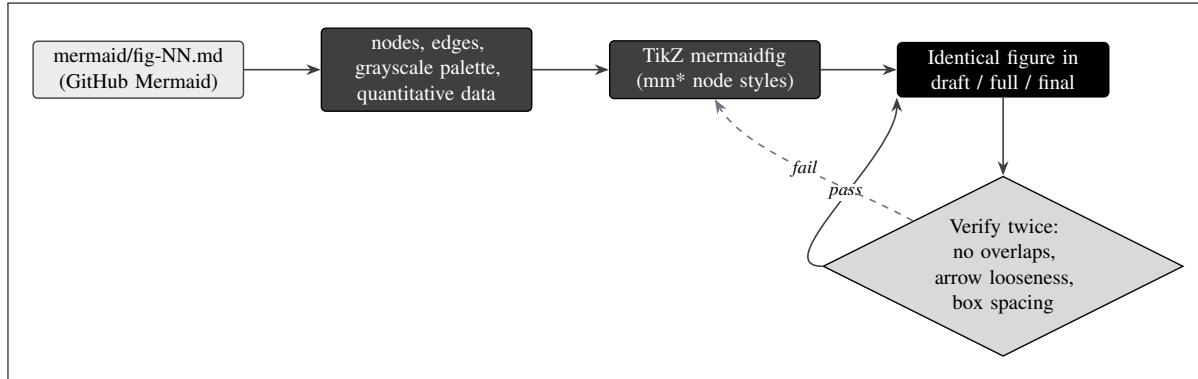


Figure 24. Figure grounding. Each Mermaid source in `mermaid/` is reproduced as a TikZ `mermaidfig` of the same complexity (the same nodes, edges, grayscale palette, and quantitative data), then verified twice for text-box and arrow overlaps, for the specified arrow looseness, and for proper box spacing, before it appears identically in the draft, full, and final stages. A figure that fails is returned to the TikZ step; only a figure that passes renders.

The practical effect of this discipline is that the figure a reviewer sees in the compiled PDF is provably the same diagram, with the same numbers, as the Mermaid source committed to the repository. There is no separate figure-rendering pipeline, no raster export, and no opportunity for a number to be transcribed differently between the source and the rendered figure. All figures are vector TikZ; the IND contains no PNG or JPG images and no `\includegraphics` calls. The grayscale ramp is fixed in the style file, so the tonal encoding of node roles (goal, process, accent, step, input, context, decision, dark) is identical across all 22 figures and across the four build stages [15].

11.3 Real-Time Auto-Commit and Auto-Pull-Request Monitorability

The build is monitorable in real time. As each section file, style file, figure source, and supporting file is written, it is committed individually and pushed to the remote, and the accumulating commits are surfaced on a single open pull request that updates continuously over the life of the build. This means that the provenance of the IND is not reconstructed after the fact from a final snapshot; it is visible while the document is being produced. A reader with access to the public repository can watch the `trial-ind/full-ind/` tree fill in section by section, can inspect the diff for any individual file, and can read the commit history as an ordered, timestamped record of how the package came together. This real-time monitorability is the documentary counterpart of the trial's own audit-trail commitments, under which the device preserves a window of records around each physical artificial intelligence adverse event, and it is consistent with the broader transparency posture that the program has argued is necessary before higher-autonomy systems are used in oncology trials [5, 6, 10]. The same single-prompt, auto-commit, auto-pull-request method has been demonstrated across prior multi-file builds in the program, including the companion Phase 2 protocol and the legislative-text repositories, which establishes that the approach is reproducible rather than specific to this submission [9, 14].

11.4 Repository, Directory, and File Name Matching

The repository, directory, and file naming is itself a control. The IND lives under `trial-ind/` in the public repository, and within it the four build stages occupy `mermaid/`, `draft-ind/`, `full-ind/`, and `final-ind/`, mirroring the mermaid to draft to full to final progression. Within each LaTeX stage the section files follow a fixed `sec-NN-<name>.tex` convention whose number and name match the REGARDDD table of contents, so that the file that holds the Chemistry, Manufacturing and Control section is named `sec-06-cmc.tex`, the file that holds the Pharmacology and Toxicology section is named `sec-07-pharmacology-toxicology.tex`, and the figure that this section reproduces first is sourced from `fig-09-five-grounding-verifications.md`. Table 18 records the directory and file-naming correspondences that make the package self-describing, so that a reviewer can move from any path named in the prose directly to the corresponding location on disk without translation, and can confirm that no section is missing and no file is misnamed.

Path in repository v4.3.0	Holds	Name-match guarantee
<code>trial-ind/</code>	The full IND across four stages	Top-level directory named for the IND it contains
<code>trial-ind/mermaid/</code>	The 22 grayscale figure sources	Stage 1 directory; <code>fig-NN-<name>.md</code> files
<code>trial-ind/full-ind/section-06-cmc.tex</code>	The full-stage section files	Stage 3 sections; one <code>sec-NN-<name>.tex</code> per top-level section
<code>sec-06-cmc.tex</code>	Chemistry, Manufacturing and Control	File name matches the CMC section it contains
<code>sec-10-relevant-information</code>	Relevant Information (this section)	File name matches the section it contains
<code>fig-09-five-grounding-verifications.md</code>	Source for Figure 23	Figure file name matches the figure it renders

Table 18. Repository, directory, and file name-matching for the IND (Rule 5). Every path named in the prose denotes the actual location on disk in repository v4.3.0, making the package self-describing and navigable.

11.5 Limitations of the AI-Generation Method

The grounding and verification controls above constrain the build but do not eliminate its limitations, and those limitations are stated plainly here so that a reviewer weighs the package accordingly. Table 19 enumerates the principal limitations together with the mitigation applied to each. The most important point is the one that no method-level control can change: the AI-generation method alters none of the clinical realities of the trial. It does not change tumor biology, it does not shorten the 28 day dose-limiting-toxicity observation window or the 30 and 90 day safety follow-up or the survival follow-up to 24 months, and it does not compress the fixed 30 day FDA review clock or the IRB meeting calendar. What the method compresses is the administrative and document-preparation bucket alone; the clinical and operational bucket and the fixed regulatory-review bucket are unchanged [7, 13]. Every generated document, this IND included, must still pass human medical, statistical, and regulatory review before use, and the build is an independent research artifact, not an active IND or investigational device exemption and not regulatory advice [3, 11].

Several narrower limitations follow from how the build operates. The method does not rely on live internet search for a project of this size, because specifying single external URLs is unreliable at scale; it

depends instead on a curated repository of appropriately sized inputs, which bounds the source material a single build can absorb. Because the model does not always have access to a PDF compiler during generation, a portion of senior-author formatting, including table column widths, figure-caption spacing, and occasional page breaks, must still be applied by hand, and the final stage performs that pass explicitly. Total input is kept under roughly 5 MB with individual files held under per-file token limits, which caps how much source material a single build can ingest. High-dimensional or approximate inputs can make outputs less deterministic, and oversized or overly complex inputs can surface server errors that signal reduced quality; the build mitigates this by keeping inputs curated and appropriately sized. These constraints are the same ones characterized for the broader single-prompt method and are disclosed here for the specific case of an IND [12, 17, 19].

Limitation	Mitigation applied in this IND
The method changes no clinical reality (tumor biology, DLT window, survival follow-up)	The 28 day DLT window, 30 and 90 day safety follow-up, and 24 month survival follow-up are preserved exactly as specified in the protocol; the build compresses only document preparation
The method cannot shorten fixed regulatory clocks	The 30 day FDA review and the IRB meeting calendar are stated as binding and unchanged; faster authoring only starts the clock sooner
Generated content can be fluent but wrong without grounding	The five name-matching and grounding verifications and the figure verify-twice checkpoint bind every figure and number to a named source file
No live internet search at scale; URLs unreliable in bulk	Inputs are a curated repository of appropriately sized files rather than ad hoc external URLs
No PDF compiler during generation; some formatting needs a manual pass	Senior-author formatting (column widths, caption spacing, page breaks) is applied explicitly in the final stage
Bounded input budget (under 5 MB total; per-file token limits)	Source material is curated and apportioned so each build stays within limits without truncation
High-dimensional or oversized inputs reduce determinism and can surface server errors	Inputs are kept appropriately sized; complex content is split so the build remains deterministic and stable
The artifact is research, not an active regulatory submission	The package is labeled an independent research artifact, not an active IND or IDE and not regulatory advice; human review is required before any use

Table 19. Principal limitations of the AI-generation method and the mitigation applied to each in this IND. The clinical and fixed-regulatory limitations are intrinsic and cannot be removed; the method-level limitations are mitigated by grounding, verification, and a human review requirement.

11.6 Compilation Statement

All LaTeX sources for this IND compile in Overleaf with pdfLaTeX. The full-stage document is built from `main.tex`, the shared style file `indstyle.sty`, the bibliography `references.bib`, and the `sections/` directory of section files, with every figure rendered as vector TikZ inside the `mermaidfig` environment and every table set full width with the ragged-right column types defined in the style file.

There are no raster images, no `\includegraphics` calls, and no external image dependencies, so the package compiles from its own sources without additional assets. The bibliography is processed with the `ieeetr` style, and the codified references throughout the package use the section symbol, for example 21 CFR §312.23, exactly as rendered by the style file. A reviewer can therefore reproduce the compiled PDF directly from the repository at version v4.3.0 by uploading the stage directory to Overleaf and compiling with pdfLaTeX, which closes the loop from grounded source to verifiable, reproducible document [7, 1].

References and Back Matter

This back matter closes Initial Investigational New Drug Application v1.0 for the LLM-Directed PDAC Robotic Daraxonrasib Phase 1 study, assembled within repository v4.3.0. The complete reference list renders below through the `ieeetr` bibliography style from `references.bib`. Every entry carries a clickable URL and, where a digital object identifier is assigned, a clickable DOI URL with the DOI text shown in full; long links break on any character so that no link runs off the right margin of the text block. The reference base supporting this application is intentionally comprehensive and spans seven evidentiary strands: the FDA Phase 1 and combination drug-device regulatory guidances; the daraxonrasib (RMC-6236) clinical and mechanistic literature underpinning the RASolute 302 and RASolve 301 programs and the June 2025 Breakthrough Therapy designation; the 21 CFR Part 312 (IND) and Part 50 (informed consent) and ICH Physical Artificial Intelligence regulatory adaptations; the five author computational works dated 2025 to 2026 (the 60-second PDAC Whipple plus daraxonrasib simulation, the artificial intelligence digital-twin PDAC simulation, the quantitative systems pharmacology metastatic-PDAC simulation with verification, validation, and uncertainty quantification, the 100,000-patient in silico Phase 3 five-arm PDAC triplicate, and the end-to-end PDAC digital-twin trial proposals); the H.R. 9510 Verification Before Generation in Physical AI Oncology Trials Act of 2026 and the national platform and clinical-trust frameworks; the sixteen-large-language-model code-generation evaluation against the FDA Adverse Event Reporting System; and the proposed clinical research protocol v1.0.0 (DOI 10.5281/zenodo.20780121) that this IND operationalizes.

The back matter elements that follow document authorship and contributor attribution, the open-weights build model used to assemble this application, the ethical and independence disclosures that bound the entire repository, the data availability and licensing terms, and the recommended citation. Table 1 records the provenance of each major content strand so that a reviewer can trace every quantitative claim in IND Sections 3 through 11 back to its grounding source.

Acknowledgments

This Investigational New Drug Application was assembled with Claude Code (Opus 4.8 Ultracode build, 1M context) as the build model, operating under the single-prompt mermaid to draft to full to final pipeline described in IND Section 11 and committing each stage in real time. The author thanks ChatGPT 5.5 (Thinking Extended) and Gemini 3.1 Pro for research-source assistance and for manual editing support. This v1.0 application was edited and approved by Kevin Kawchak and added to repository v4.3.0 under `kevinkawchak/physical-ai-oncology-trials/tree/main/trial-ind`. The repository contributors are `@kevinkawchak`, `@claude`, `@google-gemini`, and `@openai`. No funding body, sponsor, or regulatory authority directed, reviewed, or endorsed the preparation of this document.

Author and ORCID

Kevin Kawchak, Chief Executive Officer, ChemicalQDevice, San Diego, California. ORCID iD: <https://orcid.org/0009-0007-5457-8667>. Correspondence: kevink@chemicalqdevice.com. In the proposed study the author would serve as Sponsor-Investigator on behalf of ChemicalQDevice; the resulting financial interest is disclosed under 21 CFR Part 54 and is mitigated by the independent three-tier oversight structure (Data Safety Monitoring Board, Independent Safety Monitor, and Physical AI Safety Review Committee) detailed in the proposed clinical research and additional information sections.

Ethical Disclosures

This is an independent research and regulatory-authoring demonstration. No real patients were enrolled, no protected health information was used, and no Institutional Review Board approval was sought or

Table 19: Provenance map for the major evidentiary strands of IND v1.0. Each strand is traced to its grounding source and to the IND sections it supports.

Evidentiary strand	Grounding source and key facts	Supports IND sections
Daraxonrasib mechanism and class	Oral RAS(ON) multi-selective (pan-RAS) inhibitor binding GTP-bound mutant RAS in complex with cyclophilin A; once-daily dosing [1]	3.1, 8.1
Dose-escalation design	3+3 escalation across DL1 160 mg, DL2 220 mg, DL3 300 mg once daily; 28-day DLT window; RP2D at or below MTD [7]	4, 6.1
Robotic Whipple device	On-premises LLM-directed eight-arm system, 56 degrees of freedom, 640 sensor channels, force caps 3 N per arm and 18 N cumulative [2]	5, 7
Verification before generation	Ten-gate ACCEPT/BLOCK/ESCALATE suite; LLM as second-opinion advisory oracle only; full autonomy prohibited under 21 CFR §312.21(e) [5, 6]	5, 10, 11
Perioperative pause-and-restart	ISGPS fistula grade plus serum trough versus 0.5 ng/mL; 32-iteration sweep yielding T+7d 29/32, T+14d 3/32, T+21d 0/32 [8]	6.1, 9
Computational evidence base	100,000-patient in silico Phase 3, QSP metastatic-PDAC simulation, and digital-twin trials [17, 16, 15, 18]	8, 9
Regulatory framework	21 CFR Part 312 and Part 50 Physical AI adaptations; ICH harmonization [3, 11, 12]	1, 6, 10, 11

Table 1. Provenance map linking each evidentiary strand to its grounding reference and to the IND sections it supports.

obtained. The exemplar case PAT-PDAC-0001 and all quantitative simulation outputs are synthetic and illustrative. This document is not an active Investigational New Drug Application or Investigational Device Exemption, is not a submitted or approved protocol, and does not constitute medical, surgical, or regulatory advice. It is not endorsed by the United States Food and Drug Administration, the National Institutes of Health, the Department of Health and Human Services, any Institutional Review Board, the International Council for Harmonisation, or any sponsor. Any future first-in-human conduct of the design described here would require a separately submitted IND and IDE, a 30-day FDA review period without clinical hold, and prospective IRB approval.

Data Availability

All application sources, spanning the mermaid, draft-ind, full-ind, and final-ind build stages, are available in the public GitHub repository and archived on Zenodo under the Creative Commons Attribution 4.0 International (CC BY 4.0) license at DOI 10.5281/zenodo.xxxxxxxx. The underlying proposed clinical research protocol v1.0.0 is archived separately at DOI 10.5281/zenodo.20780121. No proprietary, embargoed, or patient-identifiable data are contained in or required to reproduce this application.

Rights and Permissions

This article is distributed under the terms of the Creative Commons Attribution 4.0 International License (CC BY 4.0), which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are properly credited, a link to the Creative Commons license is provided, and any modifications made are indicated. To view a copy of this license, visit <https://creativecommons.org/licenses/by/4.0/>.

Cite This Article

Kawchak K. Phase 1 PDAC IND: AI Generation. Zenodo. 2026; 10.5281/zenodo.xxxxxxxx. IND v1.0, repository v4.3.0.

References

- [1] K. Kawchak, "A Phase 2, Daraxonrasib + LLM Guided Robotic PDAC Whipple Procedure, Clinical Trial Protocol." Zenodo, June 2026. DOI: 10.5281/zenodo.20807027. URL: <https://doi.org/10.5281/zenodo.20807027>.
- [2] K. Kawchak, "On-Premises LLM-Directed Robotic Pancreaticoduodenectomy with Perioperative Daraxonrasib (RMC-6236) in KRAS-Mutated Pancreatic Ductal Adenocarcinoma." Zenodo, June 2026. DOI: 10.5281/zenodo.20780121. URL: <https://doi.org/10.5281/zenodo.20780121>.
- [3] K. Kawchak, "Adaption: 21 CFR Part 312, End-to-End Physical AI Oncology Clinical Trial Unification." Zenodo, Mar. 2026. DOI: 10.5281/zenodo.19057628. URL: <https://doi.org/10.5281/zenodo.19057628>.
- [4] K. Kawchak, "Fully Automated Sponsor: Physical AI Oncology Clinical Trial Platform." Zenodo, Apr. 2026. DOI: 10.5281/zenodo.19396256. URL: <https://doi.org/10.5281/zenodo.19396256>.
- [5] K. Kawchak, "Earning the Congress's Vote: A New Oncology Trial Framework for Enacting H. R. 9510." Zenodo, June 2026. DOI: 10.5281/zenodo.20726461. URL: <https://doi.org/10.5281/zenodo.20726461>.
- [6] K. Kawchak, "Earning the Clinician's Trust: New Framework for Verified Physical AI Oncology Trials." Zenodo, June 2026. DOI: 10.5281/zenodo.20710602. URL: <https://doi.org/10.5281/zenodo.20710602>.
- [7] K. Kawchak, "Phase 1 Pancreatic Cancer Trial: Efficient LLM Document Guidance." Zenodo, June 2026. DOI: 10.5281/zenodo.21018646. URL: <https://doi.org/10.5281/zenodo.21018646>.
- [8] K. Kawchak, "Physical AI Oncology Clinical Trial Site Complete Documentation Package." Zenodo, Mar. 2026. DOI: 10.5281/zenodo.19176370. URL: <https://doi.org/10.5281/zenodo.19176370>.
- [9] K. Kawchak, "Oncology Trial PI LLM Adoption Guide." Zenodo, June 2026. DOI: 10.5281/zenodo.20843290. URL: <https://doi.org/10.5281/zenodo.20843290>.
- [10] K. Kawchak, "National Platform for Physical AI Oncology Trials." Zenodo, Mar. 2026. DOI: 10.5281/zenodo.19244918. URL: <https://doi.org/10.5281/zenodo.19244918>.
- [11] K. Kawchak, "Adaption: 21 CFR Part 50, End-to-End Physical AI Oncology Clinical Trial Unification." Zenodo, Mar. 2026. DOI: 10.5281/zenodo.19040707. URL: <https://doi.org/10.5281/zenodo.19040707>.
- [12] K. Kawchak, "Adaption: ICH Harmonised Guideline, End-to-End Physical AI Oncology Clinical Trial Unification." Zenodo, Mar. 2026. DOI: 10.5281/zenodo.18973368. URL: <https://doi.org/10.5281/zenodo.18973368>.
- [13] K. Kawchak, "2030: 60 Second Pancreatic Ductal Adenocarcinoma Robotic Whipple Procedure & Daraxonrasib Simulation." Zenodo, May 2026. DOI: 10.5281/zenodo.20196639. URL: <https://doi.org/10.5281/zenodo.20196639>.

- [14] K. Kawchak, "Accelerated Patient Prediction in Physical AI Oncology Clinical Trials: 4 Extensive Simulations." Zenodo, May 2026. DOI: 10.5281/zenodo.19994945. URL: <https://doi.org/10.5281/zenodo.19994945>.
- [15] K. Kawchak, "Accelerating FDA Compliance and Cost Efficiency of in silico Clinical Trials via AI Digital Twin Pancreatic Cancer Simulation." Zenodo, Oct. 2025. DOI: 10.5281/zenodo.17239510. URL: <https://doi.org/10.5281/zenodo.17239510>.
- [16] K. Kawchak, "QSP Metastatic Pancreatic Cancer AI Clinical Trial Simulation From Protocol to Prediction: Code, VVUQ, and Playbook." Zenodo, Aug. 2025. DOI: 10.5281/zenodo.17001137. URL: <https://doi.org/10.5281/zenodo.17001137>.
- [17] K. Kawchak, "ChatGPT 100,000 Patient 24-Month In Silico Phase III 5-Arm Pancreatic Cancer Clinical Trial Triplicate." Zenodo, July 2025. DOI: 10.5281/zenodo.16415815. URL: <https://doi.org/10.5281/zenodo.16415815>.
- [18] K. Kawchak, "End-to-End Pancreatic Ductal Adenocarcinoma Digital Twin Clinical Trial Proposals." Zenodo, June 2025. DOI: 10.5281/zenodo.15735068. URL: <https://doi.org/10.5281/zenodo.15735068>.
- [19] K. Kawchak, "Code Generation Competition: 16 Proprietary vs. Open-Source LLMs & Iterative Learning Based on FDA Adverse Event Reporting System." Zenodo, Dec. 2025. DOI: 10.5281/zenodo.18029100. URL: <https://doi.org/10.5281/zenodo.18029100>.
- [20] K. Kawchak, "From H. R. 9510 to Federal Law: A Narrative Case for Verified Physical AI Oncology Trials." Zenodo, June 2026. DOI: 10.5281/zenodo.20685379. URL: <https://doi.org/10.5281/zenodo.20685379>.
- [21] K. Kawchak, "H. R. 9510 (Bill v5.0) 2026." Zenodo, June 2026. DOI: 10.5281/zenodo.20619762. URL: <https://doi.org/10.5281/zenodo.20619762>.
- [22] K. Kawchak, "H. R. 9510 (Bill v4.0) 2026." Zenodo, June 2026. DOI: 10.5281/zenodo.20576907. URL: <https://doi.org/10.5281/zenodo.20576907>.
- [23] K. Kawchak, "H. R. 9510 (Bill v3.0) 2026." Zenodo, June 2026. DOI: 10.5281/zenodo.20535429. URL: <https://doi.org/10.5281/zenodo.20535429>.
- [24] K. Kawchak, "Verification Before Generation Act of 2026." Zenodo, June 2026. DOI: 10.5281/zenodo.20485580. URL: <https://doi.org/10.5281/zenodo.20485580>.
- [25] K. Kawchak, "VVUQ Physical AI Oncology Trial Bill." Zenodo, May 2026. DOI: 10.5281/zenodo.20454870. URL: <https://doi.org/10.5281/zenodo.20454870>.
- [26] K. Kawchak, "Mobile Pancreatic Cancer Unitree H2 Surgical Humanoid with Priority VVUQ." Zenodo, May 2026. DOI: 10.5281/zenodo.20421754. URL: <https://doi.org/10.5281/zenodo.20421754>.
- [27] K. Kawchak, "VVUQ Oncology Clinical Trial LLM Verification Automation Priority over Existing Generated Code." Zenodo, May 2026. DOI: 10.5281/zenodo.20372501. URL: <https://doi.org/10.5281/zenodo.20372501>.
- [28] K. Kawchak, "Threefold Humanoid 24/7 Adverse Event Oncology Trial Response Team: 4-Site Rotation." Zenodo, May 2026. DOI: 10.5281/zenodo.20303281. URL: <https://doi.org/10.5281/zenodo.20303281>.

- [29] K. Kawchak, "2030: 60 Second Glioblastoma AI Robotic Surgery." Zenodo, May 2026. DOI: 10.5281/zenodo.20113157. URL: <https://doi.org/10.5281/zenodo.20113157>.
- [30] K. Kawchak, "Patient Priority of Proposed U.S. Bills for Physical AI Oncology Clinical Trials." Zenodo, May 2026. DOI: 10.5281/zenodo.20045457. URL: <https://doi.org/10.5281/zenodo.20045457>.
- [31] K. Kawchak, "A Cancer Patient's Journey Through a Regulated Physical AI Oncology Trial." Zenodo, Mar. 2026. DOI: 10.5281/zenodo.19119939. URL: <https://doi.org/10.5281/zenodo.19119939>.
- [32] K. Kawchak, "National MCP Servers for Physical AI Oncology Clinical Trial Systems." Zenodo, Mar. 2026. DOI: 10.5281/zenodo.18916731. URL: <https://doi.org/10.5281/zenodo.18916731>.
- [33] K. Kawchak, "TrialMCP: MCP Servers for Physical AI Oncology Clinical Trial Systems." Zenodo, Mar. 2026. DOI: 10.5281/zenodo.18870961. URL: <https://doi.org/10.5281/zenodo.18870961>.
- [34] K. Kawchak, "Federated Learning Physical AI Oncology Trials Unification." Zenodo, Feb. 2026. DOI: 10.5281/zenodo.18795507. URL: <https://doi.org/10.5281/zenodo.18795507>.
- [35] K. Kawchak, "Unification Standard Level for Physical AI Oncology Trials." Zenodo, Feb. 2026. DOI: 10.5281/zenodo.18778220. URL: <https://doi.org/10.5281/zenodo.18778220>.
- [36] K. Kawchak, "AI Peer Review Acceleration of LLM-Generated Glioblastoma Clinical Trial Patient Matching ML, FDA/ICH/ISO, and FastAPI." Zenodo, Dec. 2025. DOI: 10.5281/zenodo.17774560. URL: <https://doi.org/10.5281/zenodo.17774560>.
- [37] K. Kawchak, "LLM-Generated Glioblastoma Drug Synergy Machine Learning: From Rapid Code Prototypes to Project Deliverables Package." Zenodo, Nov. 2025. DOI: 10.5281/zenodo.17614396. URL: <https://doi.org/10.5281/zenodo.17614396>.
- [38] K. Kawchak, "End-to-End Oncology Clinical Trial LLM Efficiency For Industry Adoption with FDA/ICH Regulations." Zenodo, Oct. 2025. DOI: 10.5281/zenodo.17451709. URL: <https://doi.org/10.5281/zenodo.17451709>.
- [39] K. Kawchak, "10 Year Glioblastoma Clinical Trial Meta-Analyses by Autonomous AI at Scale. Survival, HR, AE, and RoB scored in AI Reports and Charts, including Verifications." Zenodo, May 2025. DOI: 10.5281/zenodo.15549830. URL: <https://doi.org/10.5281/zenodo.15549830>.
- [40] K. Kawchak, "AI revolution toward the cure of lung adenocarcinoma." Zenodo, Apr. 2025. DOI: 10.5281/zenodo.15278152. URL: <https://doi.org/10.5281/zenodo.15278152>.
- [41] K. Kawchak, "Autonomous LLM Agent and scalable Reasoning LLM for generating cancer drug industry cost solutions." Zenodo, Mar. 2025. DOI: 10.5281/zenodo.15072843. URL: <https://doi.org/10.5281/zenodo.15072843>.
- [42] K. Kawchak, "Cost containment of global monoclonal antibody drugs and cancer clinical trials via LLM focused reasoning." Zenodo, Feb. 2025. DOI: 10.5281/zenodo.14968404. URL: <https://doi.org/10.5281/zenodo.14968404>.

- [43] K. Kawchak, "Clinical decision support based on Bevacizumab cancer trials and pushing the limitations of advanced LLMs." Zenodo, Jan. 2025. DOI: 10.5281/zenodo.14968162. URL: <https://doi.org/10.5281/zenodo.14968162>.
- [44] K. Kawchak, "Cancer vs. Conversational Artificial Intelligence." bioRxiv, Dec. 2024. DOI: 10.1101/2024.12.28.630597. URL: <https://doi.org/10.1101/2024.12.28.630597>.
- [45] K. Kawchak, "mAb Bioprocess Engineering In-Context Table Forecasts using Conversational AI Literature Insight Generations." ChemRxiv, Dec. 2024. DOI: 10.26434/chemrxiv-2024-jzbj0. URL: <https://doi.org/10.26434/chemrxiv-2024-jzbj0>.
- [46] K. Kawchak, "Monoclonal Antibody Bioprocess Engineering Advancements Using Conversational Artificial Intelligence." ChemRxiv, Oct. 2024. DOI: 10.26434/chemrxiv-2024-3m7m1. URL: <https://doi.org/10.26434/chemrxiv-2024-3m7m1>.
- [47] K. Kawchak, "Paclitaxel Biosynthesis AI Breakthrough." ChemRxiv, Oct. 2024. DOI: 10.26434/chemrxiv-2024-pqjd3. URL: <https://doi.org/10.26434/chemrxiv-2024-pqjd3>.
- [48] K. Kawchak, "High Dimensional and Complex Spectrometric Data Analysis of an Organic Compound using Large Multimodal Models and Chained Outputs." ChemRxiv, Sept. 2024. DOI: 10.26434/chemrxiv-2024-06gf1. URL: <https://doi.org/10.26434/chemrxiv-2024-06gf1>.
- [49] K. Kawchak, "LMM Spectrometric Determination of an Organic Compound." ChemRxiv, Aug. 2024. DOI: 10.26434/chemrxiv-2024-qtnkj. URL: <https://doi.org/10.26434/chemrxiv-2024-qtnkj>.
- [50] K. Kawchak, "LMM Chemical Research with Document Retrieval." ChemRxiv, Aug. 2024. DOI: 10.26434/chemrxiv-2024-p91gm. URL: <https://doi.org/10.26434/chemrxiv-2024-p91gm>.